FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Automated Microscopic Image Processing and Analysis for Supporting Bladder Cancer Diagnostics via Urinary Biomarkers

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Resumo

O cancro da bexiga (BCa) é um dos tumores malignos mais prevalentes em todo o mundo, com taxas de recorrência significativas e custos de saúde elevados relacionados com a necessidade de um acompanhamento prolongado. As técnicas de diagnóstico actuais, como a cistoscopia e a citologia da urina, são invasivas ou insuficientemente sensíveis e, por conseguinte, não são ótimas para diagnóstico primário ou para o seguimento repetido. A introdução de biomarcadores não invasivos representa uma possibilidade promissora. No entanto, até à data, nenhum biomarcador molecular demonstrou um desempenho de diagnóstico suficientemente comparável para complementar ou reduzir a dependência das práticas clínicas actuais de forma fiável.

Este estudo aborda a aplicação de técnicas de Visão por Computador (VC) e de Inteligência Artificial (IA) para apoiar o diagnóstico do cancro da bexiga através da análise de imagens microscópicas de amostras de urina. Ao desenvolver um método baseado em imagens para extrair caraterísticas e classificar potenciais biomarcadores urinários, pretende-se oferecer uma técnica não invasiva e económica para a deteção e vigilância do cancro da bexiga. A estratégia proposta visa reforçar as iniciativas de medicina personalizada e melhorar os cuidados prestados aos doentes, permitindo diagnósticos mais baratos, confortáveis, precisos e atempados.

As imagens microscópicas foram pré-processadas utilizando técnicas de segmentação, nomeadamente o método de *threshold* de Otsu e uma abordagem baseada na desfocagem de Gauss, *threshold* Adaptativo e *Dilate* (GAD). Após a deteção e o refinamento dos contornos para segmentar células, foram extraídas caraterísticas das células e utilizadas para treinar classificadores de *Machine Learning* (ML) para três tarefas de diagnóstico: deteção binária de cancro, estadio do cancro e grau do cancro. Os modelos foram avaliados utilizando métricas como a exatidão, a exatidão equilibrada, a sensibilidade e a especificidade.

Os resultados mostraram que ambas as estratégias de pré-processamento produziram contornos viáveis para a extração de caraterísticas, com o GAD a oferecer uma cobertura mais ampla e melhores resultados em exatidão equilibrada, enquanto o Otsu a proporcionar uma deteção mais equilibrada entre os pacientes. Os classificadores treinados nas caraterísticas celulares geradas obtiveram um desempenho promissor, particularmente na tarefa de classificação binária, apoiando a viabilidade do quadro de diagnóstico não invasivo proposto. No entanto, são necessárias mais investigações e validações em conjuntos de dados maiores e clinicamente anotados para confirmar a sua fiabilidade e utilidade como ferramenta complementar em fluxos de trabalho de diagnóstico do mundo real.

Palavras-chave: Visão por computador, Segmentação de imagens, Extração de features, *Machine learning*, Cancro de bexiga, Imagens de citologia

Abstract

Bladder cancer (BCa) is among the most prevalent malignancies worldwide, with significant recurrence rates and high healthcare costs related to the requirement for prolonged follow-up. Current diagnostic techniques, such as cystoscopy and urine cytology, are invasive or insufficiently sensitive and are, therefore, less than ideal for primary diagnosis or repeated follow-up. The introduction of non-invasive biomarkers represents a promising possibility. However, no molecular biomarker up to this point has demonstrated diagnostic performance comparable enough to complement or reduce reliance on current clinical practices reliably.

This study addresses the application of Computer Vision (CV) and Artificial Intelligence (AI) techniques to support bladder cancer diagnosis through microscopic image analysis of urine samples. By developing an image-based method for feature extraction and classification of potential urinary biomarkers, we aim to offer a non-invasive and cost-effective technique for bladder cancer detection and surveillance. The proposed strategy aims to enhance personalized medicine initiatives and improve patient care by enabling cheaper, more comfortable, accurate, and timely diagnoses.

Microscopic images were preprocessed using segmentation techniques, namely the Otsu thresholding method and an approach based on Gauss blur, Adaptive thresholding, and Dilation (GAD). After contour detection and refinement to segment cells, cell features were extracted and used to train machine learning (ML) classifiers for three diagnostic tasks: binary cancer detection, cancer staging, and risk group classification. The models were evaluated using metrics such as accuracy, balanced accuracy, sensitivity, and specificity.

The results showed that both preprocessing strategies yielded viable contours for feature extraction, with GAD offering broader coverage and more balanced accuracy, while Otsu providing more sensitivity across patients. Classifiers trained on the cell features generated achieved promising performance, particularly in the binary classification task, supporting the feasibility of the proposed non-invasive diagnostic framework. Nonetheless, further investigation and validation on larger, clinically annotated datasets are necessary to confirm its reliability and utility as a complementary tool in real-world diagnostic workflows.

Keywords: Computer vision, Image segmentation, Feature extraction, Machine learning, Bladder cancer, Cytology images

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As this path comes to an end, I cannot help but take a moment to reflect on all the journey it encompassed. Closing this chapter is more than just finishing an academic project. It is about recognizing a transformation that has taken place within me. Along this path, I have learned to be patient with myself, to accept that knowledge is not a straight line but a path full of curves and unexpected discoveries, that there is time for many more things than we usually acknowledge, and to believe in what I can achieve by staying motivated, even when it feels like all hope is lost. I have grown in silence, among books, slides, videos, codes, and nights that seemed endless, but also in small moments of light, in which everything made sense. This thesis reflects not only my effort, but also how I have become more resilient, secure, and, above all, confident in my ability to face the unknown.

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Mariana Lima Teixeira

"Great things are done by
a series of small things
brought together"

Vincent van Gogh

Contents

1 Introduction **1**
- 1.1 Problem Definition . . . 2
- 1.2 Research Questions . . . 3
- 1.3 Contributions . . . 3
- 1.4 Sustainable Development Goals . . . 3
- 1.5 Dissertation Structure . . . 4

2 Literature Review **6**
- 2.1 Understanding BCa . . . 6
- 2.2 Context of BCa Diagnostics . . . 8
- 2.3 Comparison of current approaches . . . 11
- 2.4 Emergent methods and their limitations . . . 12
- 2.5 The Role of Computer Vision in Diagnosis . . . 14
- 2.6 Summary . . . 15

3 Computer Vision Foundational Concepts **17**
- 3.1 Image Preprocessing . . . 17
  - 3.1.1 Noise Reduction and Filtering . . . 18
  - 3.1.2 Color Space Conversion . . . 19
  - 3.1.3 Contrast Enhancement . . . 21
  - 3.1.4 Normalization . . . 22
- 3.2 Object Detection and Segmentation . . . 22
  - 3.2.1 Object Detection . . . 22
  - 3.2.2 Segmentation Techniques . . . 23
- 3.3 Feature Extraction . . . 27
  - 3.3.1 Types of Features . . . 27
  - 3.3.2 Relevance to Microscopic Imaging . . . 28
- 3.4 Feature Analysis and Classification . . . 28
  - 3.4.1 Statistical and Machine Learning Approaches . . . 28
  - 3.4.2 Dimensionality Reduction and Feature Selection . . . 29
  - 3.4.3 Interpretability and Explainability . . . 29
- 3.5 Summary . . . 29

4 Methodology **31**
- 4.1 Pipeline Description . . . 31
- 4.2 Dataset . . . 32
  - 4.2.1 Patient and Label Information . . . 33
  - 4.2.2 Data Examples and Distribution . . . 33

4.3 Preprocessing Pipeline . . . 35
4.3.1 Otsu's Thresholding . . . 36
4.3.2 Binary Thresholding . . . 37
4.3.3 Adaptive Thresholding . . . 38
4.3.4 Contrast-Colorspace-Morphological Operations (CCM) . . . 39
4.3.5 Adaptive-Gauss-Binary (AGB) . . . 40
4.3.6 Gauss-Adaptive-Dilation (GAD) . . . 41
4.3.7 Preprocessing Evaluation Strategy . . . 43
4.4 Cell Segmentation . . . 43
4.4.1 Contour Detection . . . 44
4.4.2 Contour Refinement . . . 46
4.4.3 Quality Control Metrics . . . 53
4.4.4 Implementation Considerations . . . 53
4.5 Feature Extraction and Classification . . . 53
4.5.1 Complete Feature Taxonomy . . . 54
4.5.2 Detailed Feature Specifications . . . 54
4.5.3 Normalization . . . 59
4.5.4 Classification Methods . . . 61
4.6 Summary . . . 64

5 Implementation and Results **66**
5.1 Segmentation Implementation Details . . . 66
5.2 Preprocessing and Segmentation Results . . . 68
5.2.1 Qualitative Analysis . . . 68
5.2.2 Quantitative Analysis . . . 72
5.2.3 Contour Overall Conclusions . . . 74
5.3 Classification Results . . . 76
5.3.1 Evaluation Metrics . . . 76
5.3.2 Binary Classification Results . . . 76
5.3.3 Undersampling Results . . . 80
5.3.4 Multiclass Classification – Cancer Stage and Cancer Risk . . . 81
5.3.5 Discussion of Results . . . 83
5.4 Overall Conclusions . . . 83

6 Conclusions and Future Work **85**
6.1 Conclusions and Main contributions . . . 85
6.2 Research Questions . . . 86
6.3 Future Work . . . 86

References **88**

A Confusion Matrices **98**

List of Figures

1.1 The Sustainable Development Goals. . . . 4

2.1 Bladder cancer stages of NMIBC and MIBC. . . . 7
2.2 Characterization of bladder cancer stages. . . . 7
2.3 Illustration of a cystoscopy performed with a cystoscope. . . . 8
2.4 Schematic representation of urine cytology. . . . 9
2.5 Urinary biomarkers. . . . 10
2.6 Example CV Pipeline for breast cancer detection. . . . 15

3.1 Effects of applying different blur filters on an image. . . . 18
3.2 Effects of applying different color spaces on the same image. . . . 20
3.3 Effects of applying different contrasts on the same image. . . . 21
3.4 CNN structure. . . . 23
3.5 Effects of various thresholds on an image. . . . 24
3.6 Effects of applying different contrasts on an image. . . . 25
3.7 Effects of applying edge and region-based segmentation methods on an image. . . . 27

4.1 CV pipeline structure applied. . . . 32
4.2 Images from the collected dataset. . . . 34
4.3 Distribution of image samples per diagnostic class. . . . 34
4.4 Distribution of image samples per patient. . . . 35
4.5 Otsu Thresholding result on a dataset image. . . . 37
4.6 Binary Thresholding result on a dataset image. . . . 38
4.7 Adaptive Thresholding result on a dataset image. . . . 39
4.8 CCM result on a dataset image. . . . 40
4.9 AGB result on a dataset image. . . . 41
4.10 GAD result on a dataset image. . . . 42
4.11 Contour detection steps. . . . 45
4.12 Schematic of Neubauer chamber grid structure. . . . 47
4.13 A dataset image with the Neubauer chamber grid. . . . 47
4.14 Effect of size-based filtering. . . . 49
4.15 Convex hull transformation. . . . 50
4.16 Final contour refinements. . . . 52

5.1 Segmentation results using Otsu Thresholding. . . . 69
5.2 Segmentation results using Binary Thresholding. . . . 69
5.3 Segmentation results using Adaptive Thresholding. . . . 70
5.4 Segmentation results using CCM method. . . . 70
5.5 Segmentation results using AGB method. . . . 71

5.6 Segmentation results with GAD method. 71
5.7 Total contours detected across different preprocessing algorithms. 72
5.8 Heatmap of number of contours per image per patient per preprocessing method. 73
5.9 Average number of cells per image, grouped by patient and preprocess algorithm. 74

A.1 Binary Classification Confusion Matrix. 98
A.2 Binary Classification Confusion Matrix with Undefined Patients. 99
A.3 Binary Classification Confusion Matrix with Undersampling. 100
A.4 Stage Confusion Matrix with Logistic Regression in Otsu. 101
A.5 Risk Confusion Matrix with MLP in Otsu. 102

List of Tables

2.1 Comparison of current diagnostic methods for BCa. 12

4.1 Extended Feature Categories and Their Biological Significance. 54

5.1 Binary classification results for each ML model using GAD and Otsu methods. . 77
5.2 Binary classification results including undefined stage or risk patients for each ML model using GAD and Otsu methods. 79
5.3 Binary classification results with undersampling including undefined stage or risk patients for each ML model using GAD and Otsu methods. 81
5.4 Multiclass Classification results for each ML model using GAD and Otsu methods. 82

Abbreviations and Symbols

AGB	Adaptive-Gauss-Binary
AI	Artificial Intelligence
BCa	Bladder Cancer
BTA	Bladder Tumor Antigen
C	Cancerous
CCM	Contrast-Colorspace-Morphological Operations
cfDNA	cell-free DNA
CIS	Carcinoma in situ
CLAHE	Contrast-limited adaptive histogram equalization
CNN	Convolutional Neural Network
CTCs	Circulating Tumor Cells
ctDNA	Circulating Tumor DNA
CV	Computer Vision
EV	Extracellular Vesicle
FDA	Food and Drug Administration
FN	False Negative
FP	False Positive
GAD	Gauss-Adaptive-Dilation
H	Healthy
HG	High-grade
HLS	Hue, Lightness, and Saturation
HSV	Hue, Saturation, and Value
KNN	K-Nearest Neighbors
LAB	Lightness, A, B
LG	Low-grade
MIBC	Muscle-Invasive Bladder Cancer
ML	Machine Learning
NMIBC	Non-Muscle-Invasive Bladder Cancer
NMP22	Nuclear Matrix Protein 22
ROI	Region Of Interest
RGB	Red, Green, and Blue
Tis	Tumor in situ
TN	True Negative
TP	True Positive

Chapter 1

Introduction

We are presently experiencing a true revolution in healthcare, driven by the ongoing progression of technological innovations that are transforming the diagnosis and treatment [1]. In particular, personalized medicine has seen significant advances, yet certain cancers, such as bladder cancer, continue to present unique challenges. Bladder Cancer (BCa) [2] is one of the ten most common cancers in the world, ranking seventh in the prevalence of all malignancies [3], associated with increasing incidence and high mortality rate [4]. Because of its complicated nature and multiple molecular and pathologic pathways, together with its high recurrence rate (up to 80%) [5], it admits a high morbidity rate, along with high healthcare costs [4, 6].

BCa progresses in stages, from non-muscle-invasive BCa — a generally less aggressive disease — to muscle-invasive BCa, carrying a much worse prognosis. The clinical behavior in this disease varies with tumor stage and molecular subtype. The complexity and frequent recurrence of BCa lead to a high demand for ongoing monitoring, which incurs substantial healthcare costs, making this cancer the most expensive to treat over a patient's lifetime of all cancers [2, 4, 6]. Monitoring and diagnosis of BCa involve frequent invasive tests, most notably periodic cystoscopies [7], which, though reliable, are uncomfortable to the patients and also carry associated risks of comorbidities [8] due to the invasiveness of the medical procedure [9]. Besides cystoscopy, the non-invasive process urinary cytology [9] is also performed as a diagnostic procedure; however, it often results in inconclusive diagnoses or false-negative results that further complicate effective management. Although urine cytology is inexpensive and easy to perform, its low sensitivity prevents it from being used as a primary and ideal diagnostic tool for suspected BCa [4, 10].

Biomarkers [11] have gained rising importance in healthcare, specifically in BCa, due to their potential for improving diagnostics and treatment strategies at a non-invasive level [12]. Despite representing a massive potential in BCa diagnostics and follow-up, no molecular biomarker has been, up to date, clinically adopted due to lacking the diagnostic accuracy required to replace cystoscopy [2, 13]. The use of next-generation biomarkers — alternatively to the molecular ones — represents an important step toward personalized medicine that offers the possibility to develop non-invasive diagnostics capable of early disease detection and effective follow-up of treatment

responses [12]. Such a non-invasive diagnostic technique could substantially improve patient quality of life by reducing the number of invasive interventions and enabling comfortable and timely disease monitoring [14].

Owing to the rapid development of computational resources, Artificial Intelligence (AI) and Machine Learning (ML) have undergone tremendous growth in the last couple of years. These technologies are gaining traction in medical diagnostics, particularly in image analysis. Specifically, regarding the BCa, AI algorithms have enabled important advancements, such as the precise segmentation of the bladder wall and accurate detection of bladder tumors using non-invasive 3D image-based features [15]. While the techniques of Computer Vision (CV) seem promising regarding both the diagnosis and treatment techniques, a considerable amount of research is yet to be done to unlock their potential [16]. The application of AI to microscopic image analysis has not been explored much in order to characterize cancerous cells and biomarkers. These may one day provide sensitivities and specificities higher than ever, plus non-invasive diagnostic options, although it would take further research and development to integrate these technologies into the clinical workflow for better patient care and disease monitoring [15, 16].

Hence, the motivation for this work is to fill this gap by constructing a method with comparable or better sensitivity that could enable the analysis of biomarkers in urine through a CV algorithm. Ultimately, we aim for the developed algorithm to become one of the many elements in the transformation of diagnostics and BCa follow-up. In view of the microscopic examination of urine samples, an attempt will be made to provide a different, non-invasive way of identifying relevant biomarkers but in a much more accessible manner than the current protocol used for cytological evaluation, thus contributing to effective and less burdensome management of the disease.

1.1 Problem Definition

This research points to the essential issue: the absence of reliable and non-invasive diagnostic methodologies of BCa for clinical application. That is, the present dependence on invasive procedures, like cystoscopy, implies a physical burden for the patients and restricts the possibility of frequent and comfortable monitoring. Since cytology has insurmountable drawbacks and there is a lack of widespread biomarker adoption, the need for a new approach that can make up with accuracy for patient comfort is warranted. While biomarker analysis holds great promise for the improvement of BCa detection and follow-up, its integration with new technologies may bring broader improvements in outcomes.

This research focuses on developing a CV-based solution for microscopic urine sample analysis to detect and characterize biomarkers associated with BCa. Unlike the current cytological approaches, which often rely on chemical processing of samples for the identification of different cells exfoliated in urine [17], here we aim to provide a CV algorithm capable of identifying chemically unmodified, unstained cells. The proposed tool will contribute to standard diagnosis by providing a non-invasive technique intended to complement current procedures, with the potential to eventually reduce — or even replace — the need for the actual invasive methods. Moreover, the

method will be cost-effective, with low human technical requirements, and easily deployable as it needs only a standard optical microscope, enabling diagnostic automation. Supporting biomarker analysis may also result in less traumatic monitoring for patients.

1.2 Research Questions

This research aims to develop a low-cost, automated, and non-invasive diagnostic approach for BCa by leveraging CV and ML techniques on microscopic urine samples. The overarching goal is to improve early detection and characterization of cancer-related biomarkers, ultimately supporting clinical decision-making and reducing the reliance on invasive procedures.

To address the challenges mentioned, this research will explore the following questions:

- How can microscopic image analysis techniques identify and characterize biomarkers for the early detection of bladder cancer?
- What is the role of computer vision techniques in effectively improving the sensitivity and specificity of biomarker detection in urine samples?
- In what ways do machine learning algorithms improve diagnostic accuracy and predictive capabilities for bladder cancer biomarkers?

1.3 Contributions

This work presents a CV-based approach for the analysis of microscopic urine samples with the goal of detecting and characterizing biomarkers associated with BCa. A complete image processing pipeline was developed to segment and analyze unstained exfoliated cells. A labelled dataset comprising samples from both healthy and cancerous patients was curated, with careful group-based splitting to prevent patient-level data leakage. Classification models were trained and evaluated to differentiate between healthy and cancerous cells based on features extracted from the cell images. The performance of the proposed method was assessed through quantitative metrics such as sensitivity and specificity, demonstrating its potential as a non-invasive and cost-effective diagnostic support tool.

1.4 Sustainable Development Goals

The Sustainable Development Goals (SDGs) are 17 United Nations objectives adopted in 2015 that form a plan for a more sustainable future to be achieved by the year 2030. They address global issues, such as poverty, education, healthcare, and climate change. Figure 1.1 represents the set of goals. This dissertation aligns with three of them: Goal 3 – good health and well-being; Goal 9 – Industry, Innovation, and Infrastructure; and Goal 10 – Reduce Inequalities.



Figure 1.1: The Sustainable Development Goals [18].

This work supports Goal 3 by contributing to early, non-invasive cancer diagnosis. It aligns with Goal 9 by promoting accessible digital innovation in medical diagnostics using AI and CV, and with Goal 10 by fostering equitable access to diagnostics in resource-limited settings.

1.5 Dissertation Structure

In addition to this introductory chapter, the dissertation is organized into the following chapters:

- **Chapter 2** presents the literature review in BCa diagnostics. It reviews conventional diagnostic methods, explores the development of non-invasive techniques, and highlights recent advances in the application of CV for biomarker discovery within microscopic imaging.
- **Chapter 3** introduces the fundamental concepts of CV that support the technical foundation of this work. It provides a concise overview of key techniques relevant to microscopic image analysis, including preprocessing, object detection, feature extraction, and feature analysis.
- **Chapter 4** details the methodology adopted in this research, describing the pipeline, the dataset, and the preprocessing techniques, along with the rationale behind the selected approaches. It also defines the segmentation methods and assumptions made. Finally, it describes the extracted features and ML models applied.
- **Chapter 5** describes the implementation details and presents and discusses the results obtained. It includes a comparative analysis of the preprocessing and segmentation strategies, as well as classification outcomes.

- **Chapter 6** concludes the dissertation by summarising the main contributions, identifying limitations, reflecting on the implications of the findings in the context of the research questions, and proposing directions for future work.

Chapter 2

Literature Review

This chapter presents a comprehensive overview of the literature related to the diagnostics of Bladder Cancer (BCa), addressing the ongoing challenges and the pressing need for innovative approaches in this domain. The increasing prevalence and associated economic and clinical burdens of BCa raise the need for reevaluating conventional diagnostic methods, emphasizing the potential of non-invasive techniques to improve patient outcomes.

The chapter starts by presenting a review of various historical and current diagnostic practices along with their advantages and limitations. It then goes further to describe new emerging technologies that might change the face of BCa diagnosis and management through Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision (CV). Finally, it identifies appropriate research gaps, focusing on the promising but still evolving role of urine biomarkers in enhancing cheap, accurate, and non-invasive diagnostics — especially when combined with AI-based approaches. These insights provide the needed context and relevance for the proposed study, supporting its research objectives.

2.1 Understanding BCa

Bladder Cancer (BCa) is a prevalent disease that represents a significant global health challenge, ranking as the seventh most common cancer and the most prevalent malignant tumor affecting the urinary system. The incidence of urinary BCa is higher in men [19], being the fourth most common cancer in men worldwide [3]. Prognosis and treatment choices depend on knowledge about the tumor's basic features in each patient [20]. Two crucial prognostic indicators for patients are the tumor's grade and stage. Urothelial carcinoma [19] is the predominant kind of BCa and is classified according to the 2004/2022 WHO grading system [21] into *low-grade* (LG) or *high-grade* (HG) — or *risk* —, based on the appearance of abnormal cells under a microscope, their growth rate, and risk of spread. While the LG cells resemble normal bladder cells and tend to grow slowly, having a lower risk of spread, the HG cells appear abnormal and have an aggressive growth, having a high risk of invasion or metastasis [20].

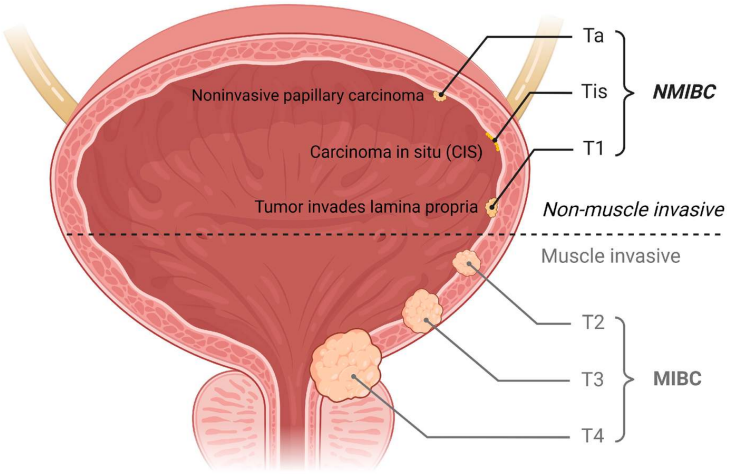


Figure 2.1: Bladder cancer stages of NMIBC and MIBC. Adapted from [22].

In terms of tumor invasion depth, BCa is further subdivided into muscle-invasive bladder cancer (MIBC) and non-muscle-invasive bladder cancer (NMIBC). This classification is closely related to tumor *stage*, as illustrated in Figure 2.1, which presents the stages of BCa for both NMIBC and MIBC.

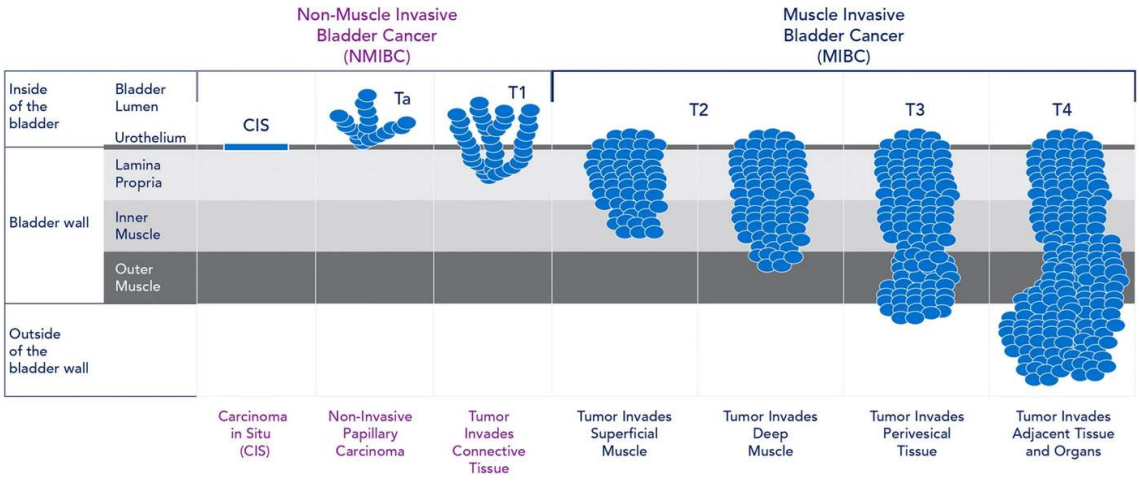


Figure 2.2: Characterization of bladder cancer stages. Adapted from [23].

Following the classification presented in Figure 2.1, Figure 2.2 illustrates the details of the progression of BCa through the tissue layers, according to the different tumor stages. Staging of BCa is based on the extent to which the tumor invades and spreads within and beyond the bladder. Ta, Tis (tumor in situ) or CIS (carcinoma in situ), and T1 are staged for NMIBC. Ta includes tumors that are limited to the urothelium with papillary architecture but do not extend into deeper tissue layers. These are flat, aggressive HG tumors limited to the urothelium. Stage T1 is the carcinoma extending to the lamina propria, the suburothelial layer of connective tissue, but not to the muscle layer. Stages T2, T3, and T4 comprise MIBC, with carcinoma extending into the

muscularis propria (muscle layer). Specifically, T2 tumors invade the musculature, T3 tumors extend outside the musculature into perivesical tissue, and T4 tumors invade nearby organs such as the prostate, uterus, vagina, or the pelvic or abdominal wall [23].

LG cancer is rarely MIBC, and NMIBC is typically LG and indolent, as only approximately one-third of NMIBC cases are HG lesions, with 20-25% of patients progressing to MIBC, which has a significantly lower 5-year survival rate. About 70-75% of newly diagnosed BCa cases are classified as NMIBC, which can be treated by transurethral resection of the bladder tumor (TURBT), aiming to improve life quality while lowering local recurrence and progression risk. However, the high recurrence rate of the disease requires continuous monitoring [24, 25].

2.2 Context of BCa Diagnostics

The current gold standard for diagnosing BCa is *cystoscopy*, an invasive procedure that allows direct visualization of the bladder interior. This examination is commonly performed for patients with hematuria (the presence of blood in the urine), both macroscopic and microscopic, to exclude BCa. During the procedure, a thin, flexible instrument called a cystoscope is inserted through the urethra, as illustrated in Figure 2.3. Equipped with a lens and light source, the cystoscope enables the doctor to examine the bladder's mucosa and identify any abnormalities, including tumors [26, 27].

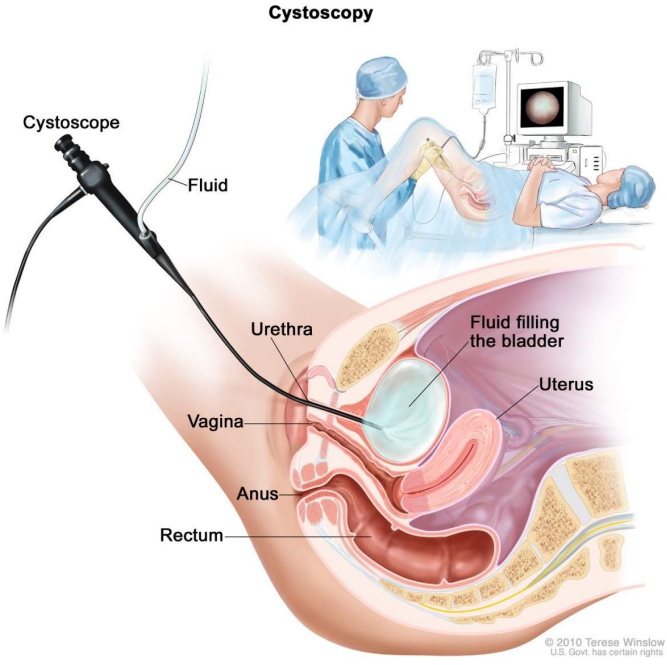


Figure 2.3: Illustration of a cystoscopy performed with a cystoscope [28].

Cystoscopy has undergone many developments in recent years. Originally a rigid procedure performed under general anesthesia, it has changed to a more flexible procedure carried out

outpatient with local anesthesia. Advanced modalities of cystoscopy include fluorescence/photodynamic cystoscopy, narrow-band imaging, and laser confocal endomicroscopy [27], some of which will be explored further in Section 2.4.

Along with cystoscopy, *urinary cytology* is one of the most common adjuvants in the diagnosis of BCa. It is a non-invasive technique by which exfoliated cells are examined through the microscope for morphological features of malignancy in urine sediment [29]. A sample of urine is collected, processed, and examined by a pathologist — a medical professional who diagnoses diseases by examining tissues and cells. During the cytological examination, the pathologist looks for abnormal cells of unusual size, shape, or nuclear structure. This may be indicative of cancer. Urinary cytology, though often employed in addition to cystoscopy, is an essential modality in diagnosing BCa [30]. It is represented in the Figure 2.4 below.

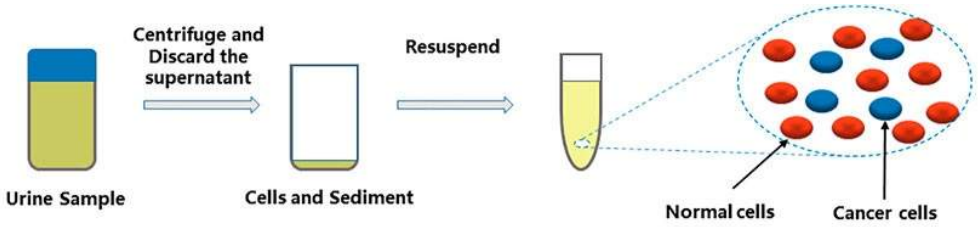


Figure 2.4: Schematic representation of urine cytology — microscopic analysis of healthy and cancerous cells from a urine sample. Adapted from [31].

Urinary cytology is the only widely used urinary marker in clinical practice. Urinary Biomarkers are tests that can be approved by the Food and Drug Administration (FDA) for the monitoring of BCa but currently are not for primary diagnosis because of their high false positive rates [32]. False positives are generated in conditions where there is increased cellular turnover, such as inflammation of the urothelial tract [33].

Urinary markers have attracted research attention for decades to reduce unnecessary cystoscopies in BCa detection. They hold potential as complementary tools for BCa diagnosis and NMIBC surveillance. Nevertheless, their clinical benefits remain to be confirmed. FDA-approved urinary markers include two protein-based urinary markers — bladder tumor antigen (BTA), and nuclear matrix protein 22 (NMP22) — and two cell-based urinary markers — Immunocyte/uCyt +, and UroVysion. While several urinary markers are FDA-approved, their clinical benefits remain inconclusive [33, 34]. In Figure 2.5, biomarkers are summarized depending on the urine specimen materials that have been used for their creation.

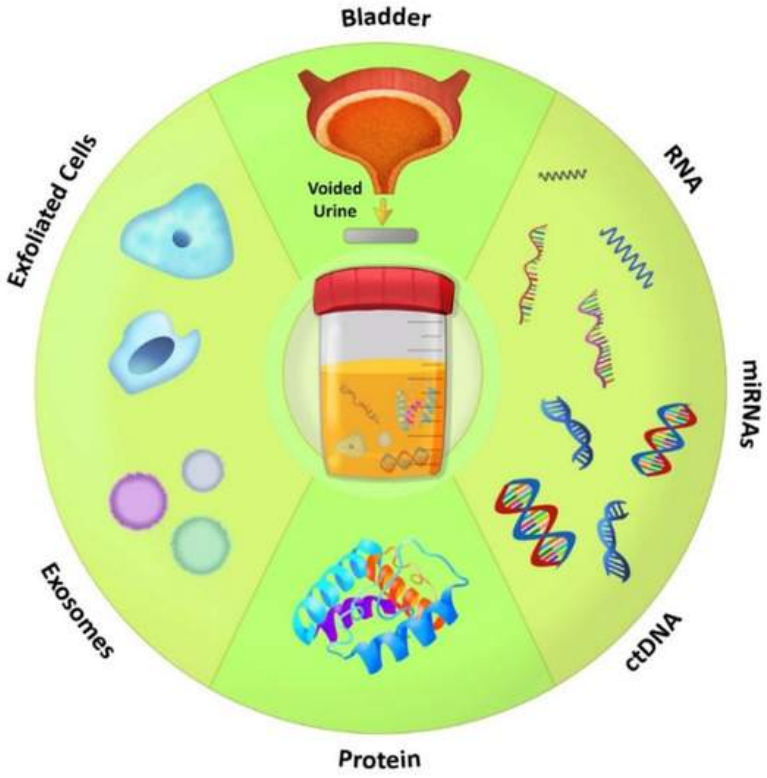


Figure 2.5: Urinary biomarkers. A diagram showing the various urine specimen materials used to create urine biomarkers [35].

Urine-based biomarkers have the potential to revolutionize BCa diagnosis by offering non-invasive and inexpensive options that may decrease dependence on frequent cystoscopies. Clinical practice may be enhanced by routine incorporation of these tests, considering that patient compliance could increase and lead to early detection of recurrence, especially in high-risk patients. Of course, further studies are needed to validate their accuracy and clinical utility.

In spite of the promising research results obtained with different FDA-approved urinary biomarkers, their use in routine diagnostics remains limited because of inconsistent sensitivity and specificity. Further, there are many limitations, including false positives, variations in sample collection, and non-standardization of test protocols, preventing the use of these markers as widespread clinical tools. The remaining barriers to overcome in this regard would be robust clinical validation and the development of more precise detection technologies.

Recent technological advances, including AI and CV-related ones, can enable better and more efficient analysis of urinary biomarkers. These technologies could improve diagnostic workflows through automated pattern detection in microscopic images, overcoming some of the shortcomings of the currently developed techniques.

2.3 Comparison of current approaches

After correlating the symptoms with BCa, urinary cytology and cystoscopy are commonly applied as the first-line approaches for diagnosis [36]. A closer examination of these techniques provides insights into their respective strengths and limitations within the diagnostic framework for BCa. The natural history of BCa necessitates regular follow-ups with cystoscopy and cytology at 3-month intervals. These procedures often involve repeated treatments due to recurrence, resulting in high morbidity and a substantial economic burden [37]. Regular cystoscopies require hospital visits, frequent use of operating rooms, and a specialized clinical team, making BCa one of the most expensive malignancies to manage across a patient's lifetime [38], particularly due to its long-term follow-up requirements. This highlights not only the health risks and patient discomfort associated with these procedures but also their broader social and economic implications.

Despite its widespread use, the diagnostic performance of cystoscopy itself has been less frequently studied, leading to inconsistent data on its accuracy, particularly in patients with hematuria. Nevertheless, cystoscopy is widely regarded as highly accurate, surpassing other imaging techniques, and remains the most reliable diagnostic procedure for BCa [27]. Yet, as an invasive method, it has significant limitations, including discomfort for the patient and potential complications such as urinary tract infections, bleeding, bladder perforation, scar tissue formation, and urinary retention. Additionally, the high costs associated with cystoscopy further underline its limitations [27, 29].

Urinary cytology, on the other hand, is the only widely used urinary marker in clinical practice. This laboratory test analyzes urine cells to detect cancerous changes and is highly sensitive and specific for HG BCa detection [39]. It is often used as an adjunct to cystoscopy to increase diagnostic sensitivity. However, its interpretation is subjective and heavily reliant on the expertise of pathologists, leading to challenges in standardization. While its specificity is superior to other established biomarkers, its low sensitivity for detecting LG tumors remains a critical drawback [29]. The limitations of urinary cytology stem from several factors. Only a small urine volume can be processed, and only a fraction of the sample is used for final analysis, reducing the likelihood of detecting tumor cells. The presence of other cells, such as red blood cells and leukocytes, can further confound the analysis. This low sensitivity limits cytology's potential for exclusive use in detection protocols [33, 40]. As such, cytology alone cannot safely reduce or replace the frequency of cystoscopies [29].

Urinary biomarkers have been explored to address these limitations [37]. Over the past decade, numerous urine-based biomarkers for BCa have been developed and commercialized. Despite outperforming cytology in sensitivity and specificity and potentially improving clinical practice by reducing cystoscopic examinations during the surveillance of NMIBC, none have been universally adopted or recommended in international guidelines [41]. This is due, for example, to limitations in performance across tumor grades and conditions. For instance, the sensitivity of NMP22 is limited, especially for LG tumors, and results can be influenced by isolated hematuria [33]. Although promising, the integration of molecular tests into routine practice requires further

validation studies [26]. Among the urinary biomarkers studied as alternatives to cystoscopy, only cytology has achieved routine clinical application, emphasizing its continued relevance despite its drawbacks.

The discussed traditional diagnostic methods for BCa are systematically compared in Table 2.1. While each technique has distinct advantages — such as cystoscopy's high accuracy or cytology's non-invasive nature — their collective limitations (like invasiveness, high costs, and low sensitivity for LG tumors) underscore the need for improved approaches.

Table 2.1: Comparison of current diagnostic methods for BCa.

Method	**Advantages**	**Limitations**	**Clinical Use**
Cystoscopy	Gold standard, real-time visualization, high accuracy [27]	Invasive, costly [29], patient discomfort, risk of complications	Primary diagnostic tool
Urinary Cytology	Non-invasive, high specificity for HG tumors	Low sensitivity for LG tumors, subjective interpretation, requires expert pathologists	Adjunct to cystoscopy
Urinary Biomarkers	Non-invasive, potential for early detection	Variable accuracy, high false positives, not FDA-approved for primary diagnosis	Monitoring and surveillance

These findings emphasize the pressing need for innovative diagnostic approaches in BCa management. Novel tests focusing on LG papillary tumor detection must be simple, effective, quick, and cost-efficient. To reduce the frequency of cystoscopies and improve the quality of life for patients, non-invasive techniques must rely on highly sensitive and specific markers [39]. Furthermore, enhancing marker sensitivity for HG cancer cases is crucial for early recurrence detection, ultimately improving patient survival rates [26]. The described drawbacks motivate the exploration of emergent technologies aiming to address these gaps while preserving clinical utility. The following sections delve into these innovations, beginning with liquid biopsy and molecular tests in the next section (Section 2.4).

2.4 Emergent methods and their limitations

Emerging methods for the diagnosis of BCa predominantly involve molecular approaches, such as liquid biopsy, molecular testing, and proteomics, which have shown great promise. However, emerging non-molecular techniques, such as those based on advanced imaging modalities, are also essential. Both groups face technical challenges and limitations that must be addressed to achieve successful clinical application [7, 42, 43].

Liquid biopsy concerns dynamic biological fluids, such as urine or blood, which present variability in biomarker concentrations according to several factors, like hydration status, renal pathologies, and medication effects. It offers a minimally invasive option with the potential for early diagnosis and real-time tumor monitoring [44]. However, its intra- and inter-patient variability complicates test result standardization. In addition, the isolation and characterization of specific extracellular vesicles (EVs) — key components for cell and tissue profiling — remain significant challenges due to their low abundance in biofluids. Circulating tumor cells (CTCs) and circulating tumor DNA (ctDNA) are also often present in low concentrations in blood and urine, complicating detection and analysis [45]. Furthermore, conventional approaches, like ultracentrifugation, are labor-intensive, expensive, and require large sample volumes, limiting clinical feasibility [46]. Ultimately, BCa tumoral heterogeneity may result in variable test outcomes because biomarkers may not represent the whole complexity of the tumor. Additionally, the lack of standardized methods for sample collection, storage, processing, and analysis of biomarkers makes it difficult to compare results between studies [43].

Early molecular urinary tests, such as NMP22 and BTA, suffer from low specificity and sensitivity, especially for LG tumors, often producing false positives and negatives due to benign conditions and cellular heterogeneity [33, 47]. More recent assays like UroVysion and CxBladder have improved diagnostic performance, but remain limited in clinical adoption because of high costs, complex workflows, and the need for specialized personnel and validation in larger patient cohorts [26, 33].

Proteomics, which enables large-scale protein profiling, offers additional opportunities for biomarker discovery but faces obstacles, including proteome complexity and interpatient variability [48, 49]. Post-translational modifications and poor standardization in analytical workflows further complicate its application [32, 43]. Specific challenges in BCa proteomics involve poor correlation between tumor tissue and urinary proteomics — likely due to tumor heterogeneity — and difficulties culturing LG tumors, which constitute most BCa cases, for research purposes [37]. These factors have slowed the translation of promising proteomic biomarkers into clinical practice.

Besides molecular diagnostics, imaging-based approaches are also being explored to improve BCa detection. For example, cystoscopy has been enhanced through modalities, such as fluorescence cystoscopy, narrow-band imaging, and laser confocal endomicroscopy, that enhance the visualization and detection of the tumor during procedures. These methods improve lesion detection sensitivity but require expensive specialized equipment and training, limiting widespread adoption [7, 50].

In summary, although molecular methods such as liquid biopsy, molecular assays, and proteomics dominate the landscape of emerging BCa diagnostics, advanced imaging techniques also offer valuable complementary insights, particularly in improving tumor visualization and detection. While all these approaches face significant technical, practical, and validation challenges, progress toward standardized methodologies, large-scale clinical validation, and technological innovation is critical for their integration into the routine clinical management of

BCa. Imaging-based strategies — especially those involving microscopic analysis — will be further explored in the next section (Section 2.5), as they offer promising avenues for addressing some of these challenges.

2.5 The Role of Computer Vision in Diagnosis

In response to the limitations of traditional diagnostic practices, Computer Vision (CV) has emerged as a powerful tool in the fight against cancer. Conventional methods like biopsy interpretation and imaging assessments often involve subjective judgment, require significant time, and depend heavily on expert availability. CV offers scalable, reproducible, and automated solutions for medical image analysis, supporting more accurate and efficient diagnostics across various clinical workflows. As cancer remains one of the most pressing global health challenges, enhancing diagnostic precision and speed through technologies like CV is crucial to improving patient outcomes.

CV techniques play a crucial role in image enhancement [51], isolating Regions of Interest (ROI), detecting abnormalities, and extracting important patterns to support clinical decisions [52]. In cancer diagnosis, these processes are particularly valuable because minor tissue structure or intensity changes can be the primary signs of malignancy. By automating core steps such as image preprocessing, segmentation, and pattern recognition, CV systems have become integral to many cancer detection workflows.

CV has been increasingly used in cancer diagnostics. Among the different imaging modalities in which CV is applied, microscopic image analysis offers unique advantages such as high-resolution visualization of cellular morphology and tissue architecture, scalability, cost-effectiveness, and automation potential. Despite these benefits, CV application to microscopy images remains unexplored, mainly in the case of BCa. Instead, most BCa CV research focuses on other modalities, such as cystoscopy imaging or molecular data analysis, which do not fully leverage the diagnostic richness and practical advantages of microscopy-based approaches [53, 54, 15].

In contrast, CV applied to microscopy images has been extensively developed and validated in other cancers, including breast, lung, and colorectal cancers [55, 56, 57]. Given the limited exploration of CV on microscopy data in BCa, this review turns to the more established CV research in microscopy for other cancer types. These examples may offer valuable insights and methodological frameworks that could be adapted to advance BCa diagnostics.

Among the many forms of cancer, breast cancer is a prime example of how CV in microscopy can be helpful in early detection and accurate diagnosis. In breast cancer, digital histopathology uses CV methods to automate tasks like cell segmentation, tissue classification, and grading, enabling more reproducible and scalable diagnostics. This microscopy-based CV leverages detailed cellular features that are inaccessible through radiological imaging.

A notable example illustrating the impact of CV and AI techniques in breast cancer diagnosis is presented in the study by Khan et al. [58]. The authors propose a comprehensive CV

pipeline (see Figure 2.6) that encompasses multiple stages, including image enhancement to reduce noise and improve contrast and feature extraction to capture breast tissue's critical textural and morphological characteristics. The study concludes that integrating AI-driven feature selection and multi-ensemble analysis significantly improves diagnostic accuracy and robustness, underscoring the promise of CV techniques in early cancer detection.

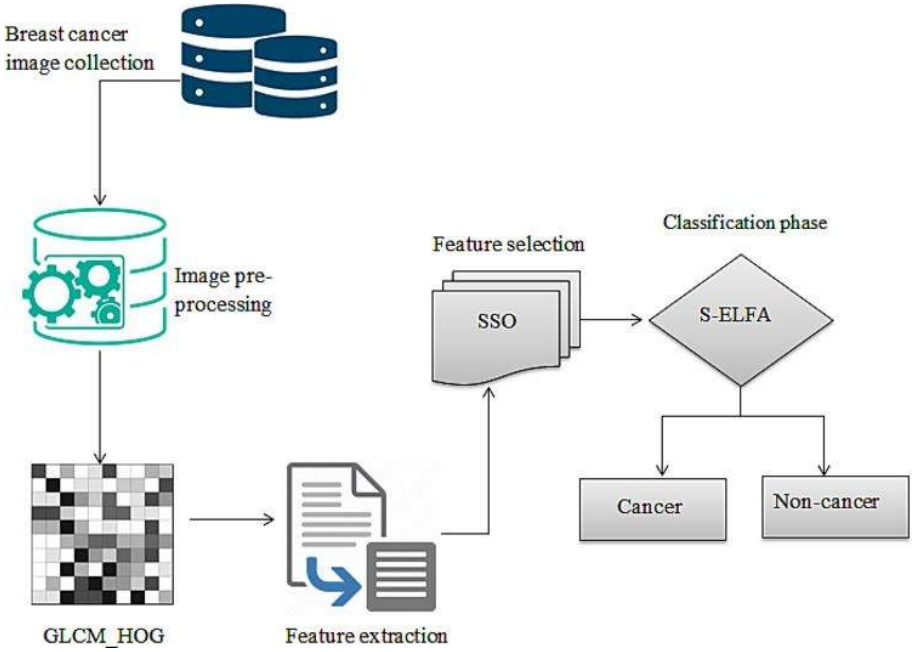


Figure 2.6: Example CV Pipeline for breast cancer detection [58].

Breast cancer is one of the most successful applications in CV for cancer, with extensive research, large public data collections, and accepted clinical utility. It illustrates the potential of CV-based systems to enhance diagnostic performance, reduce workload, and facilitate earlier detection even in low-resource settings. However, it differs from other cancers, such as bladder cancer, which have not yet benefited from the same degree of attention or technological advances, especially with microscopy images as a base dataset. Notably, while breast cancer diagnostics have benefited from these advanced CV methodologies supported by large annotated datasets, applying similarly sophisticated pipelines to microscopic images in BCa remains underdeveloped. This gap points to a valuable opportunity for future research to leverage these proven CV strategies, adapted to the microscopic scale, to enhance BCa diagnosis and patient outcomes.

2.6 Summary

CV has emerged as a game-changing technology in cancer diagnosis with the potential to automate, standardize, and optimize the interpretation of medical images. Unlike traditional diagnostic workflows that depend on manual inspection and clinical expertise, CV-driven systems

bring objectivity, scalability, and speed into image-based assessment. These aspects are especially valuable for their worth in gold in oncology, where early visualization of subtle visual characteristics significantly impacts treatment outcomes and patient survival.

Today's CV pipelines comprise several key stages, including image restoration, segmentation, and feature extraction. They enable the system to highlight clinically significant structures, isolate suspicious regions, and quantify morphological and textural features of tissues. These capabilities are generic and may be applied to a wide range of imaging modalities — several types of images are primarily valuable for assisting with the vital task of detecting cancer [59].

Many cancers, however, have not so much advancement in applying CV-based systems. NMIBC [41], for example, might require follow-up periodically by cystoscopy, an invasive and costly procedure. While other urinary biomarkers have been proposed to reduce the demand for repeated cystoscopies, consistency and accuracy limitations remain. On the other hand, the union of CV with imaging-based BCa diagnosis — e.g., cytology or histology — is also a promising avenue toward reduced patient burden and enhanced diagnostic reliability. The lack of annotated datasets and consistent imaging protocols inhibits progress. However, in the case of microscopy, it has been successfully applied in other cancers, while in BCa, a gap remains that we will fill.

Despite these variations in development, growing interest in applying CV to many cancer types is part of a broader trend: the shift toward data-driven, microscopic image analysis and image-assisted diagnostics in cancer treatment. By harnessing standard image analysis pipelines, CV can reduce inter-observer variability, speed up evaluations, and facilitate earlier intervention. As the technology continues to evolve, efforts must be directed toward lesser-explored cancer types and lower-resource health settings to spread CV advantages further.

Chapter 3

Computer Vision Foundational Concepts

Due to the interdisciplinary nature of this work, bridging cytology and engineering, this chapter presents a brief overview of the key concepts and methods in CV that serve as the foundation for subsequent analysis. Rather than delving into exhaustive technical details, it emphasizes principles and techniques relevant to a potentially more reasonable analysis of microscopic images, vital for detecting and characterizing bladder cancer (BCa) diagnostic biomarkers. The chapter addresses the main steps of CV processes, including image preprocessing, object detection and segmentation, feature extraction from segmented regions, and feature analysis for image classification [52]. Each step contributes to developing reliable and interpretable diagnostic tools, particularly for BCa. This approach aims to establish the subsequent methodological choices on solid theoretical foundations while gradually demonstrating how these concepts address the specific challenges of analyzing microscopic data in BCa diagnostics.

3.1 Image Preprocessing

Image preprocessing is a required first step in any CV pipeline [60]. Preprocessing techniques are intended to improve the raw microscopic image quality and minimize noise, illumination, or staining variability. Standard operating procedures include grayscale conversion, histogram equalization, denoising filters (e.g., Gaussian and median blurs), contrast stretching, and normalization [61]. These procedures improve the quality of images as data sources for further analysis, such as segmentation and feature extraction, making subsequent image processing easier with normalized and more meaningful data [62]. Microscopic image preprocessing is even more important due to problems like non-uniform illumination, noise, staining artifacts, low contrast between large structures and backgrounds, and other sample-related issues such as air bubbles or debris [63].

3.1.1 Noise Reduction and Filtering

Microscopy images will have noise caused by the imaging apparatus or sample preparation. Noise is removed with techniques like Gaussian filtering, median filtering, or bilateral filtering. These techniques preserve essential structures and remove high-frequency noise that hampers feature extraction or segmentation. These filtering techniques are selected based on the noise type, the imaging modality, and the importance of structural preservation for downstream tasks.

Gaussian filtering is a general blur operation that does a convolution with a Gaussian kernel to smooth the image and eliminate high-frequency fluctuations. While this operation effectively eliminates common noise, it can also blur thin edges [64].

Median filtering, on the other hand, replaces each pixel value with the median value of the local neighborhood. It effectively combats salt-and-pepper noise and better preserves edges than Gaussian blur [65].

Bilateral filtering is a generalization of Gaussian smoothing with intensity variations in the filter weights; thus, the noise gets minimized while preserving sharp edges. This is extremely useful in microscopy, where cell boundaries must be preserved to interpret analysis correctly [65].

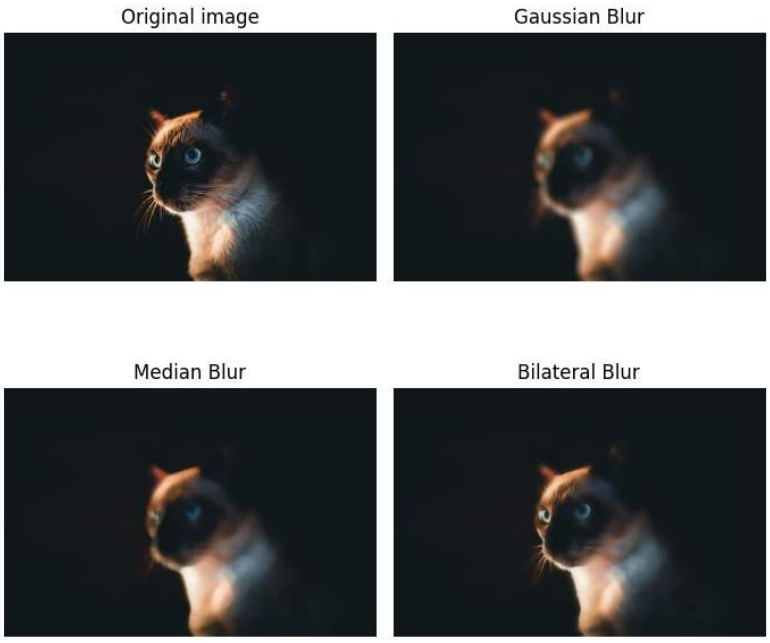


Figure 3.1: Effects of applying different blur filters on an image. Adapted from [66].

Figure 3.1 illustrates the visual effects of different filtering methods on a sample image. In Gaussian blur, the image is blurred overall due to its smoothing capability and elimination of high-frequency differences in the image. In median blur, the image is less blurred as small image segments take the median value of a local neighborhood, but it remains very different from the original image. In the case of bilateral blur, the blur is even smoother, as its edges remain sharp.

3.1.2 Color Space Conversion

Color space conversion is a critical preprocessing step in most CV processing pipelines, particularly for color image processing. It involves the transformation of an image from one color representation to another to facilitate specific analysis or accentuate certain features. Common color spaces are RGB (Red, Green, and Blue), HSV (Hue, Saturation, and Value), HLS (Hue-Lightness-Saturation), LAB (Lightness, A, B), and grayscale [67].

The RGB color space is the native form of most imaging devices and digital pictures, with colors represented as a mixture of red, green, and blue channels. Each channel typically uses 8-bit encoding, allowing intensity values to range from 0 to 255. However, such a representation can be unsuitable for image processing since it is highly correlated between channels and light sensitivity.

In most practical implementations, such as OpenCV [68], converted color spaces like HSV and HLS also use 8-bit channels, with values typically ranging between 0 and 255. These alternative color spaces such as HSV and HLS separate chromatic information from brightness, allowing algorithms to calculate color information independently of brightness. For instance, in HSV, the hue component is the type of color, saturation is the intensity, and value is the brightness. HLS is identical in this regard but replaces value with lightness, which can be helpful in certain types of contrast changes or color-based segmentation.

LAB color space separates lightness from color-opponent axes and tries to emulate human vision perception. It may enhance discrimination between colors in high-detail images, such as microscopy images with multi-stain differences.

Grayscale conversion is typically performed as an optional pre-step for most image processing algorithms, such as thresholding or edge detection because it makes computation easier at the expense of maintaining important structural information.

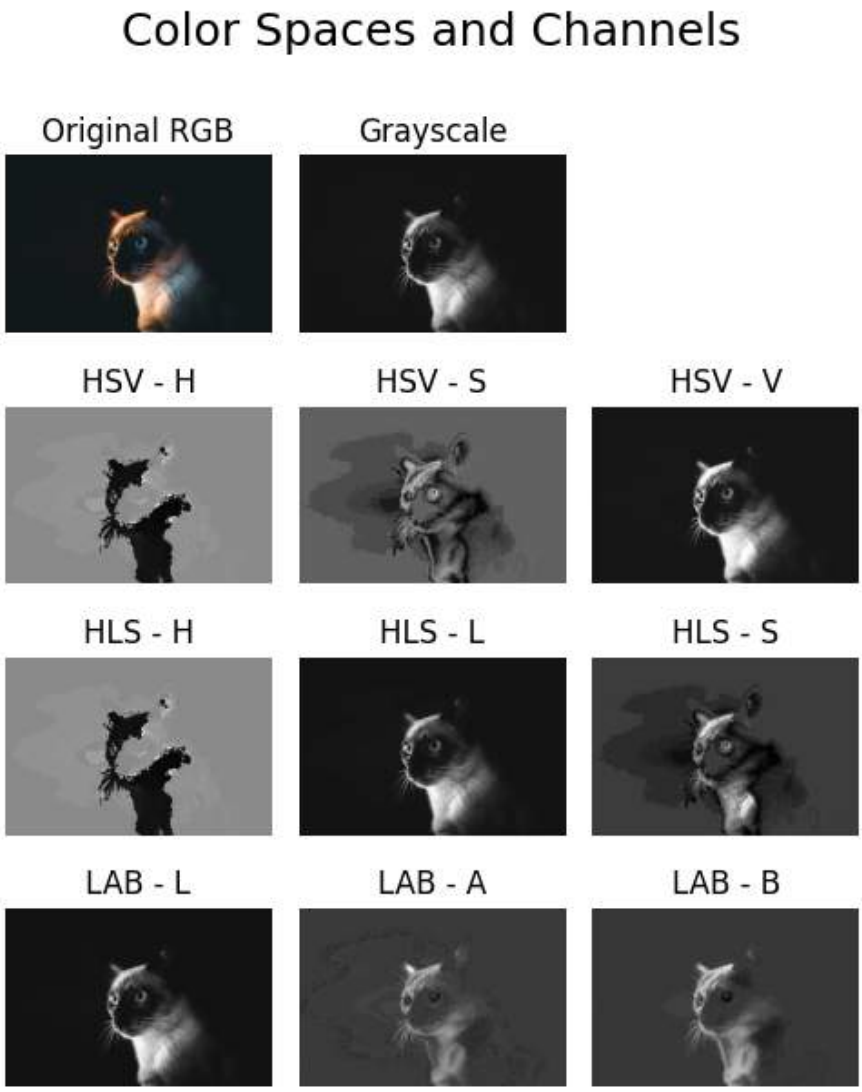


Figure 3.2: Effects of applying different color spaces on the same image. Adapted from [66].

Figure 3.2 illustrates what the same picture appears like under different color space representations. Each space separates the color information differently, with individual effects on the features that appear in their corresponding channels. For instance, subtle color changes can be more noticeable in the A and B channels in the LAB space, whereas HSV and HLS can separate hues or brightness differences between forms. Conversely, grayscale collapses all color information into intensity, striving to emphasize texture or morphological structure.

In an analysis of microscopic images, color space conversion into appropriate color spaces can significantly improve the accuracy of subsequent steps, i.e., segmentation and classification, by enhancing color differentiation between the background and biological structures or highlighting diagnostically significant features [69].

3.1.3 Contrast Enhancement

Contrast enhancement is a simple image processing operation that improves the visibility of details in an image by intensifying the contrast difference between objects and the background. Contrast enhancement is utilized extensively in photography, remote sensing, surveillance, and medical imaging to reveal important structures that would otherwise be obscure. Techniques such as histogram stretching, histogram equalization, and adaptive methods, like contrast-limited adaptive histogram equalization (CLAHE), attempt to redistribute the intensity levels to range over the whole extent of the display scale, promoting the visibility of informative content to the human or machine observer [70].

CLAHE is a type of adaptive histogram equalization, or AHE, a computer image processing algorithm that improves contrast in images. Unlike histogram equalization, this adaptive algorithm divides the image into small regions, called tiles, and computes several histograms for each, using them to redistribute the image lightness values. CLAHE introduces a contrast limiting step to prevent noise over-amplification, especially in homogeneous regions. This makes it particularly effective at enhancing local features and edge definition without introducing artifacts [70, 71, 72].

Histogram equalization and CLAHE are most frequently used to enhance their visibility. In biomedical images, poor object-background contrast between objects of interest, such as cells and biomarkers, and the surrounding medium makes it hard to identify them. However, it should be cautiously applied to avoid over-enhancing diagnostically relevant features [70, 72].

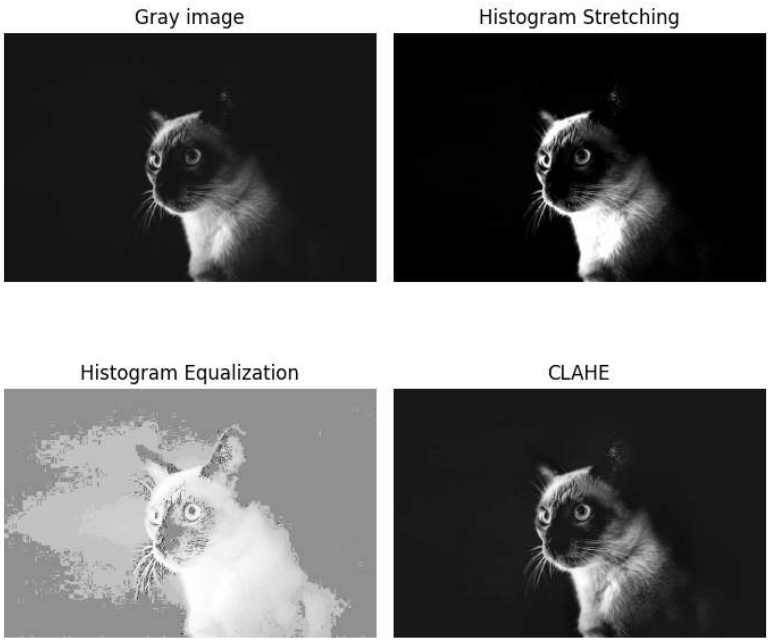


Figure 3.3: Effects of applying different contrasts on the same image. Adapted from [66].

In Figure 3.3, there is the result of applying different contrast techniques to the same original image. These techniques are applied in the grayscale version of the image. The first illustrated

technique is histogram stretching, which applies a linear transformation of intensity values to the full dynamic range available, improving global contrast [70]. Histogram equalization redistributes pixel intensities to create a flatter histogram, improving global contrast but sometimes creating unnatural artefacts in regions of flatness. CLAHE is applied to small image regions (tiles). It utilizes local histogram equalization with amplification limiting to prevent over-enhancement of the noise and over-amplification, so it is most suitable for images with mixed illumination conditions [72].

3.1.4 Normalization

Normalization techniques help ensure consistency across image datasets, especially when images are acquired under varying conditions. These may include brightness correction, intensity rescaling, or normalization of color variation. Such methods improve data comparability and can reduce biases in learning-based pipelines [73, 74].

While normalization can help mitigate inter-image variability, its practical impact in microscopy depends on the consistency of acquisition conditions. For highly heterogeneous samples such as cytology images from urine specimens, these techniques may offer limited benefits or require customized approaches [73, 74].

3.2 Object Detection and Segmentation

Object detection and segmentation are important CV problems that detect and define structures of interest in an image. Object detection focuses on classifying and localizing areas of interest — or Regions Of Interest (ROIs). Segmentation is the process of defining objects with per-pixel accuracy. In microscopic imaging analysis, a preprocessing stage is required to separate biological structures of interest, like cells, nuclei, or diagnostically meaningful structures (specific biomarkers) [75, 76].

Some methods include edge detection (e.g., Canny), thresholding, and contour-based approaches. More recent methods use deep learning. Convolutional neural networks (CNNs) for semantic segmentation and region-based CNNs (R-CNNs) for detecting instances in complex tissue structures are applied [77, 78, 79].

3.2.1 Object Detection

There are two general types of object detection approaches: classical and learning-based. Classical approaches apply manually designed features in conjunction with region-based or template-matching methods [75, 80]. Recent approaches apply deep machine learning to learn object representations from experience [80].

Convolutional Neural Networks (CNNs) are the foundation of most current top-notch object detectors like YOLO (You Only Look Once)[81], Faster R-CNN[82], or SSD (Single Shot Detector)[83]. They are applied to common natural images in their usual application but have

already proven to be successfully applied to biomedical sample cases if appropriately annotated data are available[76].

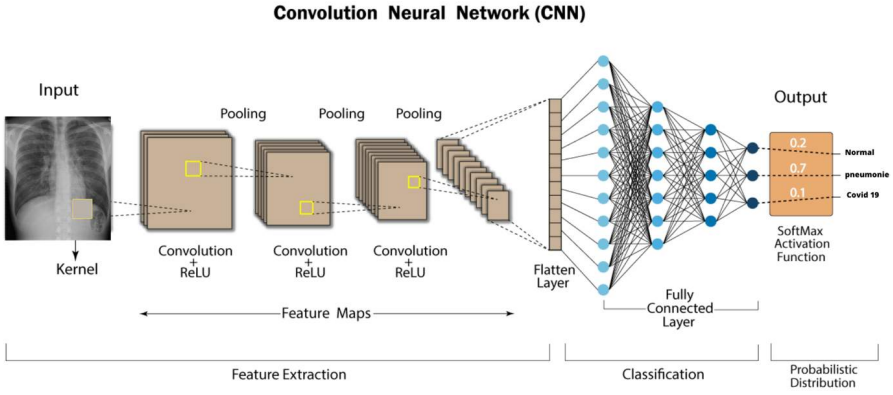


Figure 3.4: CNN structure, from [84].

Figure 3.4 illustrates a typical convolutional neural network (CNN) architecture that is utilized in image classification tasks [85]. A CNN processes the input image by propagating it through several convolutional and pooling layers that produce hierarchical features such as edges, textures, and shapes. Convolutional layers apply learned filters to identify patterns, and pooling layers shrink spatial dimensions while retaining the key features. Non-linear activation functions like ReLU introduces complexity and allow the network to learn non-linear relationships [86]. After a sequence of feature extraction layers, the result is flattened and used as input into a single or multiple fully connected layers that convert the extracted features into classification. The last output layer, commonly followed by a softmax function, gives the probability distribution over challenging classes. This tree-like structure makes CNNs particularly well suited for such endeavors as picture recognition, medical image analysis, and biomarker discovery [87, 76].

3.2.2 Segmentation Techniques

Segmentation is a preliminary process in biomedical image analysis with the goal of selecting the pixels of a Region of Interest, i.e., cells or nuclei. Based on the characteristics of the image and the object's complexity, some segmentation techniques are used:

3.2.2.1 Threshold-based Segmentation

Thresholding is a very simple image segmentation technique. It separates objects from the background by converting a grayscale image to a binary image based on the pixel intensity levels. Pixels with intensity levels outside or within a specified threshold are labelled as background or foreground. It is simple, rapid, and effective if object–background intensity contrast is noticeable and consistent. In biomedical imaging or microscopy, illumination is sometimes non-uniform, and tissue structures can exhibit high-intensity variability, making it unusable [88, 89].

A few modified thresholding methods have been suggested to overcome these issues, as can be seen from Figure 3.5:

- **Otsu's algorithm** [78] is an optimal global thresholding done automatically by minimization of intra-class variance between background and foreground. It is based on a bimodal histogram, which is common when background and structures of interest have well-separated intensity distributions. Its performance is reduced in multimodal distributions or non-uniform illumination [88].
- **Adaptive Thresholding** partitions the image into small regions and calculates a local threshold for each based on pixel statistics (e.g., mean) within a neighborhood. This is beneficial for microscopy, in which illumination and staining are non-uniform throughout the image. Global approaches like Bradley & Roth's method [90], which utilizes integral images, improve global methods by locally adjusting thresholds and processing efficiently.
- **Simple thresholding** (binary thresholding) is often used as a first step in preprocessing before applying more sophisticated segmentation algorithms or morphological operators in biomedical image processing [91].

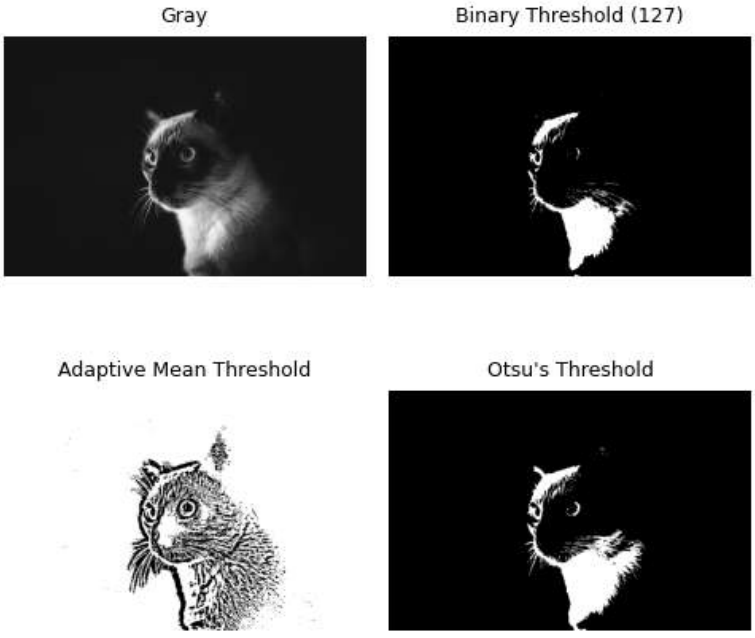


Figure 3.5: Effects of various thresholds on an image. Adapted from [66].

3.2.2.2 Morphological Operations

Preprocessing operations also prepare images for subsequent segmentation by putting their structural content in order. Morphological operations such as *dilation* and *erosion* are

automatically applied to binary images to regulate object shapes and improve segmentation quality. The operations are especially needed following thresholding, where the resulting masks tend to have noise or scattered elements [92, 93, 94].

Figure 3.6 illustrates the effect of applying different morphological transformations to a binary image obtained from the same original image, each of which has a different effect on object boundaries and shape.

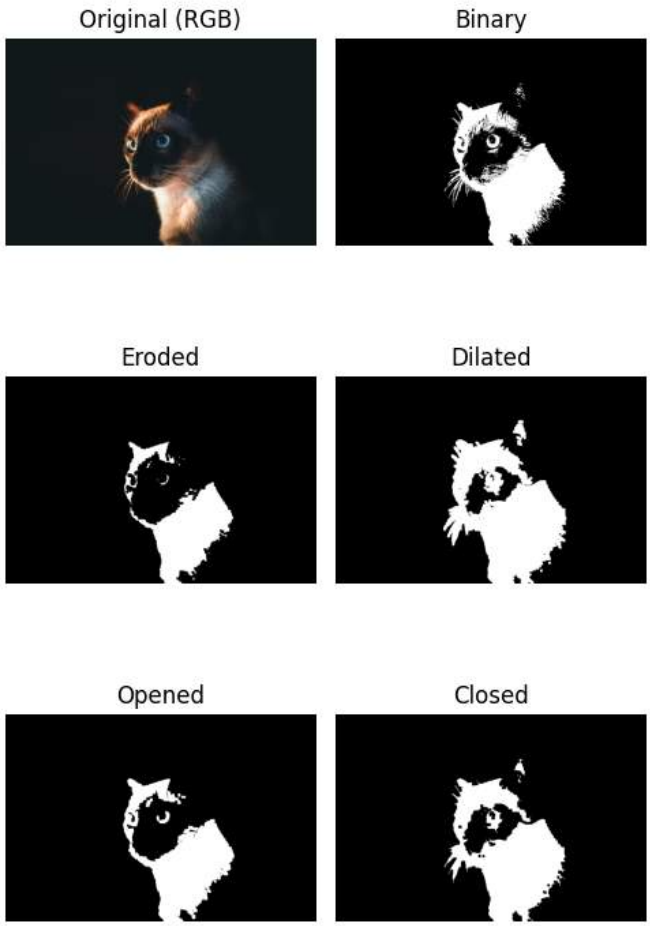


Figure 3.6: Effects of applying different contrasts on an image. Adapted from [66].

Dilation adds object boundary pixels to fill small gaps or link close structures, whereas **erosion** deletes object boundary pixels to remove small noise features or split touching components [95]. These operations are controlled by a structuring element — *kernel* — that defines the neighborhood for pixel modification and can be applied iteratively to adjust the strength of the effect. These simple operations can be combined to form more complex transformations:

- **Opening** (dilation then erosion) removes slight noise or thin features not part of significant objects without significantly altering prominent features' shape.
- **Closing** (erosion then dilation) closes small holes and gaps in or between objects, improving the solidity and continuity of broken regions.

These operations help clean up object boundaries in segmentation masks, keep important object boundaries, and delete spuriously created areas. They thus improve the robustness of subsequent processes such as feature extraction and classification [96].

3.2.2.3 Edge-based and Region-based Segmentation

Edge-based methods try to locate object boundaries by identifying intensity or gradient change at significant points between pixels. A widely used and robust method is the Canny edge detector, which integrates noise reduction, gradient calculation, non-maximum suppression, and hysteresis thresholding to identify clean edges [97, 98]. Sobel or Laplacian operators may also be used, although they are noisier and less precise in distinguishing object boundaries [88].

Once edges are detected, they may also be processed through contour extraction methods, such as OpenCV's `findContours()`, typically applied on binary thresholded images. This function gives back connected edge segments and their contours and is particularly useful in segmenting distinct objects and determining their shape and size [68].

Region-based methods, such as the watershed algorithm, adopt a different paradigm by pixel clustering based on similarity characteristics, e.g., texture or intensity. The watershed transform considers the image as a topography surface and then fills it in from labeled minima so that it can segment neighboring regions by locating ridge lines [99]. These methods are effective in images whose interior points of objects are homogeneous but may be prone to over-segmentation unless initialized or preprocessed appropriately [100].

Both region and edge-based methods produce good object boundary definition but are typically susceptible to image noise and parameter choice. They may often be used with image preprocessing techniques (e.g., smoothing or morphological operations) to improve robustness in biomedical or microscopy images under challenging conditions [101].

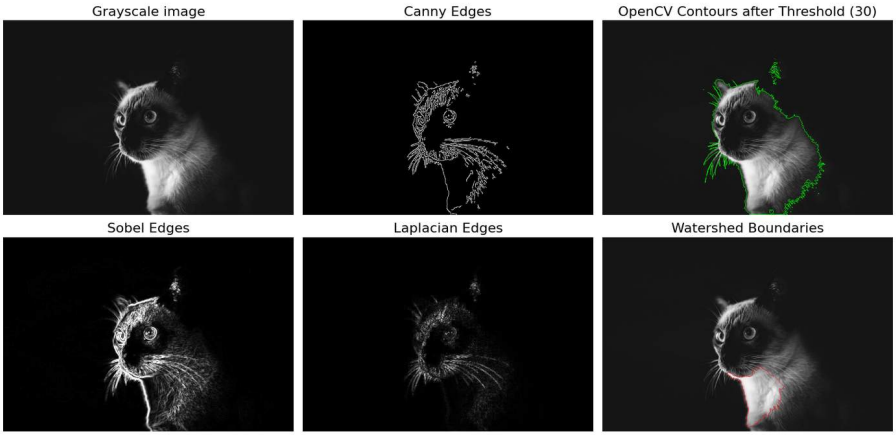


Figure 3.7: Comparison of applying edge and region-based segmentation methods on a grayscale image. Adapted from [66].

Figure 3.7 illustrates the application of these methods to a sample image. Canny and contour-based methods effectively delineate edges and object boundaries, while Sobel and Laplacian filters highlight gradients but may amplify noise. The watershed transform successfully separates homogeneous regions when sufficient preprocessing is applied, although the lack of clear regional minima may limit its performance on natural images.

3.3 Feature Extraction

After the ROIs have been detected, feature extraction techniques are used to quantify them. The features could be shape descriptors (e.g., area, boundary, eccentricity), texture features (e.g., Haralick or Local Binary Patterns), or color measurements. Other feature types, such as statistical moments (e.g., Hu moments) and multi-scale features extracted via wavelets or image pyramids, also exist but are beyond the scope of this background and not considered in this work.

The extracted features are a compact and distinctive summary of image visual patterns. Feature extraction transforms the raw pixel data of an image into numerical descriptors that describe the visual content. In microscopic image analysis, it is a crucial step to quantify the morphological, textural, and spatial features of biological objects. This makes further applications like classification, clustering, or statistical analysis possible [102, 103].

3.3.1 Types of Features

Features are typically divided into several types:

- **Shape features:** Shape features include area, perimeter, aspect ratio, compactness, and circularity, which quantify the size and shape of segmented objects (for example, cells or nuclei). Shape features are critically important in cancer diagnosis because changes in cell shape are a general indicator of malignancy [104, 105].

- **Texture features:** They describe the spatial arrangement of pixel intensities, revealing patterns like granularity, roughness, or smoothness. Common texture descriptors include Haralick features (based on the gray-level co-occurrence matrix) [102], local binary patterns (LBP), and Gabor filters [106, 107].
- **Intensity-based features:** Simple yet practical, these features summarize pixel intensity distributions using statistics like mean, variance, skewness, or histogram-based metrics [108].
- **Spatial features:** They describe how objects are aligned or spread out in a region. For example, the alignment or the distance between centroids of cells could yield valuable diagnostic information [109].

3.3.2 Relevance to Microscopic Imaging

Feature extraction is also core to structure characterization in bladder cancer microscopy images in this dissertation. Properly designed or good features, whether hand-engineered or learned, enable the discrimination of healthy from abnormal regions and among different grades or levels of malignancy [76]. Feature selection or feature design is necessary to support good diagnostic models with accuracy and interpretability [110].

3.4 Feature Analysis and Classification

Feature analysis refers to the selection, transformation, and interpretation of image-derived descriptors to support classification or decision-making tasks. These features can capture shape, texture, intensity, and spatial relationships within segmented regions of interest. Once extracted, they may be analyzed statistically or used as input for machine learning (ML) models.

Following feature extraction from microscopy images, feature analysis is the subsequent step. Feature analysis is the activity of selecting, describing, and applying these features to draw worthwhile inferences or feed machine learning algorithms. Feature analysis for BCa diagnosis is for the identification of patterns that are related to cancer occurrence, grade, or progression.

3.4.1 Statistical and Machine Learning Approaches

Statistical methods such as t-tests or ANOVA can be used to identify features that differ significantly between predefined groups. However, to model complex and multivariate relationships, machine learning algorithms are commonly employed.

There are several classical and modern classifiers, including Logistic Regression [111], Random Forests [112], XGBoost [113], and LightGBM [114], as well as neural architectures like Multi-Layer Perceptrons (MLPs) [115]. Each model presents different trade-offs in terms

of interpretability, computational cost, and ability to capture non-linear patterns. Ensemble-based methods, such as Random Forests and gradient boosting models, are particularly robust to overfitting and provide useful feature importance metrics.

3.4.2 Dimensionality Reduction and Feature Selection

In high-dimensional feature spaces, dimensionality reduction techniques such as Principal Component Analysis (PCA) [116] and t-SNE [117] are often used for visualization and redundancy reduction. Feature selection techniques like mutual information, recursive feature elimination (RFE) [118], or LASSO [119] are employed to retain only the most relevant features, improving model performance and reducing overfitting.

3.4.3 Interpretability and Explainability

Model interpretability is an essential concern when deploying predictive systems. Understanding which features contribute to a model's decision improves transparency and trust. Post-hoc explanation tools, such as SHAP (SHapley Additive exPlanations) [120] and LIME (Local Interpretable Model-agnostic Explanations) [121], are widely used to provide insight into model outputs in terms of the original input features.

3.5 Summary

This chapter has introduced the fundamental CV concepts required to process microscopic images involved in BCa diagnosis. It covered the principal tasks of a CV pipeline, i.e., image preprocessing, object detection and segmentation, feature extraction, and feature analysis, and how they are used in biomedical image analysis.

Image preprocessing techniques such as removing noise using Gaussian, median, and bilateral filtering were discussed, followed by color space transformations between RGB, HSV, LAB, and grayscale representations. Techniques such as histogram equalization and CLAHE for contrast enhancement were discussed, followed by normalization methods to normalize image data. The preprocessing processes are critical to improve image quality and prepare the data for analysis.

Learning-based and classical techniques were considered for object detection and segmentation. Classical techniques included thresholding techniques like Otsu's and adaptive thresholding, edge detection using Canny and Sobel operators, and morphological procedures like erosion and dilation. Advanced techniques involving convolutional neural networks (CNNs) and region-based methods like Mask R-CNN were also addressed for handling complex biomedical images.

Feature extraction techniques were described to quantify segmented regions using shape descriptors, texture descriptors, intensity statistics, and spatial relationships. Such features are compact numerical representations of visual patterns in microscopic images, enabling quantitative analysis of cellular morphology.

The chapter concludes with feature analysis methods, such as dimensionality reduction techniques like PCA and t-SNE, to embed and select features. Classification techniques like support vector machines and random forests were discussed, and interpretability methods like SHAP and LIME were used to explain model decisions. All these subjects together provide the basis for developing sound and interpretable diagnostic systems for BCa analysis, which will be explored in subsequent chapters.

The chapter concludes with feature analysis methods, such as dimensionality reduction techniques like PCA and t-SNE, useful for visualization. Classification approaches using machine learning models were discussed, such as support vector machines (SVM), random forests, and neural networks (MLP). Interpretability techniques like SHAP and LIME were also discussed to enhance understanding of model decisions.

Chapter 4

Methodology

This chapter outlines the comprehensive work undertaken to build a robust solution. It begins with a pipeline description and then explores the data collection methods used in this research. The chapter details the preprocessing measures implemented to enhance consistency, improve image quality, and prepare the dataset for subsequent analytical processes. Following this, it discusses the assumptions and steps involved in cell segmentation and concludes with information on feature details and classification.

4.1 Pipeline Description

The robust computer vision pipeline designed for cancer diagnosis follows a structured sequence of steps that transform raw medical images into diagnostic insights. Figure 4.1 illustrates the general workflow:

- **Data Collection:** The process begins with acquired microscopic medical images. Quality and quantity of data are essential for building robust systems.
- **Preprocessing:** Raw images are often affected by noise, illumination variation, or staining inconsistencies. Preprocessing techniques, such as normalization, contrast enhancement, or artifact removal, are applied to standardize the images and improve visual clarity.
- **Cell Segmentation:** Accurate segmentation isolates relevant anatomical or cellular structures, such as nuclei or tumor boundaries, from the background. This step is crucial for focusing the analysis on biologically meaningful regions.
- **Feature Extraction:** Visual characteristics like shape, texture, and spatial distribution are quantified from the segmented regions. These features serve as inputs for further analysis and reflect underlying tissue morphology.
- **Classification and Prediction:** The extracted features are processed by classification models, which assign diagnostic labels such as benign, malignant, or normal. The prediction output supports clinical decisions and risk assessment.

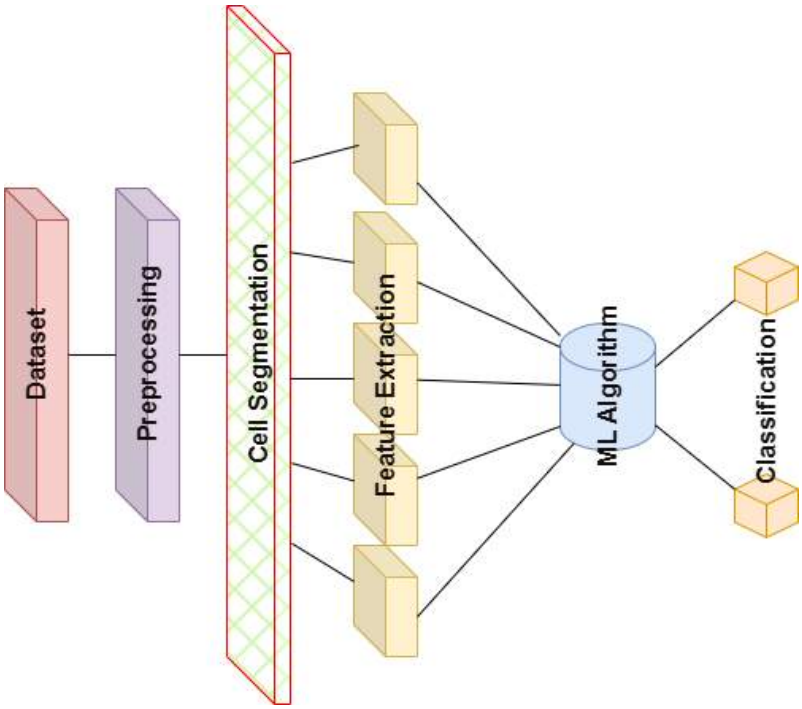


Figure 4.1: CV pipeline structure applied.

4.2 Dataset

High-quality and thoroughly curated data are the backbone of the success of any diagnostic pipeline dependent on image analysis. In bladder cancer (BCa) diagnostics, the primary data consists of microscopic images derived from urine samples of patients diagnosed with the condition and healthy control subjects. These images are critical as they represent the raw input needed for subsequent feature extraction and classification tasks explored in the following chapters. By emphasizing the significance of robust data quality and the need for careful preprocessing techniques applied, we underscore the importance of these images in driving accurate diagnostic outcomes and advancing our understanding of BCa.

The dataset used in this research consists of microscopic images of urine-derived cells collected from BCa patients diagnosed with the disease and healthy controls. It consists of images with morphologically uniform cells by photo, although sometimes morphologically different between patients, which were taken using brightfield microscopy, and they capture suspension cells, with which it is possible to observe single cellular structures. Every image is one unstained sample prepared from centrifuged urine, and was imaged under standard acquisition conditions to provide similarity throughout the dataset.

The images were obtained from CELLo Project [122], by starting to grow the cells in culture flasks under standard conditions; immediately after the trypsinization process and new cultivation, the cells remained in suspension for the first few hours, at which point the images were obtained using brightfield microscopy, without any additional staining. Therefore, the images are brightfield

microscopy scans of unstained suspension cells obtained at a suitable magnification for cytological inspection. The eyepiece's magnification was 10 times, while the objective's was 20 times, giving a total magnification of 200 times. Each image typically contains a small number of isolated or overlapping cells with round morphology. Due to the nature of suspension preparations, background variability, and debris are present in some samples. This is typical of real clinical conditions and highlights the importance of good preprocessing. The images provided have different resolutions and image formats, and are in color format, depending on the imaging protocol. Preprocessing is necessary to normalize such heterogeneity and extract useful features, as described in the following sections.

4.2.1 Patient and Label Information

The patients in the dataset form a clinically representative cohort with real-world BCa progression variation. The main types of annotations, whose details can be found in Section 2.1, are the following:

- **Diagnostic label:** The images are labeled per patient, each patient classified as either *healthy* (H) or *cancerous* (C).
- **Tumor stage and grade:** In positive BCa samples, metadata include stage grading (Tis, Ta, T1, T2, T3, T4) and risk/grade tag (low-grade (LG) vs. high-grade (HG)).
- **Patient-level grouping:** Different images are grouped by patient, varying fields of view, or time points. The dataset of images included labels assigned at the patient level.

Apart from the patient, these labels can be used for ground truth provision for supervised learning and a thorough analysis of the morphological patterns of cancer development. It enables binary classification — cancerous (C) vs. healthy (H) — and stratified analysis within the positive class, for example, low-grade (LG) vs. high-grade (HG) tumors.

For each patient, there is a set of images that eventually contains a variable number of cells. All cells of that patient are considered to have the same tags or metadata, for instance, the same condition, in every photo of that patient.

4.2.2 Data Examples and Distribution

The first step involves obtaining datasets of microscopic images; in this case, urine samples from BCa patients and healthy controls. The images in Figure 4.2 below, sourced from [122], are example images of the collected dataset.

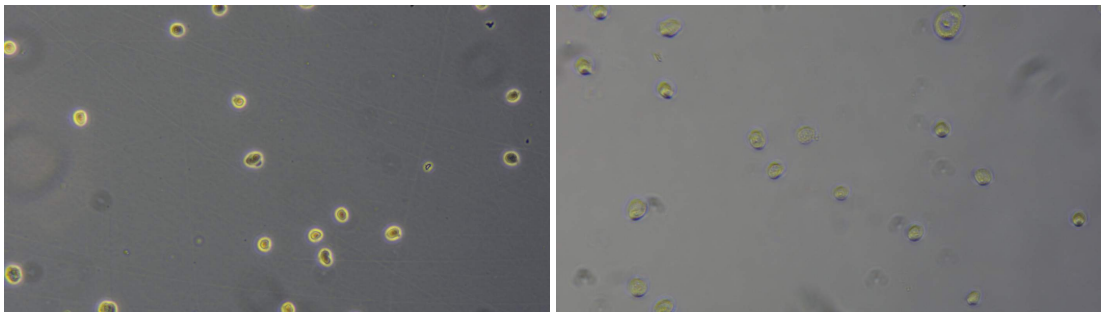


Figure 4.2: Images from the collected dataset [122].

These examples from the dataset show varied imaging characteristics, including contrast, blur, and color inconsistencies.

Even though scientists collect the data in a controllable yet random way, extensive preprocessing of the datasets about noise reduction, contrast stretching, and segmentation is required to enhance the quality and sharpness of the images. These are critical processes to ensure that the algorithm can extract relevant features from the images while ensuring that aspects like a blurry image, background noise, or other imaging artifacts are minimal and have maximum diversity in high-quality samples.

Figure 4.3 shows the number of image examples for each diagnostic class — healthy (H) and cancerous (C) — based on patient-level metadata. The class distribution is skewed, with significantly more images from cancer subjects than healthy controls. This will bias training and must be addressed when developing classification models and choosing evaluation metrics.

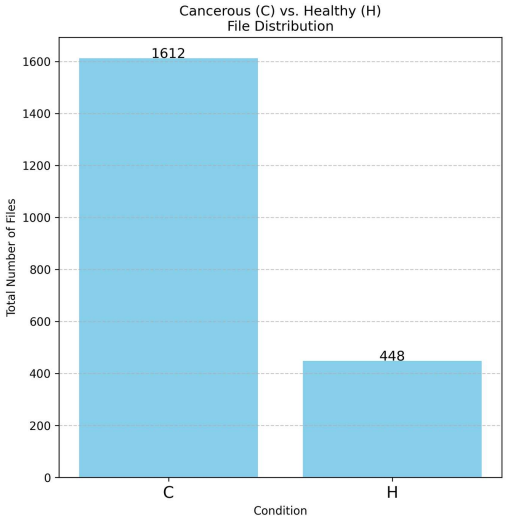


Figure 4.3: Distribution of image samples per diagnostic class.

Apart from class imbalance, there is also high variability in patient image count, as seen in Figure 4.4. Some patients have only a few images, while others have hundreds. This heterogeneity in sample size can lead to the overrepresentation of particular patients, thus requiring patient-level splitting to avoid data leakage and biased performance.

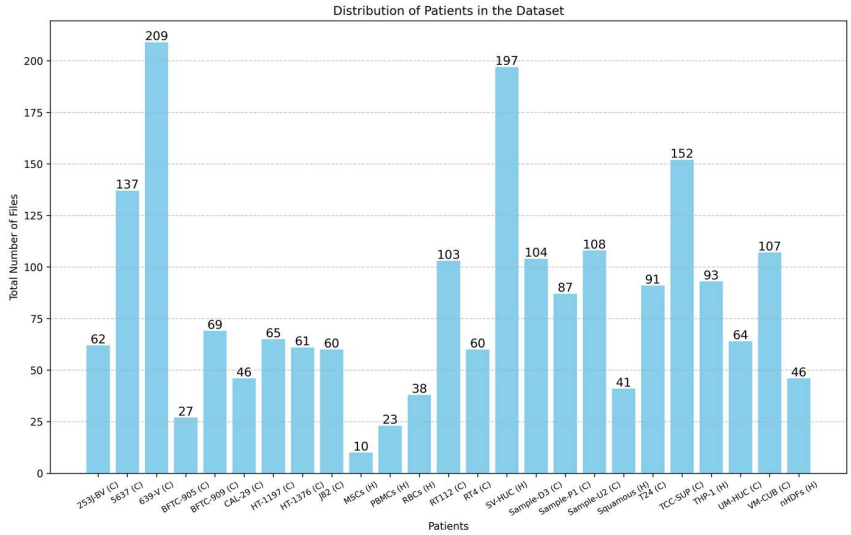


Figure 4.4: Distribution of image samples per patient.

Furthermore, the number of cells in each image is irregular and can vary considerably. Some images contain only a few well-isolated cells, while others show a denser field with overlapping or clustered cells. Although not explicitly quantified at this stage, this variability in cell count per image, and even their morphology or aggregation aspects, can influence both the difficulty of segmentation and the diagnostic relevance of each image. This reinforces the need for robust preprocessing and segmentation techniques that adapt to different cell densities and configurations.

4.3 Preprocessing Pipeline

For quality assurance, improving interpretability, and facilitating efficient feature extraction, the entire dataset of images was subjected to a systematic preprocessing pipeline. The preprocessing pipeline plays a critical role in reducing variability caused by acquisition conditions, removal of irrelevant artifacts, and elucidation of biologically relevant structures. Multiple preprocessing methods were compared and benchmarked during development to identify the most effective strategies for enhancing downstream performance. The preprocessing pipeline methods are detailed in the following subsections, along with their steps. The techniques used were explained in Chapter 3.

Each step was selected and evaluated to maximize the quality and uniformity of the input data for subsequent segmentation and classification procedures. The core preprocessing function below, in Algorithm 1, provides a unified interface for six distinct preprocessing algorithms, each targeting different image characteristics. In this function, the user must give an image in RGB format and a list of names of the available algorithms implemented, and the functions will return the preprocessed versions of the image according to each pipeline of the proposed algorithms.

Algorithm 1 Main Preprocessing Dispatcher

```
function PREPROCESS(image, algorithms)
    Initialize empty results list
    for each algorithm in input list do
        Apply selected algorithm
        Store result(s)
    end for
    return all processed images
end function
```

Each of the algorithms available in the `preprocess()` pipeline focuses on specific aspects of image enhancement or segmentation. The following sections outline each preprocessing technique, including its purpose, the operations it includes, and its relevance to the pipeline.

4.3.1 Otsu's Thresholding

Purpose: Automatic global thresholding for images with bimodal intensity distributions.

Algorithm 2 OtsuThreshold(image)

```
Convert to grayscale
Apply binary threshold with Otsu's method: cv.THRESH_BINARY + cv.THRESH_OTSU
return segmented image
```

Otsu's method, represented in Algorithm 2, automatically computes an optimal threshold by minimizing the intra-class variance of pixel intensities in bimodal histograms. It must be combined with a binary thresholding mode, such as `THRESH_BINARY`, to generate a segmented binary mask. In OpenCV, this is implemented as:

```
1 _, result = cv.threshold(gray, 0, 255, cv.THRESH_BINARY + cv.THRESH_OTSU)
```

While effective, Otsu's thresholding can sometimes produce noisy segmentations or minor artifacts, especially in cytology images with visual variability or uneven background. Figure 4.5 illustrates the result of applying Otsu's method to a representative image from the dataset.

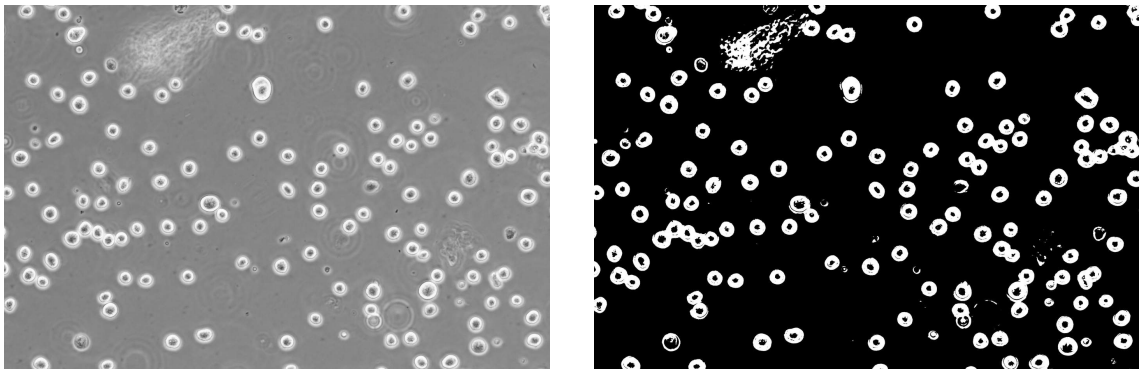


Figure 4.5: Application of Otsu Thresholding on an RGB dataset image [122] (converted to grayscale internally). **Left:** Grayscale input image. **Right:** Binary segmentation mask produced by Otsu's method.

4.3.2 Binary Thresholding

Purpose: Fast fixed-threshold segmentation for high-contrast images.

Algorithm 3 BinaryThreshold(image)

1: Convert to grayscale
2: Apply fixed threshold (200)
3: **return** binary mask

In binary thresholding, represented in Algorithm 3, threshold values such as 127 and 200 were chosen empirically through visual inspection and validation across representative image samples. These thresholds offer a good compromise between segmentation clarity and computational simplicity without requiring dataset-specific tuning. Specifically, the fixed threshold of 200/255 enables the fastest segmentation method but requires consistent lighting conditions. The approach is particularly beneficial for quick previews or quality control images, where simplicity and speed are of first importance. Figure 4.6 shows the result of applying this fixed threshold to the same input image used previously.

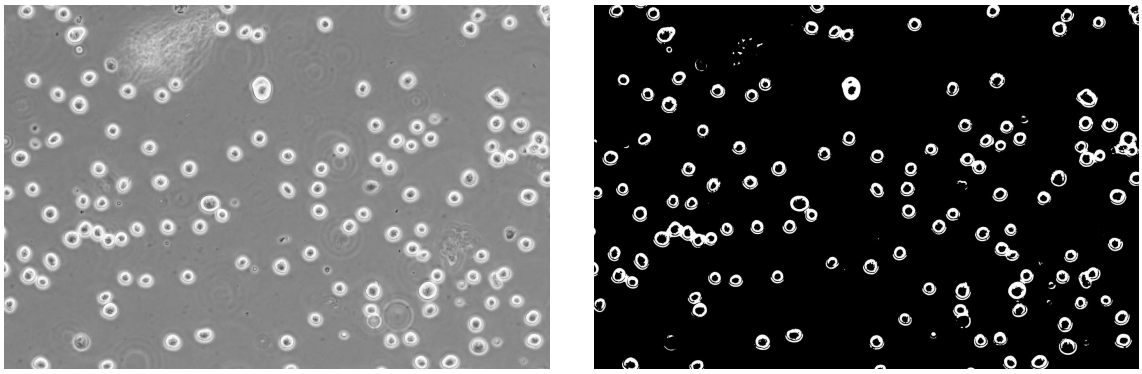


Figure 4.6: Application of binary thresholding on the same RGB dataset image [122] (converted to grayscale internally). **Left:** Grayscale input image (same as in Figure 4.5). **Right:** Binary segmentation mask produced by fixed thresholding.

4.3.3 Adaptive Thresholding

Purpose: Handle uneven illumination using local thresholding.

Algorithm 4 AdaptiveThreshold(image)

1: Convert to grayscale
2: Compute local thresholds (11x11 neighbourhood)
3: Invert result for white cells on black background
4: **return** adapted image

Adaptive thresholding, represented in Algorithm 4, works very well for images with non-uniform illumination since it calculates a different threshold for each pixel based on the local neighbourhood. It employs Gaussian-weighted local statistics within an 11×11 window, which enables more accurate segmentation of structures despite differing brightness across regions. The resulting binary image is inverted so that cell areas become white, enhancing contrast and facilitating subsequent analysis. Figure 4.7 shows the effect of applying adaptive thresholding to the same grayscale image used previously.

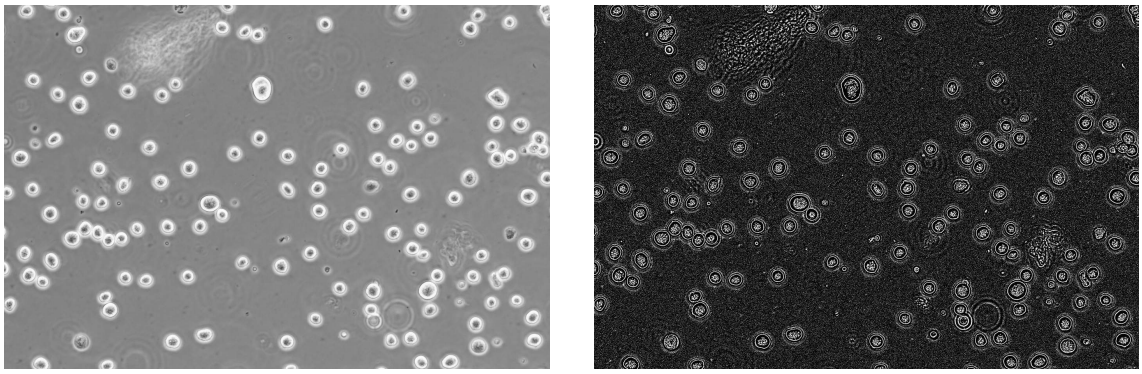


Figure 4.7: Application of adaptive Thresholding to the same RGB image from the dataset used [122] (converted to grayscale internally). **Left:** Grayscale input image. **Right:** Binary mask produced by adaptive thresholding with inversion.

4.3.4 Contrast-Colorspace-Morphological Operations (CCM)

Purpose: Comprehensive processing for challenging color images.

Algorithm 5 ContrastMorphOps(image)

1: Apply noise reduction
2: If high contrast: normalize and reduce
3: Analyze in HSV/LAB colorspaces
4: Select processing channel based on hue variance
5: Apply morphological closing
6: **return** enhanced image

This pipeline, represented in Algorithm 5, combines various preprocessing methods to address robust segmentation in complex color images, particularly when lighting, staining, or contrast inconsistency exists between areas. It begins with noise elimination and dynamic contrast normalization if high variance exists. It then converts the image into HSV and LAB color spaces and investigates channel variances — specifically hue — of segmentation cues. A selected channel is then thresholded, and morphological closure is applied to close small holes and fill in discontinuities. Although being the most computationally costly, this method is also the most adaptable under difficult staining conditions. Figure 4.8 shows the result of applying this pipeline to a challenging image sample.

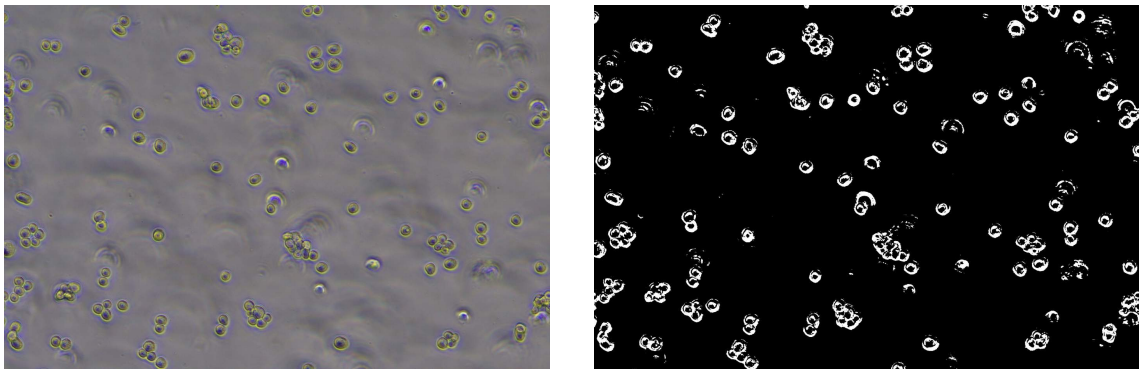


Figure 4.8: CCM preprocessing pipeline applied to a dataset image [122]. **Left:** Original RGB input image. **Right:** Final enhanced segmentation mask produced by this method.

4.3.5 Adaptive-Gauss-Binary (AGB)

Purpose: Smoothing-focused pipeline for noisy images.

Algorithm 6 AdaptiveGaussBinary(image)

1: Apply adaptive threshold
2: Gaussian blur ($\sigma = 3$)
3: Invert intensities
4: Final binary threshold (50)
5: **return** smoothed segmentation

This method, represented in Algorithm 6, combines several steps to preprocess images with medium noise or slight blur. It begins with adaptive thresholding to balance local intensity variation, followed by Gaussian blurring (standard deviation of kernel $\sigma = 3$) to blend noise artifacts. The kernel size is derived as:

$$\text{kernel_size} = 2 \times \lceil 3\sigma \rceil + 1 \tag{4.1}$$

The Gaussian blur applied introduces intermediate grayscale values, softening edges and creating faint artifacts. The resulting image is inverted to render nuclei white areas, and a fixed binary threshold of 50 (on a scale of 0–255) is applied, serving a dual purpose: it re-binarizes the image by converting these intermediate grayscale pixels back to strictly binary values, and it removes low-intensity noise or speckles caused by blurring. The threshold of 50 is chosen to ensure that true foreground regions (nuclei) remain white, while background noise and artifacts are suppressed. This final step improves the clarity and robustness of the segmentation mask, especially in images with moderate noise or slight defocus. Figure 4.9 below illustrates the result of applying this pipeline to an input image from the dataset.

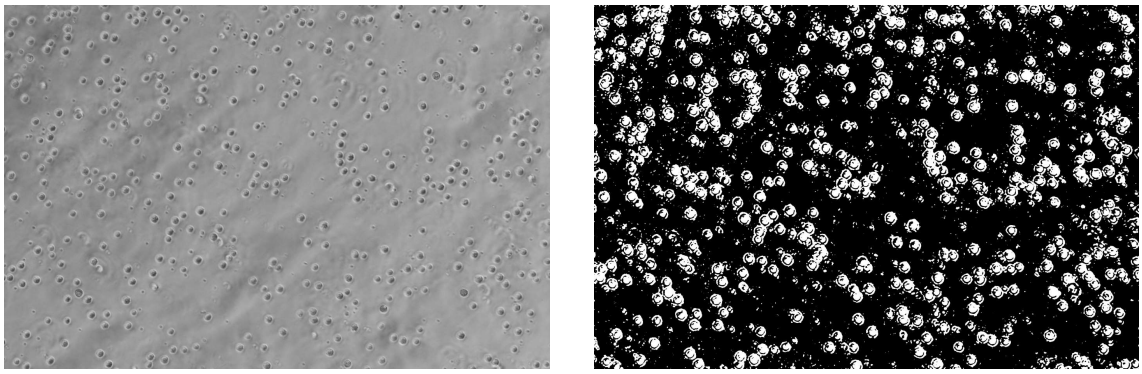


Figure 4.9: Application of the Adaptive-Gauss-Binary (AGB) pipeline to an RGB dataset image [122] (converted to grayscale internally). **Left:** Grayscale input image. **Right:** Final segmentation mask after adaptive thresholding, smoothing, inversion, and fixed thresholding.

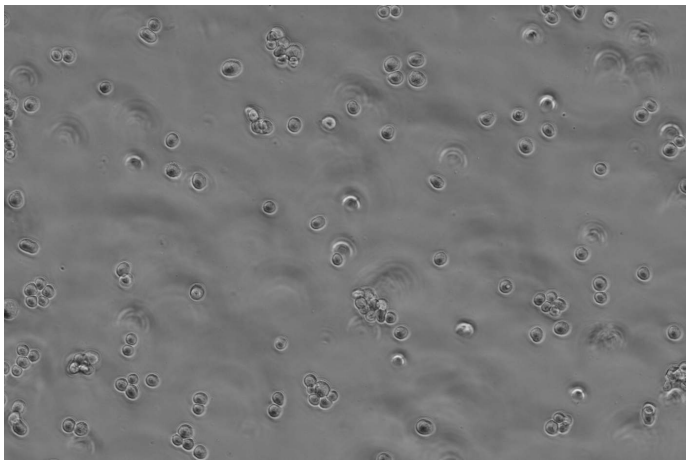
4.3.6 Gauss-Adaptive-Dilation (GAD)

Purpose: Enhance weak cell boundaries through dilation.

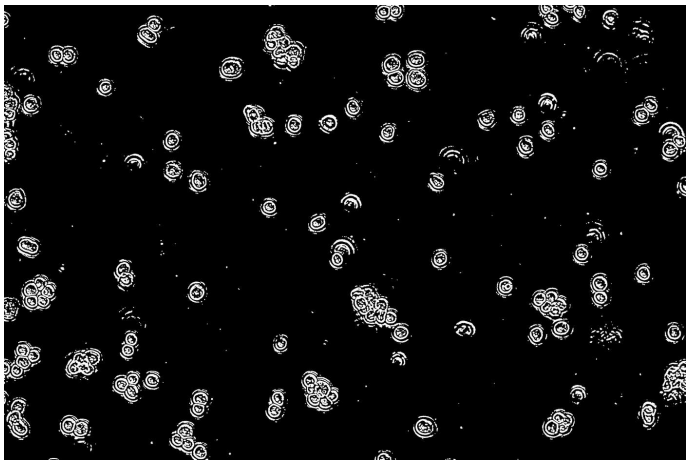
Algorithm 7 GaussAdaptiveDilate(image)

1: Convert to grayscale
2: Apply Gaussian blur ($\omega = 2$)
3: Apply adaptive thresholding
4: Generate two dilated versions:
5: Mild dilation (3x3 kernel)
6: Strong dilation (2x2 kernel, 3 iterations)
7: **return** both variants

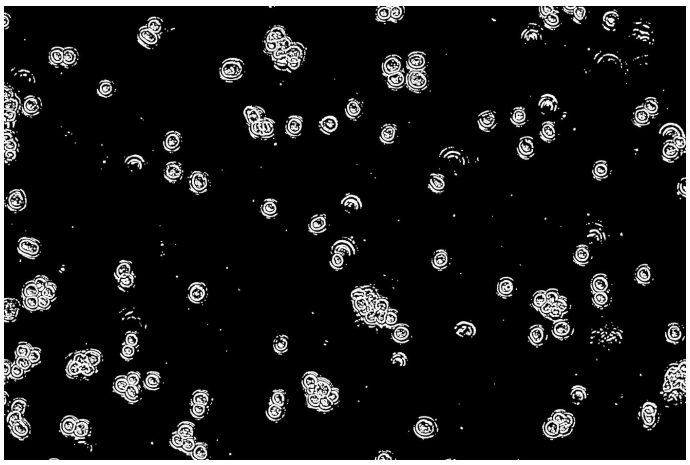
This method, represented in Algorithm 7, grounds nuclear borders as more powerful or more complete with a procedure on a combination of adaptive thresholding, Gaussian blurring, and morphological dilation. The image is initially smoothed with a Gaussian filter ($\omega = 2$) before nuclei are segmented with adaptive thresholding. The produced binary mask is then expanded using two methods of dilation: a lenient one with a 3x3 kernel (single step) and an aggressive one with a 2x2 kernel repeated over three steps. The two forms are created to respond differently under segmentation conditions — while the gentle dilation attempts to preserve detail in denser regions, the aggressive dilation performs notably well in extracting isolated or weak nuclei. However, this increased dilation makes excessive dilation possible in tightly clumped areas, sometimes merging the discrete nuclear borders. With the return of both outcomes, the pipeline allows selection based on the sample structure under investigation. Below, in Figure 4.10, we illustrate both dilation variants for a sample image from the dataset.



Original (grayscale)



Mild Dilation (3x3, 1 iter.)



Strong Dilation (2x2, 3 iter.)

Figure 4.10: Application of the Gauss-Adaptive-Dilation (GAD) pipeline to a dataset image [122]. **Top:** Grayscale input. **Middle:** Segmentation after mild dilation. **Bottom:** Segmentation after strong dilation, connecting more fragmented regions.

4.3.7 Preprocessing Evaluation Strategy

Without ground-truth segmentations, the preprocessing pipelines are evaluated through a relative comparison among the different methods themselves. The goal is to identify which preprocessing strategy most effectively enables consistent, informative cell contour extraction across the dataset. Firstly, a qualitative analysis is performed by comparing representative examples of good, poor, and intermediate segmentation results for each preprocessing method, considering the expectations for some images, aiming to gain insights into their performance characteristics.

Quantitatively, we analyze the **number of contours** detected *per image* for each preprocessing pipeline. Furthermore, the distribution of contour counts *across all images* within the dataset is assessed for each algorithm to evaluate its consistency, sensitivity, and robustness in detecting cell-like structures. It is important to note that some images, corresponding to certain patients or specific biological conditions, may naturally exhibit a low number of detectable contours. This does *not* necessarily indicate a failure of the preprocessing or segmentation pipeline; rather, if a consistently low contour count is observed across all pipelines for those images, it may reflect true underlying biological variation or image characteristics. Hence, the overall distribution of contour counts across the dataset must be interpreted with this context in mind.

This relative, self-comparison strategy enables the identification of preprocessing pipelines that are both quantitatively consistent and aligned with the visual characteristics of the dataset. Ultimately, the top-performing method(s) will be selected to preprocess the full dataset before proceeding to feature extraction and classification.

4.4 Cell Segmentation

Cell segmentation is a key process in microscopic image analysis to allow clear identification and boundary definition of single cells or cell structures. Segmentation success provides the basis for feature extraction and quantitative measurements necessary for biomarker definition in BCa diagnosis. This section describes the rationale for the segmentation approach used in this work, which transforms preprocessed microscopic images into identified cellular regions through a multi-stage pipeline, including the algorithms applied to effectively detect and segment cells. The segmentation pipeline is designed to handle the image quality variation and cell and biological heterogeneity inherent in this dataset. The processes used are described in Chapter 3. The primary function, represented in Algorithm 8, orchestrates the complete segmentation process.

Algorithm 8 Segment Cells in Processed Images

Require: Original image I, Preprocessed versions of the image $\{I^1_{proc}, \ldots, I^n_{proc}\}$
Ensure: Detected cell contours C_{cells}

$C_{all} \leftarrow \emptyset$ ▷ Initialize contour collection
for each processed image I^i_{proc} **do**
  $C^i \leftarrow \text{DetectContours}(I^i_{proc})$ ▷ Section 4.4.1
  if $C^i \neq \emptyset$ **then**
    $C_{all} \leftarrow C_{all} \cup C^i$
  end if
end for
$C_{cells} \leftarrow \text{RefineContours}(C_{all}, I)$ ▷ Section 4.4.2
return C_{cells}

The main segmentation function integrates multiple preprocessing outputs to detect cell contours robustly. It begins by initializing an empty collection to aggregate contours from all preprocessed image versions. For each preprocessed image I^i_{proc}, contours are detected using the method described in Section 4.4.1. If any contours are found, they are added to the aggregate set C_{all}. After processing all versions, the combined contours are further refined and filtered through the procedure detailed in Section 4.4.2, which improves performance by eliminating contours outside image boundaries before assessing overlap, thus streamlining the selection of valid cell contours. The final set of refined cell contours C_{cells} is then returned for subsequent analysis.

4.4.1 Contour Detection

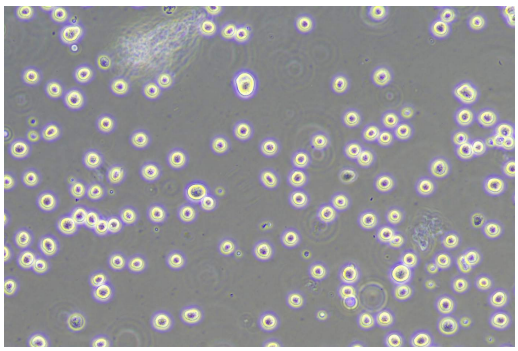
The initial contour extraction from preprocessed images is represented in Algorithm 9.

Algorithm 9 Detect Contours

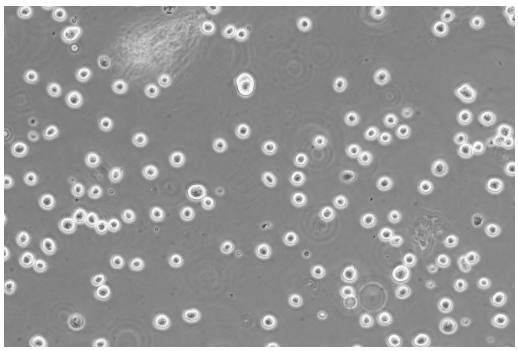
Require: Preprocessed image I_{proc}
Ensure: Contours C, Closed image I_{closed}

$C_{best} \leftarrow \emptyset$, $n_{best} \leftarrow 0$, $I_{best} \leftarrow \emptyset$
for $k \in \{1, 3, 5\}$ **do** ▷ Multiple kernel sizes
  $I_{closed} \leftarrow \text{MorphologicalClose}(I_{proc}, k \times k \text{ kernel})$
  $C_{current} \leftarrow \text{FindExternalContours}(I_{closed})$
  if $|C_{current}| > n_{best}$ **then**
    $C_{best} \leftarrow C_{current}$
    $n_{best} \leftarrow |C_{current}|$
    $I_{best} \leftarrow I_{closed}$
  end if
end for
return (C_{best}, I_{best})

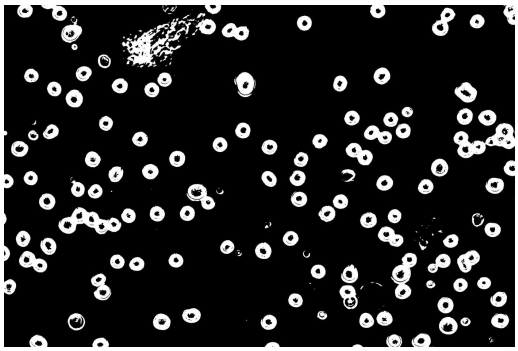
The `Detect Contours` function operates on a preprocessed image I_{proc} by applying morphological closing with multiple kernel sizes $(1 \times 1, 3 \times 3, 5 \times 5)$ in order to accommodate the variability in cell sizes. For each kernel size k, the function performs morphological closing on the image to bridge small gaps in the cell boundaries, producing a closed image I_{closed}. External contours are then detected from this closed image, focusing on extracting complete cell profiles. Among all kernel sizes tested, the set of contours C_{current} with the higher number of detected contours is selected as the best result C_{best}. Finally, the function returns this optimal set of contours along with the corresponding closed image I_{closed}. Figure 4.11 shows an example from the dataset illustrating the main steps of the contour detection pipeline, including preprocessing, thresholding, and the set of detected contours.



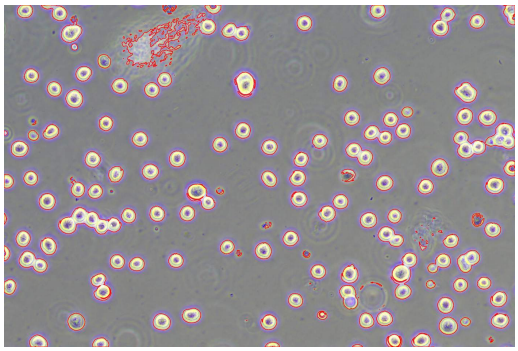
(a) Original image



(b) Grayscale



(c) Thresholded



(d) Contours drawn

Figure 4.11: Contour detection process. Starting from the original image (a), the pipeline applies grayscale conversion (b) and Otsu thresholding (c) to produce a binary mask. External contours are then detected and drawn on the image (d). Adapted from [122].

In short, this algorithm applies a multi-scale morphological closing strategy to accommodate variations in cell size better. Although morphological closing is applied during preprocessing to enhance image quality, it is also reapplied here at multiple scales to ensure robust cell boundary closure for contour detection. For each kernel size, a closing operation is performed to bridge small gaps in the cell boundaries, followed by external contour detection, which focuses on capturing complete cell profiles. Among the results, the version yielding the highest number of contours is

selected as the final output, ensuring that the most informative and complete cell structures are retained.

4.4.2 Contour Refinement

The initial contour detection step (Section 4.4.1) extracts raw cell boundaries, which may include artifacts, incomplete cells, or overlapping detections. To address these challenges, the contour refinement process applies sequential geometric and spatial validations to improve biological plausibility. First, size filtering eliminates debris using micrometer-to-pixel conversion to enforce a minimum cell diameter threshold. Next, convex hull regularization smooths irregular boundaries while preserving essential morphological features. Finally, boundary validation and overlap resolution ensure only complete, non-redundant cells within the image bounds are retained. This multi-stage refinement enhances segmentation accuracy for downstream quantitative analysis, as illustrated in Figure 4.16.

4.4.2.1 Size Filtering

The size filtering procedure in Algorithm 10 eliminates artifacts and debris based on physical dimensions.

Algorithm 10 Filter By Size

Require: Contours C, Minimum μm cell threshold t
Ensure: Filtered contours $C_{filtered}$

$C_{filtered} \leftarrow \emptyset$
$conv \leftarrow 7.88175$ px/μm
for $c \in C$ **do**
    $a_{major} \leftarrow \text{GetMajorAxisLength}(c)$
    if $a_{major} \geq t \times conv$ **then**
        $C_{filtered} \leftarrow C_{filtered} \cup \{c\}$
    end if
end for
return $C_{filtered}$

This procedure filters detected contours based on their physical size, ensuring that only those likely to correspond to real cell structures are retained. Cells usually have a minimum size of 6 μm [123], then, contours inferior to 6 μm must be excluded. Each contour is evaluated by computing its major axis length, which reflects the longest dimension of its fitted ellipse. This value is compared against a threshold provided in micrometers (μm), which is converted into pixel units using a fixed conversion factor (7.88175 px/μm, as demonstrated in section 4.4.2.1 below). Only contours whose major axis length exceeds the specified physical threshold are retained in the filtered set. This step helps remove small artefacts or noise that may have been erroneously detected during contour extraction.

Micrometer-to-Pixel Conversion

The conversion between pixel units and physical micrometers was established using images from the dataset that contained a Neubauer chamber grid (Figure 4.12 and Figure 4.13). These images were acquired with the same magnification as the experimental samples, allowing accurate calibration under the acquisition conditions.

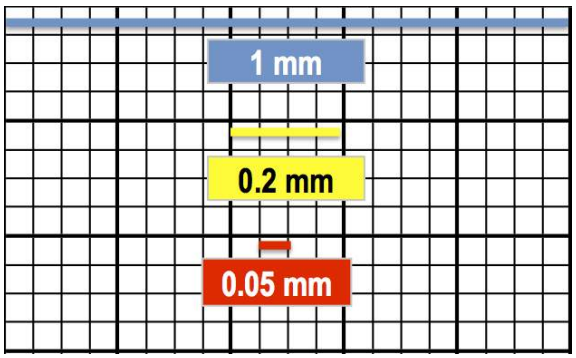


Figure 4.12: Schematic of Neubauer chamber grid structure. Addapted from [124].

The calibration involved identifying the Neubauer chamber grid present in some dataset images. The pixel dimensions of the large grid squares, known to measure 200 μm × 200 μm in the real world, were measured directly from these images. The pixel length was averaged across multiple grid squares to minimize measurement error.

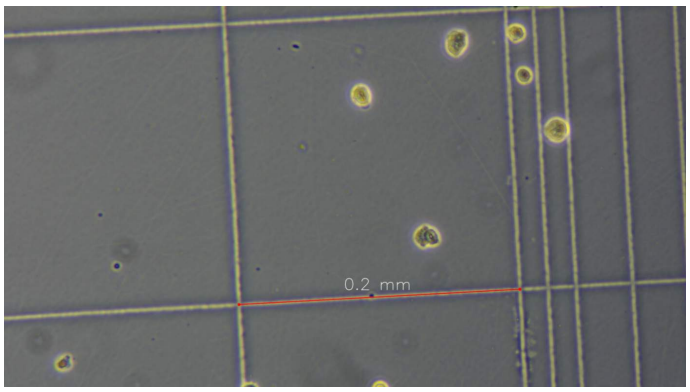


Figure 4.13: A dataset image with the Neubauer chamber grid. A line was manually drawn along one grid edge, corresponding to 0.2 mm (200 μm), and used to measure its pixel length. Adapted from [122].

The micrometer-to-pixel conversion factor f was calculated by dividing the measured length in pixels by the known physical length:

$$f\ (\text{px}/\mu\text{m}) = \frac{\text{Measured pixel length of grid side}}{200\ \mu\text{m}} \tag{4.2}$$

In our setup, one side of a grid square was measured as 1576.35 pixels, leading to:

$$f = \frac{1576.35 \text{ px}}{200\ \mu\text{m}} = 7.88175 \text{ px}/\mu\text{m} \tag{4.3}$$

The calibration process is implemented as in Algorithm 11, and then verified by comparing image references to expected sizes and consulting experienced microscopy researchers.

Algorithm 11 Micrometer-to-Pixel Conversion

Require: Grid calibration image I_{grid}
Ensure: Conversion factor f (px/μm)

```
measurements ← ∅
for i ← 1 to 5 do                                  ▷ Measure 5 grid squares
    px_length ← MeasureGridSquare(I_grid, i)
    measurements ← measurements ∪ {px_length}
end for
avg_px ← Mean(measurements)
f ← avg_px/200                                     ▷ 200μm is known grid size
return f
```

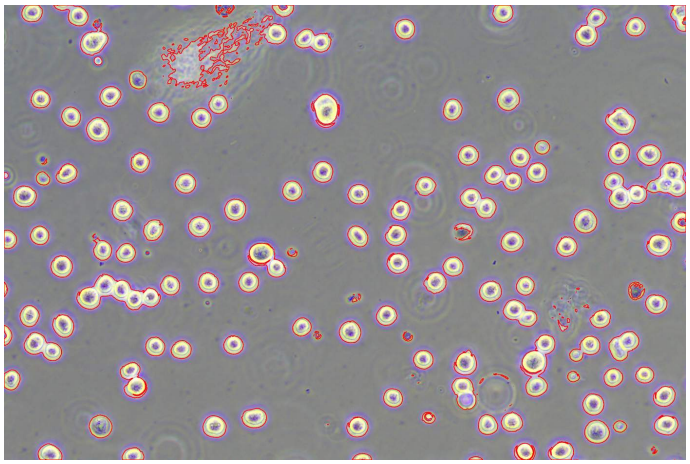
where MeasureGridSquare() calculates the pixel length of a grid square's side, Mean() computes the average of these measurements, and f represents the final conversion factor, which in our case is 7.88175 px/μm. The factor was applied in size-based filtering:

$$\text{Minimum cell size (px)} = \text{Minimum size}(\mu\text{m}) \times 7.88175 \tag{4.4}$$

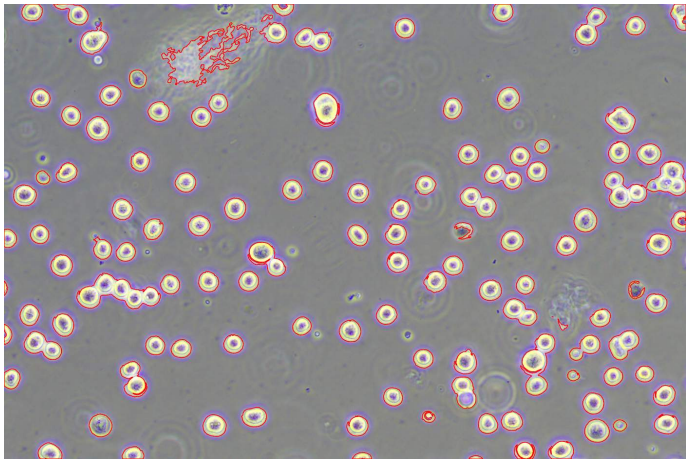
For example, our 6 μm threshold converts to:

$$6\mu m \times 7.88175 \text{ px}/\mu m = 47.29 \text{ px} \tag{4.5}$$

Figure 4.14 illustrates the effect of the size filtering step. Initially, many small contours were detected, including likely noise and debris. After filtering, only those with a major axis length above the 6 μm threshold were retained, resulting in a cleaner and more biologically plausible set of contours.



(a) Before filtering



(b) After filtering (6 μm threshold)

Figure 4.14: Effect of size-based filtering. Contours smaller than 6 μm in major axis length were removed to eliminate spurious detections and retain plausible cell regions. Adapted from [122].

4.4.2.2 Shape Regularization

Convex hulls were applied to normalize cell boundaries while preserving essential morphological characteristics. The convex hull of a contour is defined as the smallest convex polygon that completely encloses all contour points, effectively smoothing irregular boundaries and filling concave regions. This operation is particularly valuable for cell segmentation because it compensates for minor boundary irregularities caused by imaging artifacts, maintains the overall size and orientation of cellular structures, simplifies subsequent processing steps while preserving biologically relevant shape information, and reduces the impact of membrane protrusions or imaging noise on contour analysis. The algorithm processes each contour as follows in Algorithm 12:

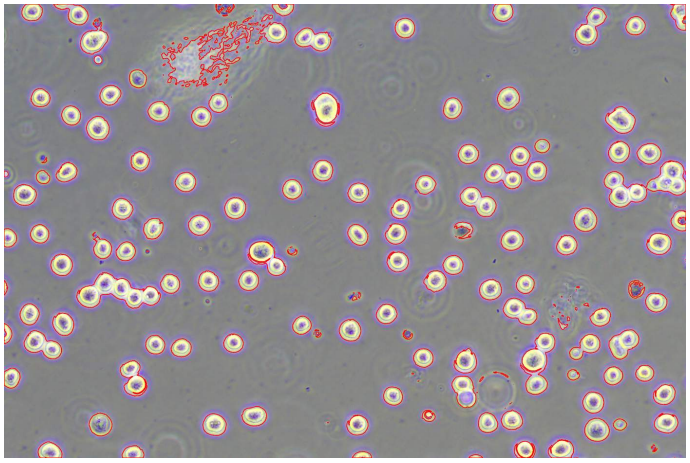
Algorithm 12 Apply Convex Hulls

Require: Contours C

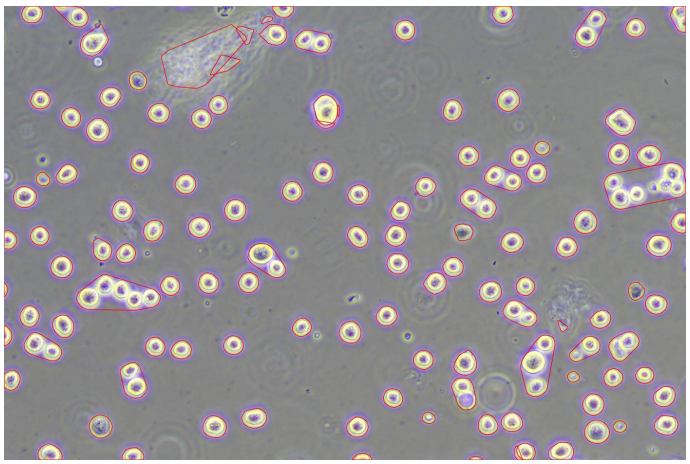
Ensure: Convex hulls H

1: **return** $\{convexHull(c) | \forall c \in C\}$

The convex hull operation provides several advantages for our application. It improves computational efficiency by reducing complex contours to their essential vertices, which speeds up later processing stages. It also reduces noise by eliminating inward depressions caused by imaging artifacts or minor cellular irregularities. Moreover, it helps standardize representations of cell shapes while preserving important metrics such as cell orientation (major and minor axes), overall size (e.g., area and perimeter), and positional relationships between cells. Figure 4.15 shows an example of the application of this procedure.



(a) Before convex hull



(b) After convex hull

Figure 4.15: Convex hull transformation: (Top) Original contours showing concave regions and irregularities, (Bottom) Regularized convex hulls maintaining key morphological features while simplifying boundaries. Adapted from [122].

4.4.2.3 Boundary Validation

This process, represented in Algorithm 13, ensures that there are only complete cells within image boundaries.

Algorithm 13 Validate Boundaries

Require: Image I, Hulls H

Ensure: Valid contours C_{valid}

1: $(w,h) \leftarrow$ dimensions(I) **return** $\{h \in H | h \text{ within } (0,0,w,h)\}$

The boundary validation step ensures that only complete cells fully contained within the image bounds are retained for further analysis. The algorithm takes the convex hulls H and image dimensions (w,h) as input, verifying that each hull lies entirely within the rectangular image boundaries defined by coordinates $(0,0,w,h)$. This critical quality control step rejects partial cells intersecting image edges (which could distort size and shape measurements) while preserving valid interior cells. The spatial validation is performed using computational geometry techniques to efficiently check polygon containment, maintaining the pipeline's biological accuracy by excluding potentially truncated cellular structures that might bias downstream analysis. This process is crucial for reliable quantitative measurements, as edge artifacts could significantly affect metrics like cell area or morphology.

4.4.2.4 Overlap Resolution

Algorithm 14 Resolve Overlaps

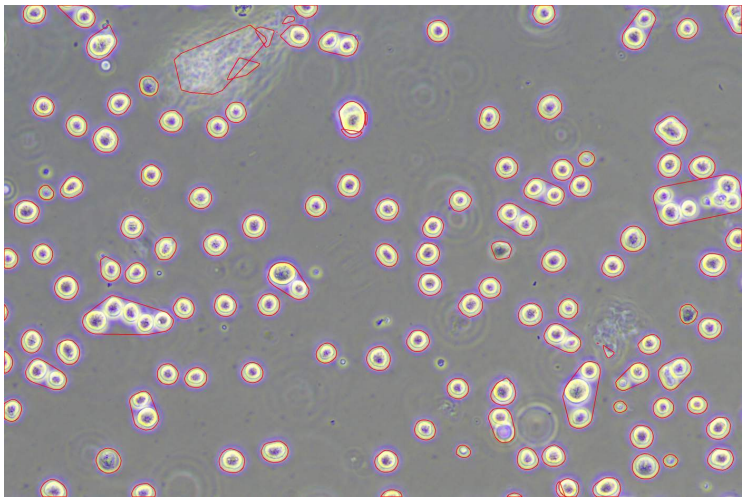
Require: Contours C, Overlap threshold θ

Ensure: Non-overlapping contours C_{final}

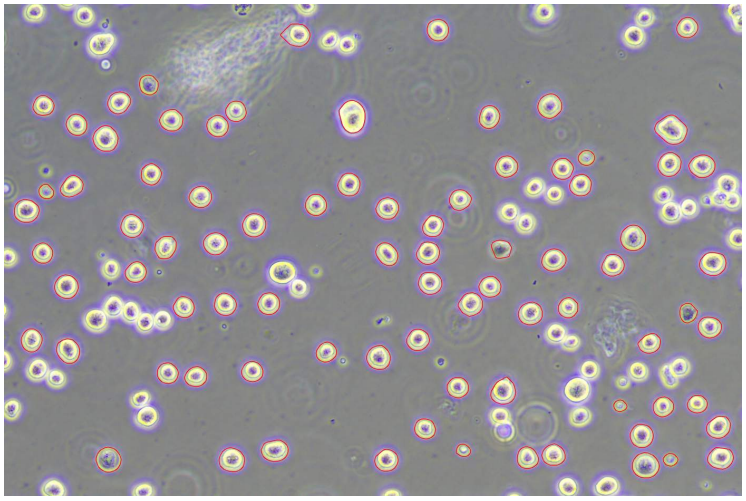
1: Sort C by descending sphericity
2: $C_{final} \leftarrow \emptyset$
3: **for** $i \leftarrow 1$ to $|C|$ **do**
4: $\quad valid \leftarrow$ True
5: $\quad$ **for** $j \leftarrow 1$ to $i-1$ **do**
6: $\quad\quad$ **if** Overlap$(C[i], C[j]) > \theta$ **then**
7: $\quad\quad\quad valid \leftarrow$ False
8: $\quad\quad\quad$ **break**
9: $\quad\quad$ **end if**
10: $\quad$ **end for**
11: $\quad$ **if** $valid$ **then**
12: $\quad\quad C_{final} \leftarrow C_{final} \cup \{C[i]\}$
13: $\quad$ **end if**
14: **end for**
15: **return** C_{final}

The method represented in Algorithm 14 handles potentially overlapping cells. The overlap resolution algorithm eliminates redundant cell detections by processing contours in order to decrease sphericity (most circular first). For each candidate contour, it checks against previously accepted contours and rejects those exceeding a specified overlap threshold θ (typically 30%). This priority-based approach favors biologically plausible, regularly-shaped cells while efficiently resolving clusters of potential duplicates. The nested loop structure ensures computational tractability, with early termination when detecting overlaps. The algorithm outputs a final set of non-overlapping contours suitable for quantitative analysis, preventing the double-counting of individual cells in crowded microscopic fields.

Figure 4.16 below is an example of the application of these two last processes (from Sections 4.4.2.3 and 4.4.2.4) on a dataset image, to ensure full cell visibility and retain the most plausible cells.



(a) Before final refinements.



(b) Contours touching image borders are discarded, and overlapping contours are resolved by priority, such as sphericity.

Figure 4.16: Final contour refinement stages. Adapted from [122].

4.4.3 Quality Control Metrics

Two key metrics employed to ensure the biological plausibility of detected cells were sphericity and size.

Sphericity Index

This metric quantifies the circularity of detected cells and is defined as:

$$\psi = \frac{4\pi A}{P^2} \in (0,1) \tag{4.6}$$

where A denotes the cell area and P the perimeter. Cells with a sphericity index $\psi > 0.93$ are considered valid, reflecting near-circular morphology.

Size Distribution

To maintain physiological relevance, detected cells are constrained to a minimum diameter of 6 μm (equivalent to approximately 47 pixels), ensuring the exclusion of minor artifacts or debris.

4.4.4 Implementation Considerations

The processing pipeline balances computational efficiency with biological accuracy by incorporating early rejection of non-cellular artifacts, parallel execution of independent stages, memory-efficient contour representations, and multi-threaded overlap resolution. While visual validation checkpoints were included during development to verify intermediate results, these were turned off in the production environment to optimize runtime performance.

Additionally, texture analysis was performed on image patches surrounding each cell, including a margin of extra space to capture cell regions potentially not identified by segmentation. In contrast, shape-based features were computed directly from the cell contours extracted during segmentation, ensuring accurate morphological characterization.

4.5 Feature Extraction and Classification

In computational pathology, raw pixel data from cytology images contains vast amounts of information that is not directly interpretable by machine learning (ML) algorithms. Our comprehensive feature extraction framework addresses this challenge through multiple complementary approaches. First, dimensionality reduction techniques transform millions of pixels into hundreds of quantitative descriptors while preserving diagnostically relevant information. Next, multi-scale analysis captures features spanning from subcellular details to broader cellular organization. The framework also incorporates biologically meaningful representations by mathematically encoding nuclear morphology, chromatin patterns, and cellular architecture. Finally, the design emphasizes technical robustness, ensuring consistency across variations in sample preparation and image acquisition.

4.5.1 Complete Feature Taxonomy

Table 4.1: Extended Feature Categories and Their Biological Significance.

Category	**Count**	**Pathological Correlation**
First-Order Statistics (FOS)	16	Nuclear staining intensity distribution
Gray-Level Co-occurrence (GLCM)	28	Local chromatin organization patterns
Run-Length Matrices (GLRLM)	11	Linear chromatin arrangements
Neighborhood Differences (NGTDM)	5	Nuclear heterogeneity and granularity
Shape Features	5	Cellular morphology characteristics
Local Binary Patterns (LBP)	6	Micro-texture variations at multiple scales
Histogram Features	128	Multi-region intensity distributions
Gabor Transform	16	Oriented texture patterns
Hu Moments	7	Rotation-invariant shape descriptors
Clustering Features	2	Cellular spatial distribution
Total	$\sim$220+	

Table 4.1 presents an extended taxonomy of the extracted feature categories along with their biological significance and pathological correlations. These features comprehensively characterize nuclear intensity, texture, shape, spatial distribution, and other relevant attributes critical for diagnostic interpretation.

4.5.2 Detailed Feature Specifications

Our feature extraction pipeline includes biologically and mathematically grounded descriptors that characterize cellular morphology and texture at multiple scales, capturing both nuclear details and overall cell architecture.

4.5.2.1 First-Order Statistics (FOS)

First-Order Statistics capture the overall intensity distribution of the nuclei in the image, providing insight into nuclear staining and chromatin density variations. These statistics summarize pixel intensity information without considering spatial relationships. These computed features provide a succinct quantitative summary of the intensity characteristics within the nuclear region, serving as foundational descriptors for subsequent texture and morphology analysis. It was calculated as represented in Algorithm 15.

Algorithm 15 First-Order Statistics (FOS) Computation

Require: Grayscale image I, binary mask M

Extract masked pixels: `masked_pixels =` $I[M]$

Compute normalized histogram: `hist, bins = np.histogram(masked_pixels, bins=32, density=True)`

Initialize features dictionary:

```
features = {
  'FOS_Mean': np.mean(masked_pixels),
  'FOS_Variance': np.var(masked_pixels),
  'FOS_Energy': np.sum(hist**2),
  'FOS_Entropy': -np.sum(hist * np.log2(hist + 1e-10)),
  ... }                ▷ Additional statistics can be added here
```

return `features`

4.5.2.2 Gray-Level Co-occurrence Matrix (GLCM)

This set of features quantifies spatial relationships between pixel intensities, capturing chromatin texture patterns at micrometer resolution. It is especially useful for describing how pixel intensities vary with distance and direction, reflecting the local organization of nuclear material. The process to obtain the features is represented in the Algorithm 16.

Algorithm 16 GLCM Feature Extraction

Require: Grayscale image I, offsets $\Delta = \{(1,0),(0,1),(1,1),(-1,1)\}$

Initialize feature accumulator: `features = {}`

for each offset $\delta \in \Delta$ **do**

  Compute GLCM: `glcm = graycomatrix(I, offsets=[`δ`], levels=256, symmetric=True)`

  for property `prop` in `['contrast','homogeneity','energy','correlation']` **do**

    `features[f'GLCM_{prop}_Mean'] += graycoprops(glcm, prop)[0,0]`

  end for

end for

Average features across offsets

return `features`

These features capture second-order statistics by summarizing how often pairs of intensity values occur at specific spatial offsets. They are particularly sensitive to local texture patterns and directional chromatin arrangements.

4.5.2.3 Gray-Level Run-Length Matrix (GLRLM)

GLRLM features, represented in Algorithm 17, measure chromatin organization by quantifying linear texture patterns across different directions. These descriptors highlight the distribution of consecutive pixels with the same grey level, helping differentiate between fine and coarse chromatin textures. The key features are SRE (Short Run Emphasis), which is sensitive to fine chromatin, and LRE (Long Run Emphasis), that detects coarse chromatin.

Algorithm 17 GLRLM Feature Extraction

Require: Grayscale image I, binary mask M

Compute GLRLM features: `glrlm = pyfeats.glrlm_features(I, M, Ng=256)`
return {
  `'GLRLM_SRE': glrlm[0],`
  `'GLRLM_LRE': glrlm[1],`
  `...` ▷ `9 additional features`
}

By encoding the frequency and length of consecutive intensity runs, these features capture directionally coherent texture patterns often seen in chromatin strands. They are useful in detecting both subtle and coarse textural variations within the nucleus.

4.5.2.4 Neighborhood Gray-Tone Difference Matrix (NGTDM)

The NGTDM algorithm, represented in Algorithm 18, quantifies local intensity differences between each pixel and its neighborhood, providing a measure of local chromatin heterogeneity. It captures subtle variations in tone that may reflect uneven chromatin distribution or irregularities in nuclear staining.

Algorithm 18 NGTDM Feature Extraction

Require: Grayscale image I, binary mask M, neighborhood radius $d = 1$

Initialize difference array: `diff = np.zeros(256)`
for each pixel $(x,y) \in M$ **do**
  Extract neighborhood: `nb` $= I[x-d : x+d+1, y-d : y+d+1]$
  Compute tone difference: `diff[`$I(x,y)$`]` += $|I(x,y) - \texttt{mean}(nb)|$
end for
return {
  `'NGTDM_Coarseness':` $(\sum p(i)s(i))^{-1}$,
  `'NGTDM_Contrast':` $\frac{1}{N}\sum_{i,j} p(i)p(j)(i-j)^2$,
  `...` ▷ `3 additional features`
}

These features are particularly effective in capturing fine-grained irregularities in nuclear texture, offering complementary information to co-occurrence and run-length matrices in characterizing chromatin complexity.

4.5.2.5 Local Binary Patterns (LBP)

Local Binary Patterns, represented in Algorithm 19, encode micro-texture information by thresholding neighborhood pixels around each pixel in the image, effectively capturing the distribution of chromatin granularity. This descriptor is particularly suited for identifying repetitive textural motifs and local irregularities in nuclear chromatin.

Algorithm 19 Multi-Scale LBP Feature Extraction

Require: Grayscale image I, binary mask M

Initialize `features = {}`
for each scale (P,R) in `[(8,1), (16,2), (24,3)]` **do**
    `lbp = local_binary_pattern(I, P=P, R=R, method='uniform')`
    `hist = np.histogram(lbp[M], bins=P+2)[0]`
    `features[f'LBP_{P}_{R}_energy'] = np.sum(hist**2)`
    `features[f'LBP_{P}_{R}_entropy'] = -np.sum(hist * np.log2(hist + 1e-10))`
end for
return `features`

By combining multiple radii and sampling points, multi-scale LBP features provide a robust description of chromatin architecture, supporting discrimination of nuclei based on subtle textural variations.

4.5.2.6 Histogram Features

Histogram-based features, represented in Algorithm 20, capture the distribution of pixel intensities across different regions of the nucleus, reflecting variations in chromatin condensation and nuclear staining heterogeneity. By analyzing both global and successively eroded regions, these features provide a layered representation of nuclear content at different spatial scales.

Algorithm 20 Multi-Region Histogram Computation

Require: Grayscale image I, binary mask M, erosion levels $L = 3$

Initialize features dictionary: `features =`

Compute global histogram: `hist = np.histogram(I[M], bins=32, range=(0,255))[0]`

Store features: `features.update('Hist_global_bin_'+str(i): hist[i] for i in range(32))`

for level in 1 to L **do**

    Erode mask: `M_eroded = binary_erosion(M, iterations=level)`

    Compute regional histogram: `hist = np.histogram(I[M_eroded],bins=32)[0]`

    Store features: `features.update('Hist_erode'+str(level)+'_bin_'+str(i): hist[i] for i in range(32))`

end for

return `features` ▷ 128 features total (32 bins × 4 regions)

The use of concentric erosion levels enables a multi-region intensity analysis, capturing both core and peripheral nuclear intensity distributions and supporting finer discrimination of textural phenotypes.

4.5.2.7 Gabor Transform Features

Gabor filters, represented in Algorithm 21, detect oriented texture patterns in nuclear regions, capturing fine details that relate to chromatin organization and irregularities in the nuclear membrane. These features are particularly useful for identifying directional structures within the nucleus.

Algorithm 21 Gabor Filter Bank Implementation

Require: Grayscale image I, mask M, orientations *[0°,45°,90°,135°]*, freq=0.1

Initialize features: `features =`

for theta in orientations **do**

    Create Gabor kernel: `kernel = cv2.getGaborKernel((9,9), sigma=3, theta=theta, lambd=1/freq, gamma=0.5)`

    Filter image: `filtered = cv2.filter2D(I, cv2.CV_32F, kernel)`

    Extract masked responses: `responses = filtered[M]`

    Store features: `features[f'Gabor_thetamean'] = np.mean(responses)`

    Store features: `features[f'Gabortheta_std'] = np.std(responses)`

end for

return `features` ▷ 16 features total: mean and std for each of 8 filters

The directional sensitivity of the Gabor filters allows the model to capture subtle anisotropic textural cues, which are often linked to underlying chromatin fiber orientations and nuclear

architectural changes.

4.5.2.8 Hu Moments

Hu moments, represented in Algorithm 22, provide rotation-invariant shape descriptors that quantify nuclear morphology independent of cell orientation. These features are particularly robust to translation, scale, and rotation, making them suitable for capturing global nuclear shape patterns.

Algorithm 22 Hu Moments Calculation

Require: Binary mask M of cell region

Ensure: Dictionary of 7 Hu moments with keys `Hu_Moment_1` to `Hu_Moment_7`

1: Compute spatial moments: `moments = cv2.moments(`M`)` ▷ OpenCV calculates raw/central moments

2: Derive Hu moments: `huMoments = cv2.HuMoments(moments).flatten()`

3: Label moments: `labels = [f'Hu_Moment_i+1' for i in range(7)]`

4: **return** `dict(zip(labels, huMoments))`

The resulting seven Hu moments provide a compact and invariant summary of nuclear shape, useful for differentiating cells based on global morphological distortions.

4.5.2.9 Clustering Features

These features quantify how cells are spatially distributed, revealing potential patterns related to tumor growth or microenvironment organization. Two custom features were implemented to characterize clustering behavior based on intercellular distances. Firstly, the centroid of each segmented cell is computed. Then, a neighborhood radius is defined for each cell as twice the length of the largest major axis among all cells in the image. Cells whose centroids fall within this radius are considered neighbors.

The first feature computes, for each cell, the ratio between the number of its neighbors and the total number of cells in the image. This quantifies how locally clustered a cell is relative to the global cell count. The second feature computes, for each cell, the ratio between the number of neighbors and the image area (in pixels), divided by 1.000.000, to ensure manageable numeric values. This second feature reflects local cell density adjusted for image size. The two features aim to describe both relative clustering and spatial concentration of cells across the image.

4.5.3 Normalization

Cell-level variations require careful normalization to bring the extracted features to a common level across cells. Histogram bin distributions require per-cell normalization to remove the impact of variations in intensity ranges. Shape features require normalization with respect to the cellular

area to ensure equal comparison of various-sized cells. Spatial locations are also pre-conditioned with area-based standardization to minimize position and scale biases in cells. The normalization process is depicted in Algorithm 23.

Algorithm 23 NormalizeCellFeatures

procedure NORMALIZECELLFEATURES($features$)
  for $i \leftarrow 0$ to 31 **do**
    $bin_key \leftarrow$ 'Histogram_bin_' $+ i$
    if $bin_key \in features$ **then**
      $total \leftarrow \sum_{j=0}^{31} features[\text{'Histogram_bin_'} + j]$
      if $total > 0$ **then**
        $features[bin_key] \leftarrow features[bin_key]/total$
      end if
    end if
  end for
  if $features[\text{'SHAPE_area'}] > 0$ **then**
    for $shape_feat$ in {'SHAPE_perimeter', 'SHAPE_XcoordMax', 'SHAPE_YcoordMax'} **do**
      if $shape_feat \in features$ **then**
        $features[shape_feat] \leftarrow features[shape_feat]/features[\text{'SHAPE_area'}]$
      end if
    end for
  end if
  return $features$
end procedure

The procedure `NormalizeCellFeatures` normalizes features at the single-cell level through shifting histogram bins and shape-related parameters. It iterates through 32 histogram bins named `Histogram_bin_0` through `Histogram_bin_31`. For each histogram bin of the feature set, it calculates the total of all 32 histogram bins. If the sum is greater than zero, each value in the bins is divided by it, effectively normalizing the histogram to reflect relative proportions rather than absolute frequencies. Then the procedure checks if the `SHAPE_area` attribute exists and is greater than zero. If it does, it normalizes shape attributes such as `SHAPE_perimeter`, `SHAPE_XcoordMax`, and `SHAPE_YcoordMax` by dividing them by the area of the cell. This normalization accounts for cell size differences, allowing one to compare shape descriptors regardless of cell size. The function returns the normalized features such that both intensity distribution and shape measurements become standardized at the cell level. A returned normalized data frame is finally ready for downstream analysis or ML processing.

4.5.4 Classification Methods

Following feature extraction, we compare four diverse classification approaches to determine the most suitable method for identifying bladder cancer from cytology images. This comparative framework enhances robustness against algorithmic biases while leveraging the complementary strengths of each method. The classifiers selected represent different paradigms of ML, chosen for their complementary strengths in medical image analysis. K-Nearest Neighbors (KNN) is an instance-based model that excels when similar instances cluster together in feature space. Support Vector Machine (SVM) provides robust high-dimensional separation based on maximum margin principles. Logistic regression offers interpretable probabilistic outputs, beneficial for clinical decision-making. Random forests combine ensemble learning with intrinsic feature selection, particularly useful for heterogeneous patterns in cytology images.

4.5.4.1 Random Forest

Random Forests help address several key challenges in cytology-based classification through ensemble learning. By constructing multiple decision trees on bootstrap samples (bagging) using random subsets of features at every node, the method achieves improved generalizability and less tree overfitting. The *Gini* impurity splitting criterion performs satisfactorily on categorical and continuous features, common in cytological image analysis.

The intrinsic feature importance score of the algorithm, in terms of average impurity decrease across all trees, reflects which morphological features are primarily accountable for diagnostic decisions. Out-of-bag error estimation offers an efficient internal validation procedure. Random forests display exceptional robustness to label noise and missing data — common issues in medical data sets — by ensemble prediction averaging over numerous heterogeneous trees. The approach's intrinsic parallelism enables efficient training with high-dimensional spaces of features. It is represented in Algorithm 24.

Algorithm 24 Random Forest Training

Require: Data (X, y), trees B, features per node m

for $b \leftarrow 1$ to B **do**
    Bootstrap sample (X_b, y_b)
    Grow tree with feature subset size m:
    while node impurity $> \varepsilon$ **do**
        Find best split among random m features
        Partition node
    end while
end for
return Ensemble $T_{b\,b=1}^{B}$

4.5.4.2 Logistic Regression

Logistic regression, represented in Algorithm 25, provides a probabilistic model of specific clinical decision-making relevance to diagnosing bladder cancer. Through the encoding of the log-odds of malignancy as a linear combination of features, it affords direct interpretability of risk factors as coefficient analysis. The sigmoid activation function ensures outputs as valid probabilities in the range of 0 to 1, allowing clinicians to assess levels of diagnostic confidence.

L1 (lasso) or L2 (ridge) regularization prevents overfitting but enables different forms of feature selection - L1 creates sparsity by zeroing uninformative feature coefficients, while L2 distributes weights more evenly across correlated features. Convex optimization of the approach guarantees global minima convergence, and robust training even for big datasets with stochastic gradient descent variants. For multiclass classification, we employ multinomial extension with cross-entropy loss.

Algorithm 25 Logistic Regression Training

Require: Features X, labels y, learning rate η, epochs T

1: Initialize weights $w \leftarrow \mathbf{0}$
2: **for** $t \leftarrow 1$ to T **do**
3: Compute probabilities: $p_i \leftarrow \sigma(w^T x_i)$
4: Update gradient: $w \leftarrow w - \eta X^T (p - y)$
5: **end for**
6: **return** w

4.5.4.3 XGBoost (Extreme Gradient Boosting)

XGBoost, represented in Algorithm 26, is a state-of-the-art ensemble method that employs gradient boosting with high-level regularization processes. Its efficiency in medical image analysis can be explained by several significant characteristics: sequential tree building correcting previous errors, L1/L2 regularization to prevent overfitting and intrinsic support for missing values. The weighted quantile sketch of the algorithm is effective in handling our high-dimensional cytological features, whereas its parallel tree building makes it faster for training.

Algorithm 26 XGBoost Training

Require: Data (X, y), trees T, learning rate η, max depth d

1: Initialize model $f_0(x) = \text{argmin}_\gamma \sum_i L(y_i, \gamma)$
2: **for** $t \leftarrow 1$ to T **do**
3: Compute pseudo-residuals: $r_{it} = -\left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f=f_{t-1}}$
4: Fit tree h_t to (X, r_t) with depth $\leq d$
5: Find optimal weights: $w_j = \text{argmin}_w \sum_{i \in I_j} L(y_i, f_{t-1}(x_i) + w)$
6: Update model: $f_t(x) = f_{t-1}(x) + \eta \sum_j w_j \mathbf{1}(x \in R_j)$
7: **end for**
8: **return** Ensemble $\{h_t\}_{t=1}^T$

The method employs a zero-inflated, sparsity-aware split-finding algorithm that automatically handles zero-inflated features common in cellular morphology research. XGBoost's gain-based feature importance yields more reliable biomarker identification compared to that of standard random forests. The algorithm also includes early stopping during training, which automatically determines the optimal number of boosting rounds via cross-validation — particularly beneficial given the computational complexity of large cytology datasets.

4.5.4.4 LightGBM (Light Gradient-Boosting Machine)

LightGBM, which is represented in Algorithm 27, is a gradient boosting framework that builds an ensemble of decision trees sequentially to improve predictive accuracy. It learns complex nonlinear relationships by optimizing an objective function through gradient descent, making it well-suited for cytology data where feature interactions and subtle patterns are critical.

LightGBM grows trees leaf-wise with depth constraints, focusing on the most significant splits, resulting in fast training and high efficiency on large datasets. It inherently handles missing values and categorical features, and supports regularization to reduce overfitting, a common risk in biomedical datasets. Hyperparameters such as learning rate, number of trees, and maximum depth are tuned to balance model complexity and generalization, typically via cross-validation.

Algorithm 27 LightGBM Classification

Require: Training set (X_{train}, y_{train}), test sample x_{test}, trained LightGBM model $\mathscr{M}$

1: $\hat{y} \leftarrow \mathscr{M}.predict(x_{test})$
2: **return** $\hat{y}$

4.5.4.5 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP), represented in Algorithm 28, is a feedforward artificial neural network composed of input, hidden, and output layers. It learns nonlinear mappings between features and class labels through iterative weight updates via backpropagation.

In cytological image analysis, MLP models intricate relationships between cell features by stacking multiple layers with nonlinear activation functions such as ReLU. Training minimizes a loss function (e.g., cross-entropy) using gradient-based optimization methods like Adam or SGD. Hyperparameters such as network architecture, learning rate, and regularization are carefully tuned to avoid overfitting, especially given the limited size of biomedical datasets.

Algorithm 28 MLP Classification

Require: Training set (X_{train}, y_{train}), test sample x_{test}, trained MLP model $\mathcal{N}$

1: $\hat{y} \leftarrow \mathcal{N}.predict(x_{test})$

2: **return** $\hat{y}$

4.6 Summary

This chapter presented the end-to-end approach tailored for bladder cancer diagnosis from microscopic cell images in urine. The proposed pipeline has four key stages: data preparation and collection, image preprocessing, cell segmentation, and feature extraction and classification.

The data set includes microscopy images of bladder cancer patient urine-derived cells and healthy controls captured at 200× total magnification. The images had natural inconsistencies in cell density, focus quality, and background composition and required robust preprocessing solutions. Six preprocessing algorithms were tested and compared, ranging from elementary techniques like Otsu's thresholding and adaptive thresholding to sophisticated pipelines that include contrast enhancement and morphological processing (CCM, AGB, and GAD variants).

Cell segmentation is a multi-step procedure beginning with the detection of contours, followed by biologically inspired filtering with a minimum threshold of 6 μm diameter. The method involves precise micrometer-to-pixel conversion calibrated using Neubauer chamber grid measurements to deliver a conversion factor of 7.88175 pixels/micrometer. Other processing steps include convex hull regularization for shape standardization and overlap resolution to tackle overlapping cells.

Feature extraction generates a considerable set of over 220 quantitative features across different categories. First-order statistics (FOS) provide 16 major intensity distribution features. Texture analysis provides 28 features based on gray-level co-occurrence matrices (GLCM), 11 from run-length matrices (GLRLM), and 5 from neighborhood difference matrices (NGTDM). Moreover, there was shape characterization with 5 geometric features, 128 histogram features, and other features, as 7 Hu moments and 2 developed clustering features. Multi-scale pattern analysis yields 6 local binary patterns (LBP) features, and 16 Gabor filter responses and spatial distribution features yield cellular organization patterns.

For classification, five distinct algorithms are employed to evaluate and compare performance: Logistic Regression for probabilistic modeling, Random Forests as an ensemble of decision trees, XGBoost and LightGBM for gradient-boosted tree-based learning, and Multi-Layer Perceptron

(MLP) representing neural network approaches. This diverse set balances interpretability, computational efficiency, and predictive power for cytological image analysis.

The method favors biological plausibility by size filtering and shape regularization but guarantees computational effectiveness via parallel processing and early rejection of non-cellular artifacts. The remainder of this book outlines the implementation and validation of this pipeline on clinical data, demonstrating its efficacy for bladder cancer detection.

Chapter 5

Implementation and Results

This chapter presents the specifications of the proposed solution functions and analyzes the results obtained at the various pipeline stages, including segmentation and preprocessing, and classification. We define the parameters used in the implemented functions, referred to in Chapter 4. Subsequently, preprocessing and segmentation are evaluated through a qualitative analysis of their outputs, followed by a quantitative assessment of the detected contours. Finally, we report on the classification performance, highlighting how the extracted features and normalization procedures contribute to BCa label prediction, whether binary, stage, or risk-related.

5.1 Segmentation Implementation Details

The implementation followed a modular pipeline design, where each workflow stage could be independently modified or tested. Parameters were selected based on empirical trials over representative images from the dataset across patients. Each segmentation method used in this study was implemented with carefully selected parameters tailored to enhance performance across the diverse dataset. The main parameter choices for each algorithm are described below.

Otsu's Thresholding

Otsu's method was implemented using OpenCV's `cv.threshold()` [68] function with the following parameters: the threshold value `thresh` was set to 0 (ignored due to the automatic nature of Otsu), `maxval` was set to 255, and the thresholding type was defined as `cv.THRESH_BINARY + cv.THRESH_OTSU`. This ensured automatic binarization based on histogram analysis of the input grayscale image.

Binary Thresholding

For simple global thresholding, a fixed intensity threshold was applied using the same `cv.threshold()` function, with `thresh` set to 200 and `maxval` to 255. The thresholding type

used was `cv.THRESH_BINARY`. This method assumes relatively uniform illumination across the image, which makes it less robust under varying lighting conditions.

Adaptive Thresholding

Adaptive thresholding was implemented using OpenCV's `cv.adaptiveThreshold()` [68] function. The maximum value was defined as 255, with the adaptive method set to `cv.ADAPTIVE_THRESH_GAUSSIAN_C`. The binary thresholding mode (`cv.THRESH_BINARY`) was applied, with a block size of 11 and a constant `C` value of 2 subtracted from the calculated mean. This configuration allowed for better adaptation to local variations in pixel intensity, which is particularly useful in heterogeneous background conditions.

Contrast-Colorspace-Morphological Operations (CCM)

The CCM pipeline combines several preprocessing steps for enhanced segmentation robustness. A median blur with a 3×3 kernel was applied to reduce noise, followed by contrast normalization when the image's contrast exceeded a threshold of 50. Color space transformations to HSV were employed to isolate features based on hue variations. Morphological closing operations were applied using variable kernel sizes. Specifically, a 30×30 rectangular kernel was used when appropriate, while a reduced 1×1 or 20×20 kernel was selected if the standard deviation of the hue channel was above 38. These heuristics were tuned to suppress fragmented contours and bridge small gaps in cellular boundaries.

Adaptive-Gaussian-Binary (AGB)

This hybrid method incorporated Gaussian smoothing, adaptive thresholding, and a final fixed threshold. A Gaussian blur was first applied using a **zero kernel size** $((0,0))$, which **does not mean no blurring** but instead automatically computes the kernel dimensions based on the provided standard deviation ($\sigma = 3$). The kernel size, derived by the equation 4.3.5, is a **19×19 kernel** for $\sigma = 3$.

The adaptive thresholding step followed the same parameterization described in Section 5.1. Finally, a binary thresholding step was applied with a fixed value 50 to consolidate the mask. This approach aimed to reduce noise while preserving well-defined contours in images with moderate variability.

Gauss-Adaptive-Dilation (GAD)

The GAD method begins with a Gaussian blur (zero kernel size, $\sigma = 2$) to reduce high-frequency noise. It is followed by adaptive thresholding and a dilation operation to enhance segmentation continuity. Two dilation strategies were explored: a mild dilation using a 3×3 kernel for one iteration, and a stronger variant with a 2×2 kernel applied over three iterations. These

options were chosen based on visual inspection to assess the optimal balance between contour completeness of isolated cells and cell separation (detection of some aggregated cells).

Implementation Interface

Each of the preprocessing strategies described above was implemented as an independent function. The selection interface for these methods is shown below:

```
preprocessed_images = {
    "Otsu": otsu_threshold(image),
    "Binary": binary_threshold(image),
    "Adaptive": adaptive_threshold(image),
    "CCM": contrast_colorspace_morph(image),
    "AGB": adaptive_gauss_binary(image),
    "GAD": gauss_adaptive_dilate(image)
}
```

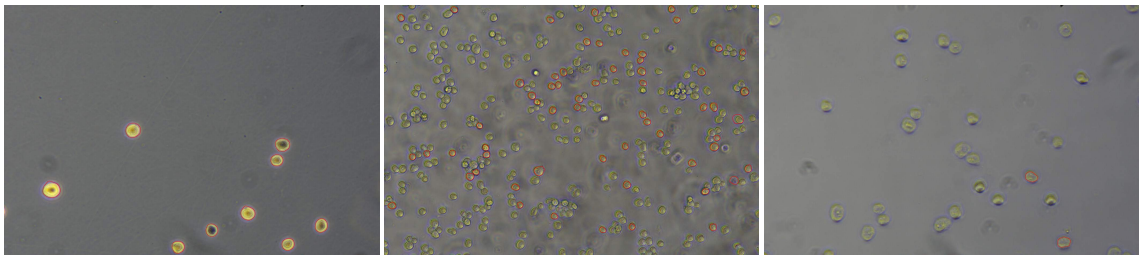
5.2 Preprocessing and Segmentation Results

We regularly assess our segmentation approach alongside our preprocessing methods, as every segmentation requires a preprocessing method, even the simplest, such as thresholding the image to create a mask for the contours. Our segmentation demonstrates variant quantity and quality performance across patients, depending on the preprocessing algorithm applied. In the following subsections, we qualitatively analyze our results through examples from each method, describing situations in which the output was considered successful, acceptable, or failed.

5.2.1 Qualitative Analysis

The qualitative analysis aims to provide an intuitive understanding of how well the segmentation algorithms perform under various imaging conditions. Since the input images vary greatly in terms of colors, lighting, and background complexity, segmentation results also exhibit a broad spectrum of quality. Rather than relying solely on quantitative metrics, this section uses representative examples to illustrate typical successes, borderline cases, and clear failures for each method. This approach helps highlight the strengths and limitations of each preprocessing and segmentation pipeline and guides future decisions by identifying challenging scenarios.

Otsu's Thresholding Algorithm

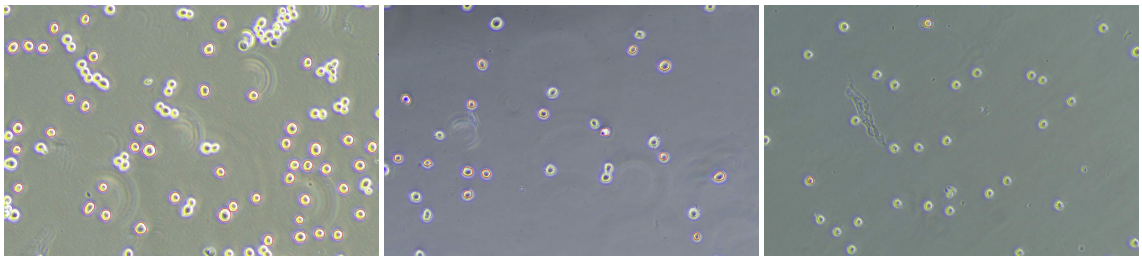


(a) Good result (b) Medium result (c) Poor result

Figure 5.1: Segmentation results using Otsu's method. (a) Accurate separation of cells; (b) partial detection with noisy contours; (c) failure in complex background. Adapted from [122].

Otsu's method consistently performed well when cell contrast was high, successfully segmenting the majority of visible cells. It was less effective in the presence of uneven artifacts or when background textures mimicked cellular shapes, which occasionally led to false positives or fragmented contours. On Figure 5.1, it's possible to check the result of this approach in three different cases of performance: good, medium, and poor results.

Binary Thresholding Algorithm

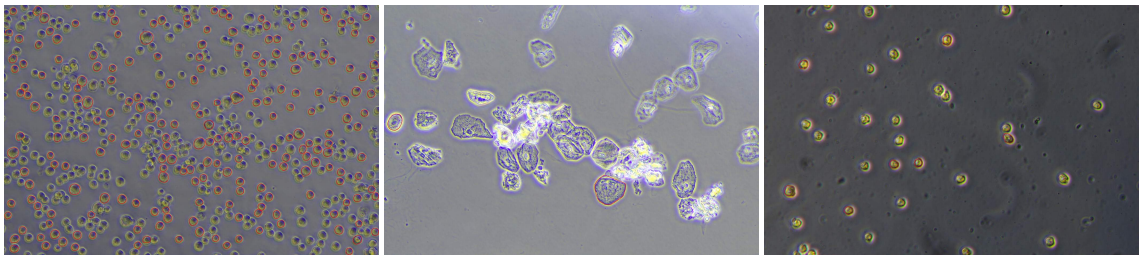


(a) Good result (b) Medium result (c) Poor result

Figure 5.2: Segmentation results using Binary Thresholding. Performance drops under heterogeneous staining or background texture. Adapted from [122].

This method achieved reasonable segmentation under uniform illumination but struggled significantly in images with heterogeneous backgrounds. While computationally efficient, its inability to adapt locally to varying intensities resulted in either missed detections or excessive noise, limiting its practical usability. On Figure 5.2, it's possible to check the result of this approach in three different cases of performance: good, medium, and poor results.

Adaptive Thresholding Algorithm

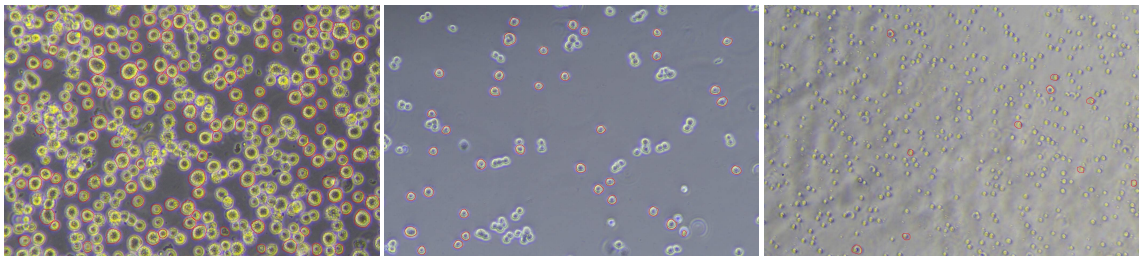


(a) Good result (b) Medium result (c) Poor result

Figure 5.3: Results from adaptive thresholding: effective in variable lighting but susceptible to noise in crowded or poorly stained regions. Adapted from [122].

Adaptive thresholding showed strong performance in cases where illumination was inconsistent. However, the finer sensitivity to local variations also made it susceptible to over-segmentation, particularly in dense cellular regions. Some outputs revealed noisy artifacts along tissue edges or background textures. In Figure 5.3, it's possible to check the result of this approach in three different cases of performance: good, medium, and poor results.

Contrast-Colorspace-Morphological Operations (CCM) Algorithm

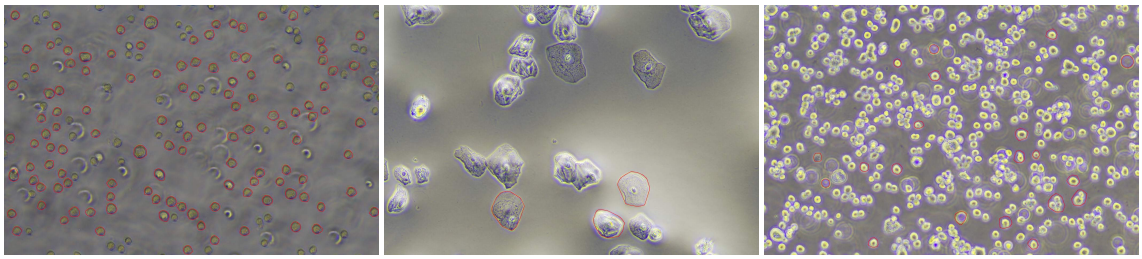


(a) Good result (b) Medium result (c) Poor result

Figure 5.4: Segmentation results using contrast and morphological preprocessing. Capable of handling difficult backgrounds, but may over-smooth boundaries. Adapted from [122].

The CCM algorithm was particularly effective in challenging cases. Enhancing contrast and operating in alternative color spaces (such as HSV) could better isolate cellular regions. Morphological operations helped refine masks and eliminate noise. However, performance remained sensitive to parameter selection and image characteristics. In Figure 5.4, it's possible to check the result of this approach in three different cases of performance: good, medium, and poor results.

Adaptive-Gaussian-Binary (AGB) Algorithm

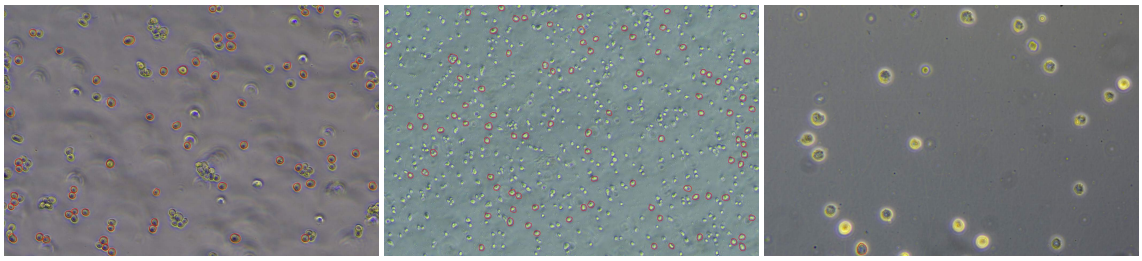


(a) Good result (b) Medium result (c) Poor result

Figure 5.5: Segmentation results using Adaptive thresholding, Gauss blur, and Binary thresholding (AGB). Effective smoothing, though fine details may be lost. Adapted from [122].

Combining smoothing with adaptive thresholding and binary postprocessing, this hybrid method produced robust segmentations in most scenarios. It reduced background noise effectively and clarified cellular contours, although in some instances, small or faintly stained cells were lost due to the blurring step. In Figure 5.5, it's possible to check the result of this approach in three different cases of performance: good, medium, and poor results.

Gauss-Adaptive-Dilation (GAD) Algorithm



(a) Good result (b) Medium result (c) Poor result

Figure 5.6: Segmentation with Gauss blur, Adaptive thresholding and Dilation (GAD). Fills disconnected contours but may merge adjacent cells. Adapted from [122].

The GAD pipeline was very valuable in clustered areas, improving cell contours individually, even when they were in clusters. While it increased clustered cells' segmentation, this smaller dilation introduced the risk of keeping isolated cells fragmented and undetected, especially when contours were initially incomplete. In turn, using a bigger dilation would merge adjacent cells into a single object. That is why the two kernels with their iterations were used: to keep both clustered and isolated cells detectable. This algorithm is, therefore, worthwhile for enhancing completeness in weaker contoured cell zones. In Figure 5.6, it's possible to check the result of this approach in three different cases of performance: good, medium, and poor results.

5.2.2 Quantitative Analysis

In addition to visual examples, a quantitative analysis of preprocessing approaches was carried out. Assessing how each preprocessing method impacts the number of contours detected per image across the dataset is important. It provides insight into whether a method produces a consistent and plausible number of regions, which is crucial for downstream analysis.

To objectively assess the segmentation performance of the preprocessing methods, we analyzed the average number of contours per image per algorithm in order to understand the impact of the various preprocesses tested on the dataset as a whole.

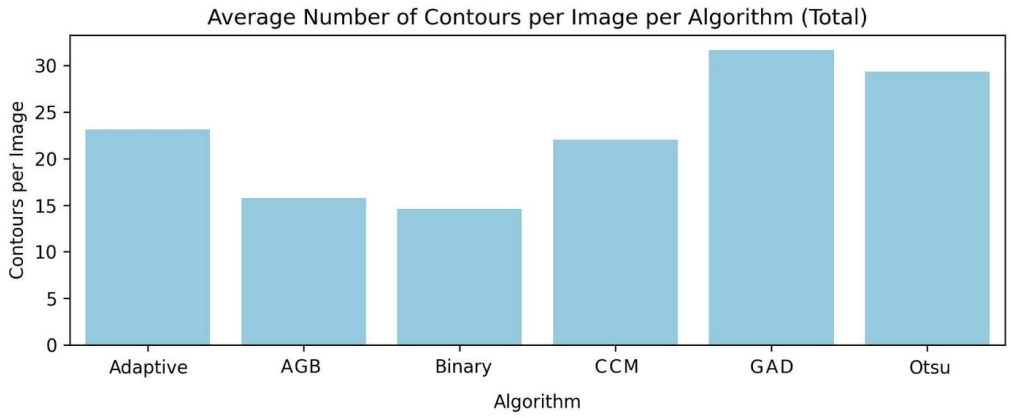


Figure 5.7: Total contours detected across different preprocessing algorithms.

Figure 5.7 shows the distribution of the contours detected per image for each preprocessing algorithm evaluated. While methods like Otsu's thresholding and GAD tend to detect more cells on average, this may reflect a better performance in patients with a higher number of images or inherently more cells per image. However, this variation is a biological factor we cannot fully control. Attempting to normalize it could introduce bias by underrepresenting patients who naturally exhibit more cellular clustering. The goal remains to detect as many cells as possible per patient while ensuring the method performs consistently and reliably across the entire dataset.

Thus, it is important to consider the distribution of the number of detected contours per image in each algorithm and each patient. Figure 5.8 shows each preprocessing method's cell distribution per patient.

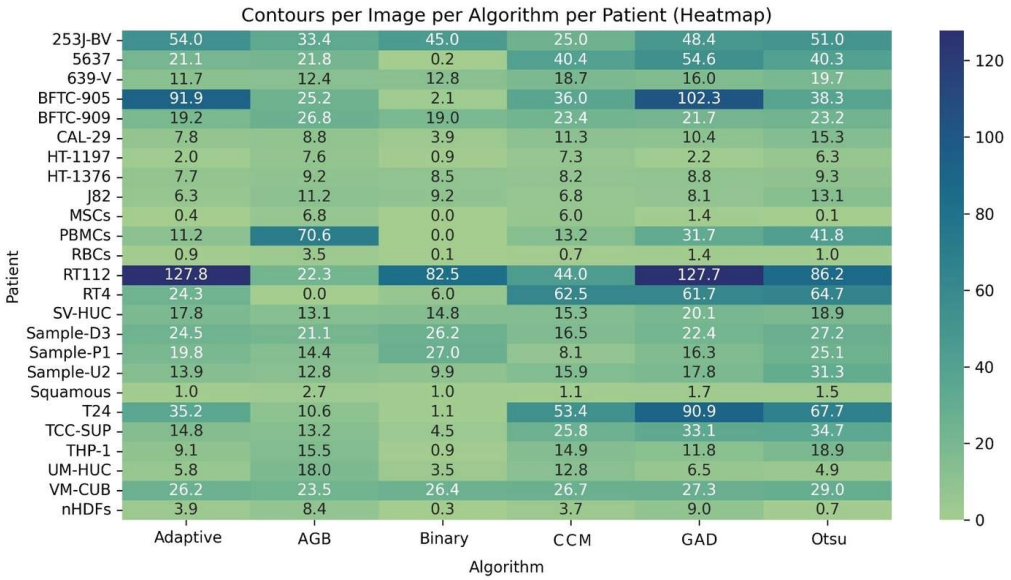


Figure 5.8: Heatmap showing the number of contours detected per image per patient for each preprocessing method. Rows correspond to individual patients (identified by anonymized internal codes), while columns represent the different algorithms developed.

This visualization highlights the variability in cell detection across patients and methods, revealing that algorithms like Otsu, GAD, or Adaptive, besides being the ones with the most detected cells on average by image (Figure 5.7), generically cover all the dataset, avoiding near-zero values generally. On the one hand, at the patient level, GAD avoids the frequent near-zero detections observed in other methods, suggesting robustness in managing different image conditions without failing in more complex or lower-quality cases. On the other hand, both Otsu and Adaptive thresholding have a near zero value in the same patient but are highly capable of adapting to different image conditions, being the second and third algorithms with more cells per patient, respectively.

These metrics provide a rough but informative indication of whether a method tends to detect cells well in various patients. Regardless, it is important to note whether there is a general trend in algorithms to detect few cells in each patient and confirm whether patient images contain very few cells or the images are of debatable quality.

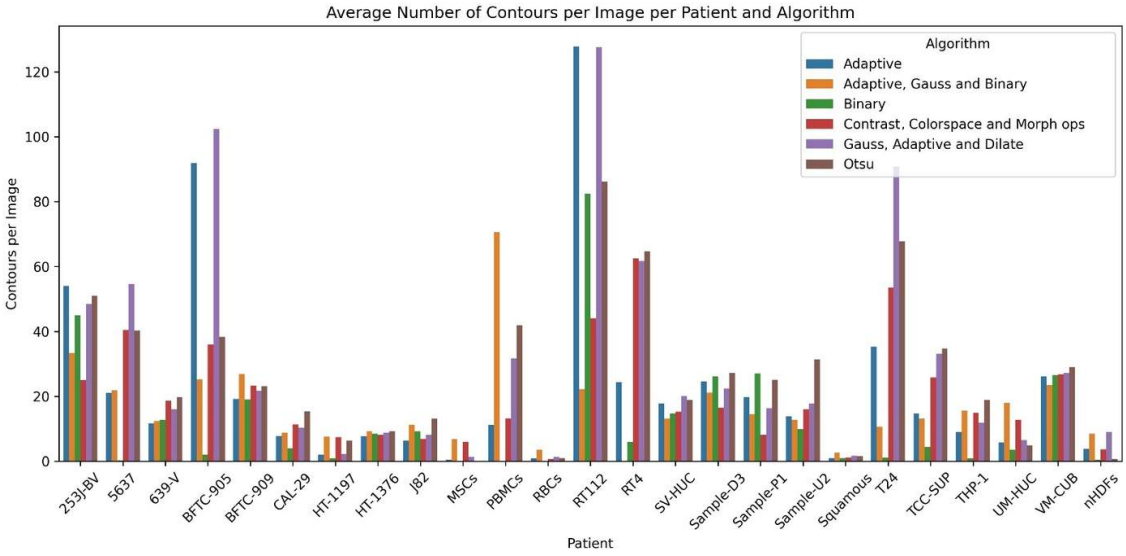


Figure 5.9: Bar chart showing the average number of detected cell contours per image, grouped by patient and preprocessing algorithm.

In Figure 5.9, each bar represents the mean number of contours found per image for a given patient, considering all images associated with that patient. While Figure 5.8 helps determine whether specific algorithms consistently detect fewer or more cells across different patient samples, this visualization particularly supports the identification of general trends, such as a tendency for all algorithms to detect few cells in some patients, for instance, in patients RBCs and Squamous. This could reflect either a biologically lower-than-average number of cells or raise concerns about image quality, prompting further investigation. This makes it plausible that an algorithm does not become a failure due to bad results in patients where, generically, there are few cells.

The last three figures (Figures 5.7, 5.8, 5.9) highlighted that GAD, Otsu, and Adaptive thresholding performed reliably overall. However, GAD and Otsu were selected for the classification step. GAD already uses an adaptive approach similar to Adaptive thresholding and performs well even in cases where Adaptive shows stronger results. At the same time, Otsu detects more cells on average per image, making both methods strong and complementary choices. They are considered the most suitable preprocessing approaches for subsequent analysis, offering both robustness and interpretability in a biologically diverse dataset.

5.2.3 Contour Overall Conclusions

The segmentation methods were analyzed after applying the previous preprocessing techniques, as any segmentation process requires a preprocessing method, at least a very minimal one, like thresholding the image to obtain a mask for the contours. Our segmentation performed differently in quantity and quality by patient, depending on which preprocessing algorithm is being used.

The methodology was characterized qualitatively by examples of each approach, detailing conditions under which the output succeeded, was acceptable, or failed. The qualitative assessment

demonstrates that, accepting that each algorithm can have few outliers, some tend to under-segment by missing relevant cellular structures. The quantitative analysis showed that three methods, the GAD algorithm (Gaussian blur, Adaptive thresholding, and Dilation), Otsu, and Adaptive thresholding, consistently achieve a balanced and reliable performance across the dataset. Nevertheless, the two chosen for classification were GAD and Otsu, as GAD contains the same adaptive technique as Adaptive thresholding, and already performs well in the patients where Adaptive has its peaks. Furthermore, Otsu finds more cells per image, on average. From the methods tested, GAD and Otsu were the most successful and most consistent with varying samples. Otsu's algorithm worked perfectly when the cell contrast with the background was good and clear, and it provided clear and stable borders in most cases. It was a good reference because it was easy to implement and efficient but not very efficient for images with clutter that had artifacts or superposed textures.

The segmentation process was able to separate ROIs within the cell areas across the dataset. In spite of some noise, especially in complex or textured background images, most of the algorithms could produce viable segmentations as a good basis for subsequent feature extraction and classification activities. Notably, pipeline performance was also preprocessing algorithm — and sample intrinsic complexity — sensitive, with some patients giving more reproducible responses than others. Such variability is attestation to the fact that the data were non-trivial: images contained significant heterogeneity in collection conditions, background noise, illumination, and cell morphology. It was, therefore, challenging in principle to develop a single one-size-fits-all segmentation strategy.

GAD, on the other hand, was a preferable choice in more challenging cases, especially for cells compacted closely or cells with poor contours. Background noise was diminished through Gaussian blur, while local intensity gradients were still preserved via adaptive thresholding. Dilation, with kernel sizes and iterations suitably tuned, effectively avoided the twin limitation of keeping loose cells' contours continuous without fusing too much in the dense cell areas. This made GAD particularly valuable in providing high completeness and accuracy, even in changing or poor imaging conditions.

Together, these results justify the application of Otsu and GAD as the best segmentation techniques to be employed downstream in the pipeline. While none was confirmed as a general best due to the intrinsic unconformity of the dataset, both of these represented the best compromises between cell structure sensitivity and noise resistance, enabling consistent analysis in most instances.

The capability of adapting to different image characteristics is particularly relevant in clinical datasets, where biological variability and differences in image acquisition naturally occur. Attempting to normalize these differences artificially can lead to biased interpretations and obscure meaningful pathological variation.

5.3 Classification Results

This section presents the results of the classification tasks using the features mentioned in Chapter 4, which were extracted from contours detected with the GAD and Otsu preprocessing methods. We evaluated performance across multiple classification problems — binary cancer detection, cancer staging, and risk level — under different dataset configurations. The metrics used to evaluate the classifiers are introduced first, followed by a structured analysis of the results.

5.3.1 Evaluation Metrics

To assess classification performance, we used several evaluation metrics commonly applied in medical image analysis, taking into account that TP corresponds to True Positives, FP to False Positives, TN to True Negatives and FN to False Negatives.

Accuracy (Acc) measures the overall correctness of the model and is defined as:

$$Acc = \frac{TP+TN}{TP+TN+FP+FN}$$

Balanced Accuracy (BAcc) accounts for class imbalance by averaging the true positive rate and true negative rate:

$$BAcc = \frac{1}{2}\left(\frac{TP}{TP+FN}+\frac{TN}{TN+FP}\right)$$

Sensitivity, also known as recall or the true positive rate, reflects the proportion of actual positive cases correctly identified:

$$Sensitivity = \frac{TP}{TP+FN}$$

Specificity, or the true negative rate, quantifies the proportion of negative cases correctly classified:

$$Specificity = \frac{TN}{TN+FP}$$

Finally, the **Confusion Matrix** was used to provide a detailed breakdown of predictions, showing the distribution of true versus predicted labels for each class.

5.3.2 Binary Classification Results

This section compares the classification performance using features extracted after GAD and Otsu preprocessing. Results are reported separately depending on the conditions of the analysis: evaluation without patients that have risk or stage as undefined (uncertain) or dual diagnosis; evaluation that includes those previously excluded patients; undersampling experiment, without previously excluding any patients group.

Binary Classification – Cancer vs. Non-cancer

Table 5.1 evaluates the models' ability to distinguish between cancerous and non-cancerous samples. Patients with dual diagnoses — in this case, classified with both Tis and T1 stages — or with an undefined stage or risk were excluded from the initial analysis, although a separate analysis was conducted later. There was a split based on the patients, so that the model was not biased on that, resulting in no patient being present in both train and test. Also, two splits are performed within the cancerous cells group or the healthy cells group, to form a training and a test set. In each split, 60% of patients would go to the training set, while 40% would go to the test set. This does not necessarily mean 60% of the rows will be on train, as it depends on the number of cells detected in each patient. Finally, the random state variable in the split functions was always 42, a commonly used seed, which ensures reproducibility of the data and consistency of the results obtained. Furthermore, it summarizes the five machine learning models — Random Forest, Logistic Regression, XGBoost, LightGBM, and MLP — on three segmentation techniques: GAD, Otsu, and their combination. The performance of all models was compared on four vital metrics rounded to unity: Accuracy, Balanced Accuracy, Sensitivity, and Specificity.

Table 5.1: Binary classification results for each ML model using GAD and Otsu methods.

Method	**ML Model**	**Accuracy**	**Balanced Accuracy**	**Sensitivity**	**Specificity**
GAD	Random Forest	82%	60%	88%	31%
	Logistic Regression	67%	72%	66%	78%
	XGBoost	79%	67%	81%	52%
	LightGBM	71%	72%	71%	73%
	MLP	78%	63%	82%	45%
Otsu	Random Forest	84%	57%	93%	20%
	Logistic Regression	75%	68%	77%	60%
	XGBoost	83%	62%	89%	35%
	LightGBM	81%	70%	85%	56%
	MLP	83%	66%	89%	42%
Both	Random Forest	83%	67%	89%	45%
	Logistic Regression	69%	69%	69%	68%
	XGBoost	81%	61%	84%	59%
	LightGBM	76%	75%	77%	73%
	MLP	80%	72%	83%	61%

As the segmentation techniques were used, the Otsu method delivered slightly higher Accuracy and Sensitivity measures in most models. For instance, Random Forest's highest Sensitivity of 93% and Accuracy of 84% using Otsu was superior to using the same model on GAD, with 88% Sensitivity and 82% Accuracy. Nevertheless, the Otsu approach underperformed in the context

of Specificity, where Random Forest only achieved up to 20%, which tends to over-predict the positive class and yield more false positives.

Even with slightly lower accuracy, the GAD method tended to come up with more equilibrated models with higher Specificity. Logistic Regression with GAD, for example, had a Specificity of 78% and balanced accuracy of 72%, highlighting its consistency in detecting both classes.

The combination of both GAD and Otsu showed the most stable and equilibrium results. LightGBM, in particular, was outstanding with 76% Accuracy, 75% Balanced Accuracy, 77% Sensitivity, and 73% Specificity, yielding a more stable diagnostic accuracy across all scopes. MLP also showed competitive performance under this combined configuration with 80% Accuracy and 72% Balanced Accuracy.

Upon model comparison, Random Forest and MLP were both very Sensitive to all segmentation techniques, which would be desirable in the clinical environment where a lost case of cancer is expensive. These models were, however, often low on Specificity, particularly with the Otsu technique, and hence would decrease their overall reliability in detecting negative cases. Logistic Regression consistently yielded higher Specificity scores, especially when using GAD and combined segmentation, but this sometimes came at the expense of lower Sensitivity.

Combined segmentation in LightGBM yielded the most interpretable and balanced results, maximizing the trade-off between detecting positive cases and limiting false alarms. This reveals that segregating approaches may increase classification stability and relieve limitations experienced in monomethod solutions. These findings highlight the importance of selecting a strong classification algorithm and an optimal segmentation approach, as both play an equal role in diagnostic performance in bladder cancer biomarker detection.

Examples of the best-performance cases' confusion matrices are present in Figure A.1 in the Appendix.

Binary including Undefined Stage or Risk Patients

In this variation, we included patients with undefined stage or risk annotations as a separate category, both in training and evaluation. Satisfying the inclusion of undefined-stage or undefined-risk patients in the data is an important test for the robustness and generalization capacity of built classification models. These samples represent real-world ambiguity where clinical labels are missing or uncertain, constituting a severe challenge to automated systems. Table 5.2 illustrates the binary classification performance obtained with the GAD, Otsu, and combined segmentation methods on five machine learning models. Measurements such as Accuracy, Balanced Accuracy, Sensitivity, and Specificity are cited to provide comprehensive picture of performance.

The classification performance is improved in raw accuracy terms for nearly all techniques and models compared to experiments where cases that were not defined were excluded. However, this increase can be accounted for partially by the skewed effect of high-sensitivity scores, reflecting a bias toward the positive class, particularly when working with imprecise data. For this reason, Balanced Accuracy is especially important here since it adjusts for class imbalance and reflects whether a model is performing equally well in both classes.

Table 5.2: Binary classification results including undefined stage or risk patients for each ML model using GAD and Otsu methods.

Method	ML Model	Accuracy	Balanced Accuracy	Sensitivity	Specificity
GAD	Random Forest	90%	61%	96%	26%
	Logistic Regression	85%	75%	87%	64%
	XGBoost	89%	65%	93%	37%
	LightGBM	87%	79%	88%	70%
	MLP	88%	64%	93%	34%
Otsu	Random Forest	87%	60%	95%	26%
	Logistic Regression	81%	70%	85%	55%
	XGBoost	86%	63%	94%	32%
	LightGBM	85%	70%	89%	52%
	MLP	85%	61%	92%	31%
Both	Random Forest	88%	63%	98%	27%
	Logistic Regression	84%	72%	89%	56%
	XGBoost	88%	65%	97%	34%
	LightGBM	87%	76%	92%	60%
	MLP	87%	68%	94%	41%

LightGBM consistently demonstrated the highest Balanced Accuracy among the models, particularly with the GAD method (79%) and the combined method (76%). This indicates that LightGBM is more effective in balancing detecting positive and negative classes despite uncertain cases. Logistic regression was also performed relatively consistently, particularly with GAD (75%) and the combined method(72%), showing its ability to handle general trends and avoid overfitting noise.

On the other hand, models like MLP and Random Forest tended heavily to high sensitivity (for instance, up to 98% in the combined case for Random Forest). They, therefore, labeled a large proportion of samples as positive, including probable misclassifications of undefined or ambiguous cases. While this tendency is valuable in not missing potential positives, it is paid for through low specificity. For instance, Random Forest was as low as 26% specific with Otsu and GAD and 27% when combining the two preprocessing outputs. This confirms a general difficulty in correctly classifying negative instances of the class, especially when label certainty is not guaranteed.

Comparing preprocessing techniques, GAD regularly performed the best classification scores, with an excellent sensitivity/specificity trade-off. Its segmentation algorithm, which captures a more refined and complete cellular structure, is the explanation for generating more reliable features for classification. Otsu's faster and simpler method underperformed in most configurations by a narrow margin, with lower Balanced Accuracy and specificity values generally.

Appreciating undefined samples allows us to analyze how far the models generalize to

ambiguous or uncertain examples. The results show that while most of the models can maintain high sensitivity, only some — most significantly LightGBM and Logistic Regression — are the ones that obtain a reasonable balance and do not overpredict. We also discovered that undefined samples are predicted to the positive class, especially by lower-specificity models. This triggers significant issues regarding deployment within clinical environments, where care needs to be exercised so as not to over-predict pathological results in ambiguous situations.

Finally, this evaluation reaffirms the value of the GAD preprocessing pipeline as being the most optimal and reliable method in dealing with an ambiguous and heterogeneous data set. It also indicates that model choice considerably affects how ambiguity is treated and that sensitivity vs. specificity trade-offs must be effectively managed depending on the potential clinical application.

Examples of the best-performance cases' confusion matrices are present in Figure A.2 in the Appendix.

5.3.3 Undersampling Results

Table 5.3 presents binary classification results for predicting patient groups using undersampling to address class imbalance. In contrast to earlier experiments where the dataset was naturally imbalanced (often dominated by high-risk or cancerous samples), this version of the pipeline enforced a balanced distribution (50% each of healthy and cancerous patients) in both the training and test sets. Due to this enforced parity, **balanced accuracy becomes equivalent to regular accuracy**, and thus it is omitted in this analysis.

To further expand the learning capacity beyond traditional machine learning classifiers, we also explored a basic Convolutional Neural Network (CNN) to assess the feasibility of leveraging spatial representations directly from the images. However, due to the limited dataset size and lack of strong data augmentation or regularization strategies, CNN results are not included in the final comparative table, as the model underperformed relative to the ML baselines.

From the table, we observe that all models performed better under the GAD preprocessing pipeline than under Otsu, reinforcing earlier findings that GAD better preserves meaningful cell boundaries while suppressing irrelevant noise. The top-performing models using GAD were Random Forest, XGBoost, and LightGBM, all achieving an accuracy of 81% and maintaining a healthy balance between sensitivity (84%–85%) and specificity (77%–79%).

Otsu-based models consistently lagged behind, with overall lower specificity. For instance, Logistic Regression with Otsu preprocessing showed high sensitivity (84%) but poor specificity (55%), reflecting a tendency to over-predict the cancerous class, even under balanced conditions.

When combining both preprocessing approaches (GAD and Otsu), results were slightly less stable. Although sensitivity remained high — up to 90% in the MLP model — the specificity dropped, indicating a possible dilution of signal due to mixed feature origins. That said, the combined method still offered competitive performance, with Random Forest, XGBoost, and LightGBM all reaching 79% accuracy and balanced sensitivity–specificity trade-offs.

Overall, this experiment demonstrated that undersampling effectively levels classifier attention across classes, improving specificity without sacrificing sensitivity. GAD remains the preferred preprocessing strategy under balanced conditions, and while CNN exploration was insightful, further work is required to make deep learning viable in such low-data scenarios. Examples of the best-performance cases' confusion matrices are present in Figure A.3 in the Appendix.

Table 5.3: Binary classification results with undersampling including undefined stage or risk patients for each ML model using GAD and Otsu methods.

Method	ML Model	Accuracy	Sensitivity	Specificity
GAD	Random Forest	81%	84%	79%
	Logistic Regression	74%	86%	62%
	XGBoost	81%	85%	77%
	LightGBM	81%	84%	78%
	MLP	79%	86%	73%
Otsu	Random Forest	78%	82%	73%
	Logistic Regression	70%	84%	55%
	XGBoost	75%	83%	66%
	LightGBM	76%	83%	68%
	MLP	70%	86%	53%
Both	Random Forest	79%	89%	70%
	Logistic Regression	73%	88%	57%
	XGBoost	79%	89%	70%
	LightGBM	79%	89%	69%
	MLP	76%	90%	62%

5.3.4 Multiclass Classification – Cancer Stage and Cancer Risk

Table 5.4 presents multiclass classification results for stage and risk prediction tasks with five machine learning models under three preprocessing settings: GAD, Otsu, and a combination of both. All settings were evaluated with Accuracy and Balanced Accuracy metrics. To create a stratified and balanced evaluation, the test set was constructed to include 15% of the patients from each *defined* stage class and, separately, 30% of the patients from each *defined* risk class. Importantly, all patients with *undefined* labels — those with multi-tags or ambiguous labels — were *forcibly included* in the test split. This forced inclusion was done to challenge the capacity of models to generalize to uncertain or borderline clinical cases, similar to real-world diagnosis scenarios.

For stage classification, all models and approaches performed equally poorly. Accuracy typically ranged from 18% to 35%, balanced accuracy no greater than 28%. This poor

performance reflects the task's inherent challenge and morphological overlap between adjacent stages. Surprisingly, the Logistic Regression model trained with Otsu preprocessing achieved the best stage classification (35% accuracy and 26% balanced accuracy), which indicates that fewer preprocessing steps may prevent some segmentation mistakes from damaging class boundaries. A confusion matrix example is present in Figure A.4 in the Appendix.

By contrast, risk classification was more successful and more stable. Accuracies ranged from 45% to 59%, and balanced accuracy was stable between 50% and 55% across all settings. This suggests that the image features capture more discriminative patterns regarding patient risk than the stage. Somewhat unexpectedly, performance was not significantly affected by the preprocessing pipeline, although MLP and LightGBM, in conjunction with Otsu, yielded marginally better results. For example, MLP, in conjunction with Otsu preprocessing, obtained 59% accuracy and 55% balanced accuracy. A confusion matrix example is present in Figure A.5 in the Appendix.

It is also notable that all the values in Table 5.4 were *rounded to whole units* for the sake of readability and easier method-to-method comparisons. This rounding does not affect the underlying trends, but interpretability is improved.

Table 5.4: Multiclass Classification results for each ML model using GAD and Otsu methods.

Method	**ML Model**	**Stage**		**Risk**	
		Accuracy	**Balanced Accuracy**	**Accuracy**	**Balanced Accuracy**
GAD	Random Forest	20%	18%	54%	54%
	Logistic Regression	28%	22%	45%	54%
	XGBoost	20%	24%	53%	54%
	LightGBM	22%	25%	50%	55%
	MLP	18%	18%	50%	53%
Otsu	Random Forest	24%	23%	58%	52%
	Logistic Regression	35%	26%	52%	55%
	XGBoost	27%	28%	56%	52%
	LightGBM	28%	28%	54%	55%
	MLP	26%	27%	59%	55%
Both	Random Forest	21%	20%	53%	51%
	Logistic Regression	31%	25%	46%	53%
	XGBoost	23%	26%	50%	50%
	LightGBM	25%	28%	47%	52%
	MLP	21%	25%	50%	52%

5.3.5 Discussion of Results

The experimental results demonstrate several key insights about the performance of different segmentation approaches and machine learning models for medical image classification tasks. The comparative analysis reveals a fundamental trade-off between sensitivity and specificity across methods. While some approaches achieved high detection rates for positive cases, this often came at the expense of increased false positives. The hybrid segmentation method consistently provided the most balanced performance, suggesting that combining complementary techniques can improve overall classification robustness.

Class imbalance emerged as a significant factor affecting model performance. The undersampling experiments showed that balancing class distributions can help mitigate bias, though the optimal strategy depends on the specific clinical requirements of the application. Models that maintained reasonable performance across both majority and minority classes proved most reliable for general use.

The multiclass classification results were notably less accurate than binary classification, highlighting the inherent challenges of fine-grained medical categorization. The performance gap between different clinical categories suggests that some diagnostic distinctions may be more difficult to capture from imaging features alone.

Several practical considerations emerge from these findings. First, the choice of segmentation method should be guided by the clinical priorities of the specific use case. Second, model selection requires careful evaluation of both overall accuracy and class-specific performance metrics. Finally, the handling of ambiguous or borderline cases remains an important challenge that warrants further investigation.

These results provide a foundation for future work in medical image analysis, particularly in optimizing the balance between detection sensitivity and diagnostic specificity. The findings also underscore the importance of tailoring technical approaches to specific clinical requirements and constraints.

5.4 Overall Conclusions

This chapter presented a complete implementation and evaluation of our computer vision pipeline for bladder cancer detection from urine cytology images. Through systematic experimentation and analysis, we reached several significant conclusions that advance the field of automated cancer diagnostics.

The segmentation and preprocessing evaluation demonstrated that GAD (Gauss-Adaptive-Dilation) and Otsu's thresholding consistently outperformed other methods. While Otsu excelled in sensitivity by detecting 93% of cancerous cells, GAD demonstrated a superior balance between sensitivity and specificity. Both methods proved remarkably effective, with LightGBM achieving 76% accuracy and 75% balanced accuracy, suggesting that hybrid approaches may offer optimal performance in clinical settings.

Regarding classification capabilities, the pipeline demonstrated strong binary classification performance, with Random Forest and MLP models achieving up to 98% sensitivity - crucial for medical applications where false negatives carry severe consequences. However, multiclass classification proved more challenging, with stage prediction accuracy peaking at 35% and risk classification reaching 59%, reflecting the inherent difficulty in distinguishing subtle pathological gradations through image analysis alone.

The inclusion of undefined-stage patients in testing revealed important limitations in model generalization. While sensitivity remained high at 96% for Random Forest with GAD preprocessing, specificity dropped to 26% for the same configuration, highlighting the need for careful clinical interpretation of ambiguous cases and suggesting areas for future improvement in handling diagnostic uncertainty. Our experiments with undersampling strategies showed that balanced datasets improved specificity without compromising sensitivity, with Random Forest and XGBoost maintaining 81% accuracy under GAD preprocessing.

The comprehensive evaluation across multiple preprocessing methods and classification algorithms provides valuable insights for clinical implementation. The choice between GAD and Otsu preprocessing should be guided by clinical priorities - favoring GAD for balanced performance or Otsu for maximal sensitivity. These findings collectively demonstrate that while our pipeline shows strong potential for automated bladder cancer detection, particularly in binary classification, challenges remain in handling diagnostic ambiguity and subtle pathological gradations that will require further research.

Chapter 6

Conclusions and Future Work

6.1 Conclusions and Main contributions

This study investigated the potential of image-based computational techniques for supporting Bladder Cancer (BCa) diagnosis through the analysis of microscopic urine samples. Motivated by the clinical limitations of current diagnostic methods, such as cystoscopy and urine cytology, this work aimed to propose a non-invasive, low-cost system that could contribute to the early detection and follow-up of BCa.

Two segmentation approaches, Otsu thresholding and Gauss-Adaptive-Dilation (GAD), were selected based on their quantitative consistency and robustness across patient samples. These preprocessing methods provided viable contours for subsequent feature extraction, enabling the use of Machine Learning (ML) algorithms for various classification tasks. The features derived from segmented regions were used to train standard ML models targeting three diagnostic objectives: cancer detection, cancer staging, and cancer risk classification.

The quantitative evaluation demonstrated encouraging performance, particularly in binary classification for cancer detection, where the best balanced accuracy reached 79%. This performance highlights the practical potential of using computer vision (CV) techniques on cytological images to assist in diagnostic workflows. Although multiclass classification tasks presented greater challenges — likely due to class imbalance and biological variability — the results still indicated that morphological and textural patterns extracted from urine samples hold clinically relevant information.

While the proposed pipeline is not intended as a replacement for existing diagnostic tools, it may serve as a complementary method to support clinical decision-making. The approach developed in this work contributes to ongoing efforts in applying artificial intelligence and computer vision to non-invasive cancer diagnostics and reinforces the potential of urine cytology as a source of informative biomarkers.

6.2 Research Questions

To consolidate the contributions of this work, the final chapter of this dissertation addresses the key research questions posed at the outset. Based on the findings, implementation results, and overall pipeline development, we now reflect on each question and provide a synthesized response grounded in the evidence gathered throughout the project.

How can microscopic image analysis techniques identify and characterize biomarkers for the early detection of bladder cancer?

This study demonstrates that microscopic image analysis, when combined with appropriate preprocessing and segmentation techniques, allows for the extraction of morphological and textural patterns in cell structures that serve as potential biomarkers. These patterns, once quantified, offer valuable indicators of early-stage malignancy in urine cytology, supporting non-invasive diagnostic approaches.

What is the role of computer vision techniques in effectively improving the sensitivity and specificity of biomarker detection in urine samples?

Computer vision techniques enhance diagnostic performance by systematically isolating and analyzing cellular features with greater consistency than manual interpretation. Through image enhancement, region segmentation, and feature extraction, the pipeline reduces variability, increases contrast of relevant structures, and allows subtle differences between normal and abnormal cells to be more easily identified, ultimately improving both sensitivity and specificity.

In what ways do machine learning algorithms improve diagnostic accuracy and predictive capabilities for bladder cancer biomarkers?

The application of machine learning models to the extracted features enables effective classification of samples into diagnostic categories. These models learn complex patterns from training data and generalize to unseen samples, leading to high prediction accuracy. Their integration within the CV pipeline allows the system to support clinical decision-making by providing objective, data-driven assessments of cancer likelihood based on cellular imagery.

6.3 Future Work

Future research should prioritize the expansion and refinement of the dataset, ideally including more healthy patients, to allow for a binary balanced analysis, and well-annotated samples covering all cancer stages and risk profiles. This would allow more robust training and evaluation of classification models, particularly for underrepresented classes.

Additionally, improvements to segmentation could be explored by integrating deep learning-based methods capable of learning cell boundaries directly from data. End-to-end pipelines combining segmentation, feature extraction, and classification in a unified framework would further streamline the process and potentially improve accuracy.

The use of more advanced classifiers, such as ensemble learning methods or convolutional neural networks (CNNs), also warrants further investigation. In particular, incorporating shape, texture, and contextual information more explicitly may enhance the discriminative power of the extracted features.

Finally, collaboration with clinical experts and validation in real-world clinical settings are essential steps before such methods can be considered for integration into diagnostic practice. This would ensure the system's interpretability, usability, and compliance with clinical standards.

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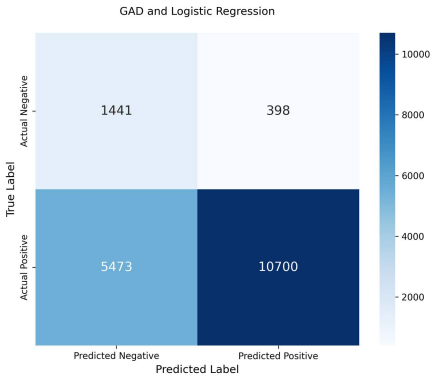
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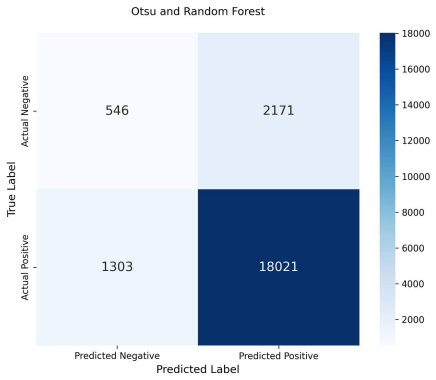
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Appendix A

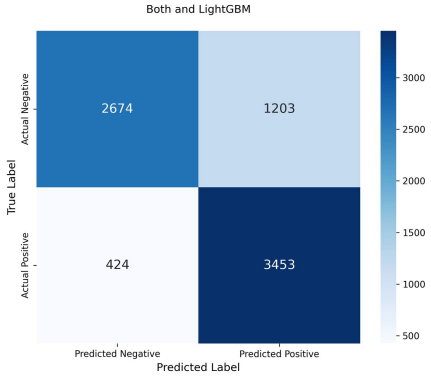
Confusion Matrices



Logistic Regression (GAD).

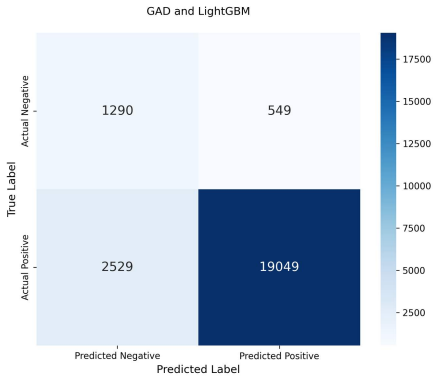


Random Forest (Otsu).

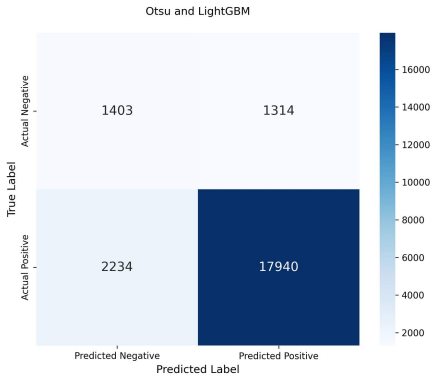


LightGBM (Both).

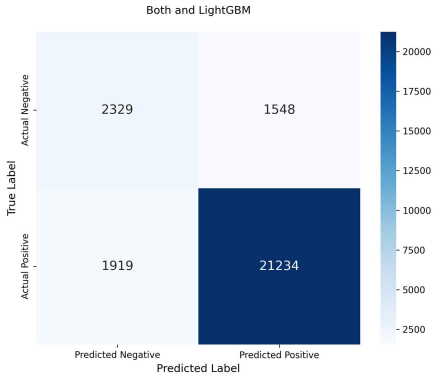
Figure A.1: Binary Classification Confusion Matrix.



LightGBM (GAD)

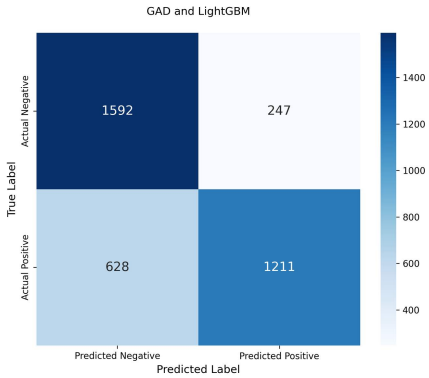


LightGBM (Otsu)

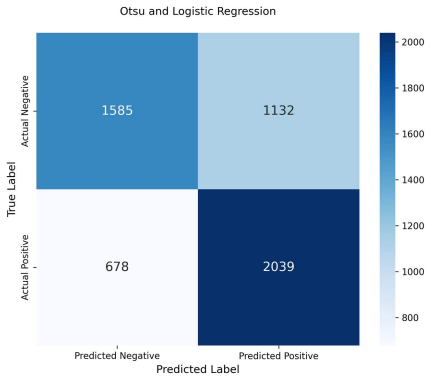


LightGBM (Both)

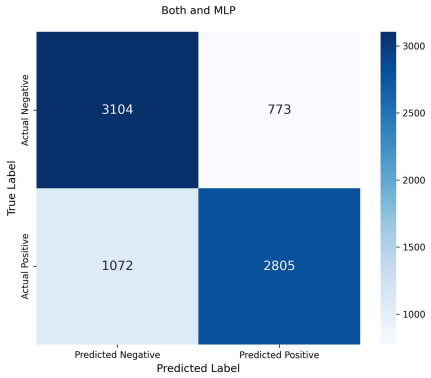
Figure A.2: Binary Classification Confusion Matrix with Undefined Patients.



LightGBM (GAD)



Logistic Regression (Otsu)



MLP (Both)

Figure A.3: Binary Classification Confusion Matrix with Undersampling.

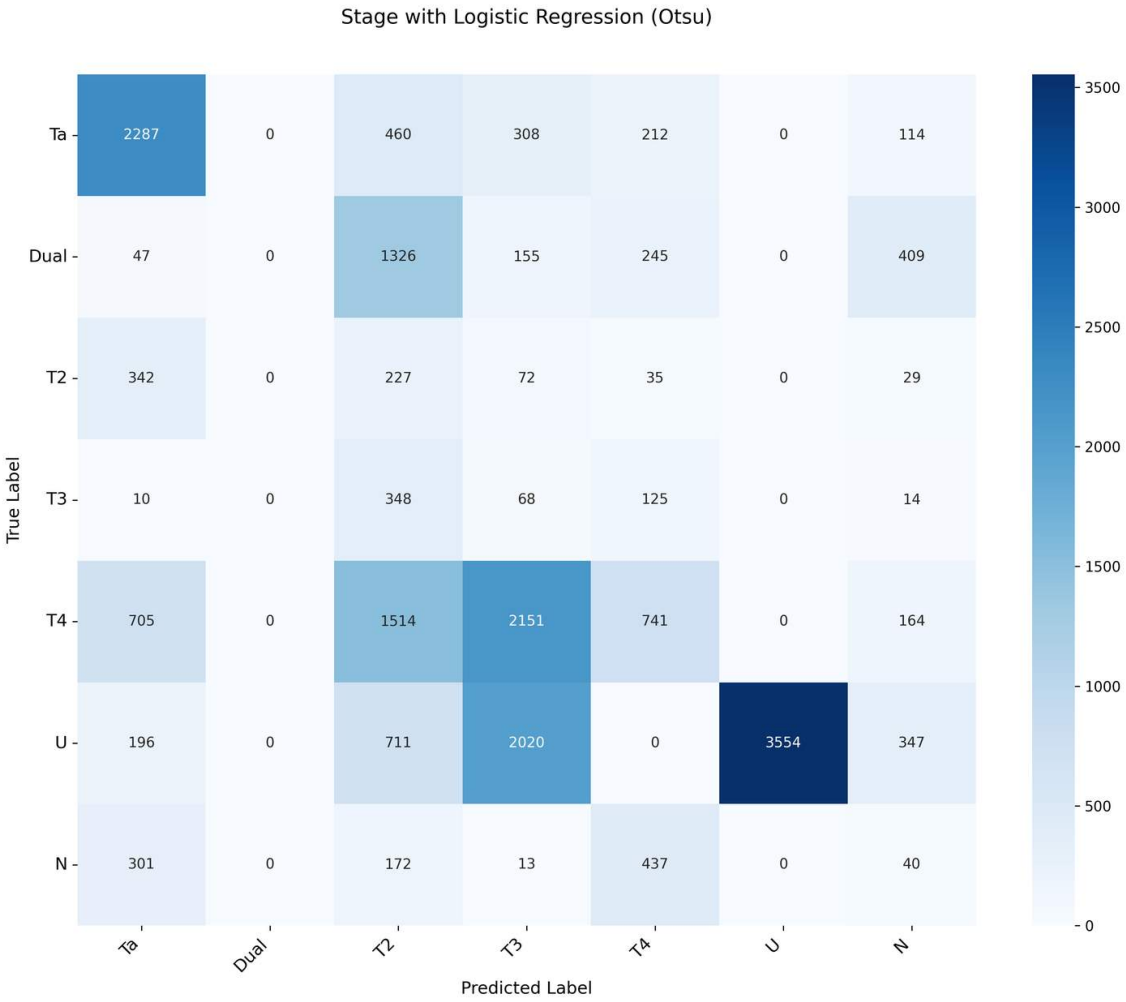


Figure A.4: Multi-Classification Confusion Matrix of Stage with Logistic Regression in Otsu.

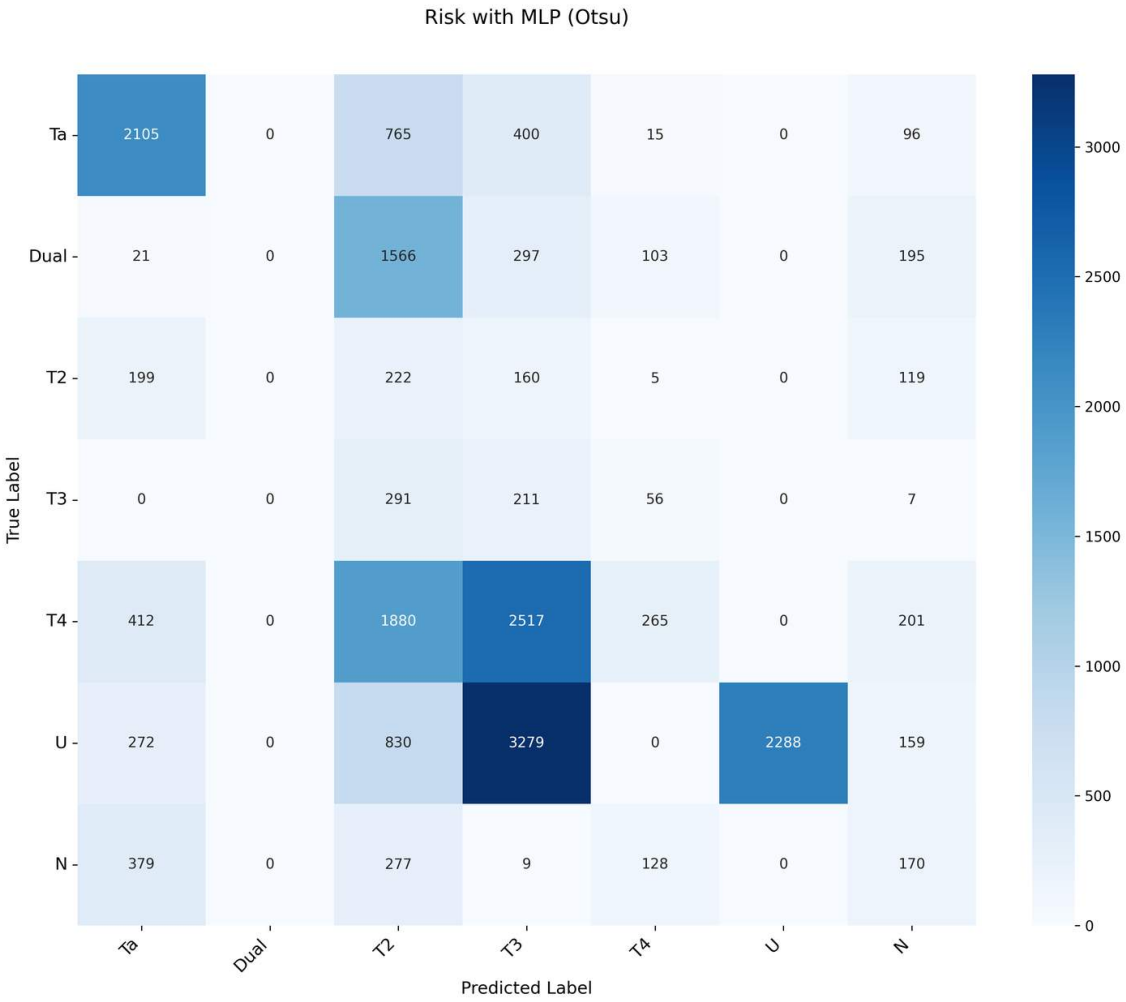


Figure A.5: Multi-Classification Confusion Matrix of Risk with MLP in Otsu.